

Chapter 3

Mobile Radio Propagation

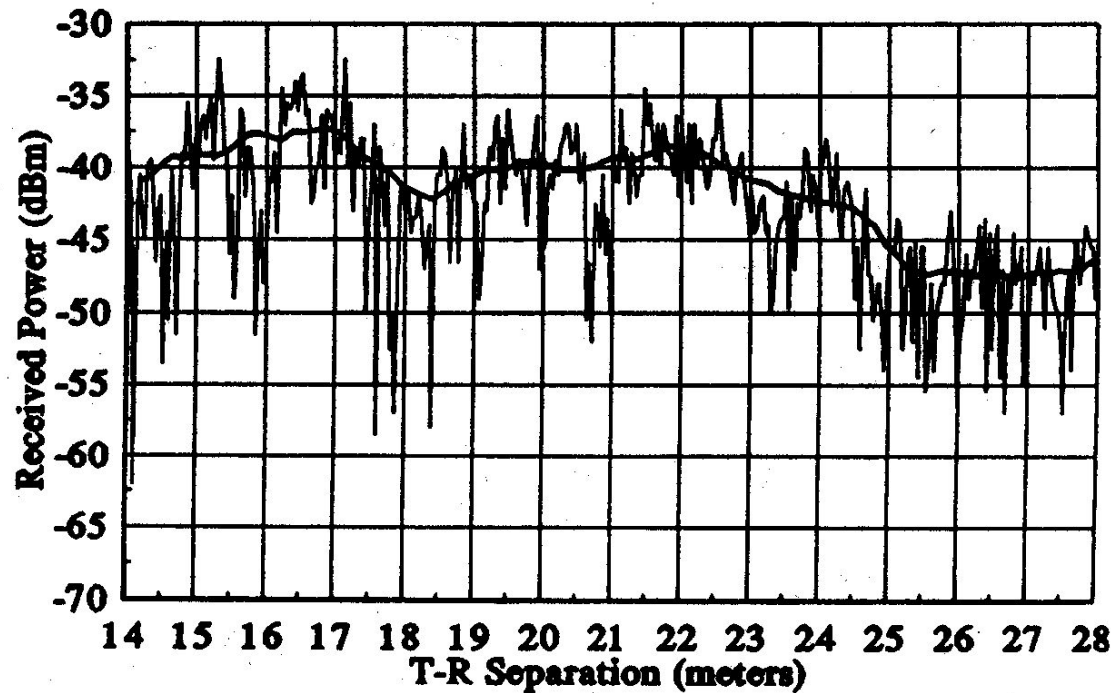
Large-Scale Path Loss

Large scale path loss

3.1 Introduction to Radio Wave Propagation

- Electromagnetic wave propagation
 - reflection
 - diffraction
 - scattering
- Urban areas
 - No direct line-of-sight
 - high-rise buildings causes severe diffraction loss
 - multipath fading due to different paths of varying lengths
- Large-scale propagation models predict the mean signal strength for an arbitrary T-R separation distance.
- Small-scale (fading) models characterize the rapid fluctuations of the received signal strength over very short travel distance or short time duration.

- Small-scale fading: rapidly fluctuation
 - sum of many contributions from different directions with different phases
 - random phases cause the sum varying widely. (ex: Rayleigh fading distribution)
- Local average received power is predicted by large-scale model (measurement track of 5λ to 40λ)



3.2 Free Space Propagation Model

- The free space propagation model is used to predict received signal strength when the transmitter and receiver have a clear line-of-sight path between them.
 - satellite communication
 - microwave line-of-sight radio link
- Friis free space equation

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

P_t : transmitted power

$P_r(d)$: received power

G_t : transmitter antenna gain

G_r : receiver antenna gain

d : T-R separation distance (m)

L : system loss

λ : wave length in meters

- The gain of the antenna

$$G = \frac{4\pi A_e}{\lambda^2}$$

A_e : effective aperture is related to the physical size of the antenna

- The wave length λ is related to the carrier frequency by

$$\lambda = \frac{c}{f} = \frac{2\pi c}{\omega_c}$$

f : carrier frequency in Hertz

ω_c : carrier frequency in radians

c : speed of light (meters/s)

- The losses L ($L \geq 1$) are usually due to transmission line attenuation, filter losses, and antenna losses in the communication system. A value of $L=1$ indicates no loss in the system hardware.

- Isotropic radiator is an ideal antenna which radiates power with unit gain.
- Effective isotropic radiated power (EIRP) is defined as

$$EIRP = P_t G_t$$

and represents the maximum radiated power available from transmitter in the direction of maximum antenna gain as compared to an isotropic radiator.

- Path loss for the free space model with antenna gains

$$PL(dB) = 10 \log \frac{P_t}{P_r} = -10 \log \left(\frac{G_t G_r \lambda^2}{(4\pi)^2 d^2} \right)$$

- When antenna gains are excluded

$$PL(dB) = 10 \log \frac{P_t}{P_r} = -10 \log \left(\frac{\lambda^2}{(4\pi)^2 d^2} \right)$$

- The Friis free space model is only a valid predictor for P_r for values of d which is in the far-field (Fraunhofer region) of the transmission antenna.

- The far-field region of a transmitting antenna is defined as the region beyond the far-field distance

$$d_f = \frac{2D^2}{\lambda}$$

where D is the largest physical linear dimension of the antenna.

- To be in the far-field region the following equations must be satisfied

$$d_f \gg D \quad \text{and} \quad d_f \gg \lambda$$

- Furthermore the following equation does not hold for $d=0$.

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

- Use close-in distance d_0 and a known received power $P_r(d_0)$ at that point

$$P_r(d) = P_r(d_0) \left(\frac{d_0}{d} \right)^2 \quad d \geq d_0 \geq d_f$$

or

$$P_r(d) \text{ dBm} = 10 \log \left(\frac{P_r(d_0)}{0.001 \text{ W}} \right) + 20 \log \left(\frac{d_0}{d} \right) \quad d \geq d_0 \geq d_f$$

The pathloss for the 2 Ray model is given by

$$PL(dB) = 40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

To calculating the received signal strength and path loss exponent.

$$P_r(d) = P_r(d_0) + 10n_p \log_{10}\left(\frac{d_0}{d}\right) \dots\dots\dots(1)$$

$P_r(d)$ = received signal strength for a certain Tx- Rx separation distance,

d = certain Tx -Rx separation distance in meters,

$P_r(d_0)$ = received signal strength at a close-in-reference-distance,

n_p = path loss exponent

Advanced Formula:-

$$\bullet PL = 10n_p \log_{10}(d) + 7.8 - 18 \log_{10}(h_{BS}) - 18 \log_{10}(h_{UT}) + 20 \log_{10}(f_c)$$

Where,

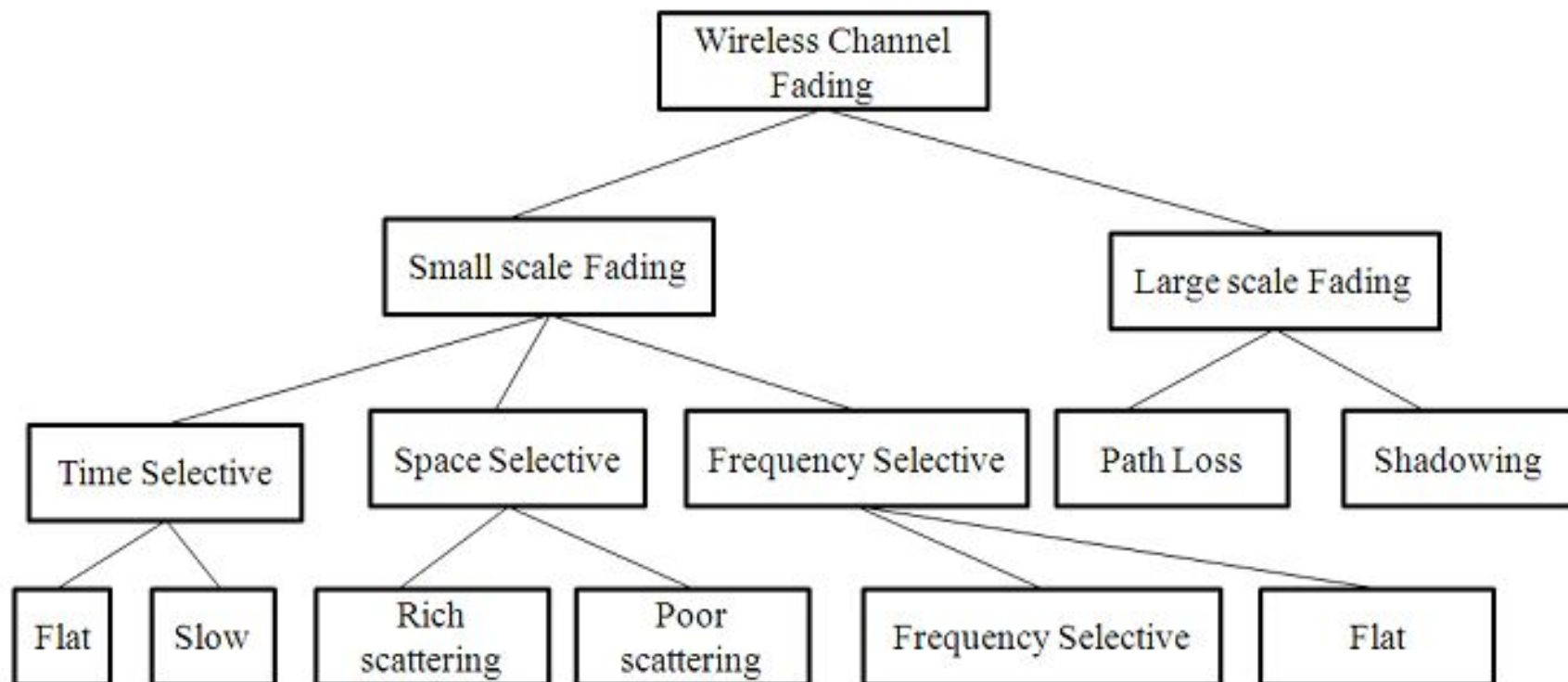
d = Tx – Rx, i.e. ; Tx and Rx separation distance in meters.

h_{BS} = the base station antenna height in meters.

h_{UT} = the user terminal i.e. receiver antenna height in meters.

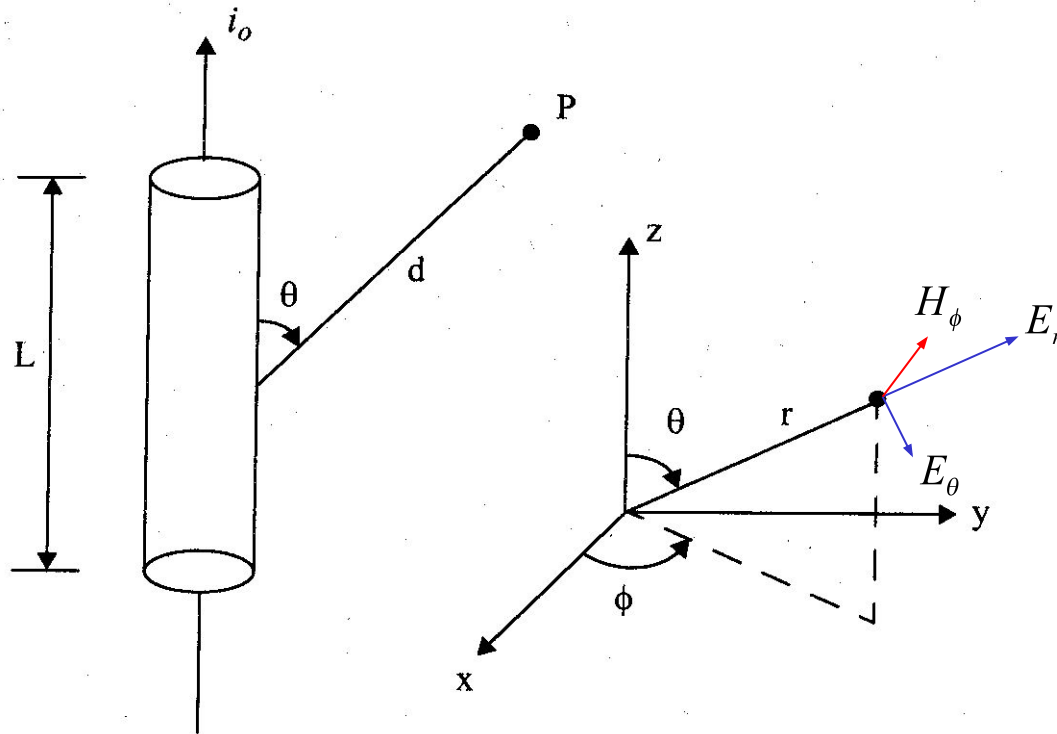
f_c is the carrier frequency in GHz.

Environment	Pathloss Slope n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line of sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3



3.3 Relating Power to Electric Field

- Consider a small linear radiator of length L



- Electric and magnetic fields for a small linear radiator of length L

$$E_r = \frac{i_0 L \cos \theta}{2\pi\epsilon_0 c} \left\{ \frac{1}{d^2} + \frac{c}{j\omega_c d^3} \right\} e^{j\omega_c(t-d/c)}$$

$$E_\theta = \frac{i_0 L \sin \theta}{2\pi\epsilon_0 c} \left\{ \frac{j\omega_c}{d} + \frac{c}{d^2} + \frac{c^2}{j\omega_c d^3} \right\} e^{-j\omega_c(t-d/c)}$$

$$H_\phi = \frac{i_0 L \sin \theta}{4\pi c} \left\{ \frac{j\omega_c}{d} + \frac{c}{d^2} \right\} e^{j\omega_c(t-d/c)}$$

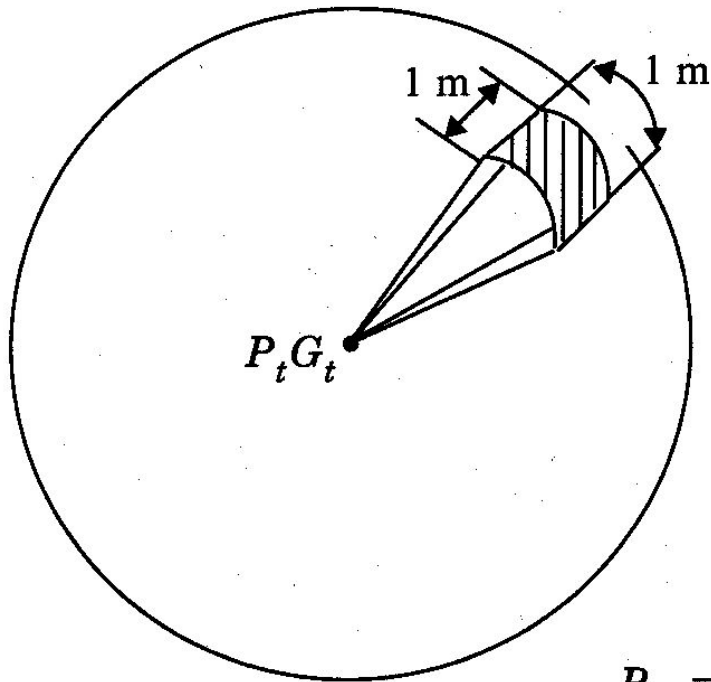
with $E_\phi = H_r = H_\theta = 0$

- At the region far away from the transmitter only E_θ and H_ϕ need to be considered.
- In free space, the power flux density is given by

$$P_d = \frac{EIRP}{4\pi d^2} = \frac{P_t G_t}{4\pi d^2} = \frac{|E|^2}{R_{fs}} = \frac{|E|^2}{\eta} \text{ W / m}^2$$

- where R_{fs} is the intrinsic impedance of free space given by $\eta = 120\pi \text{ } \Omega$

$$P_d = \frac{|E|^2}{377 \text{ } \Omega} \text{ W / m}^2$$



(a)

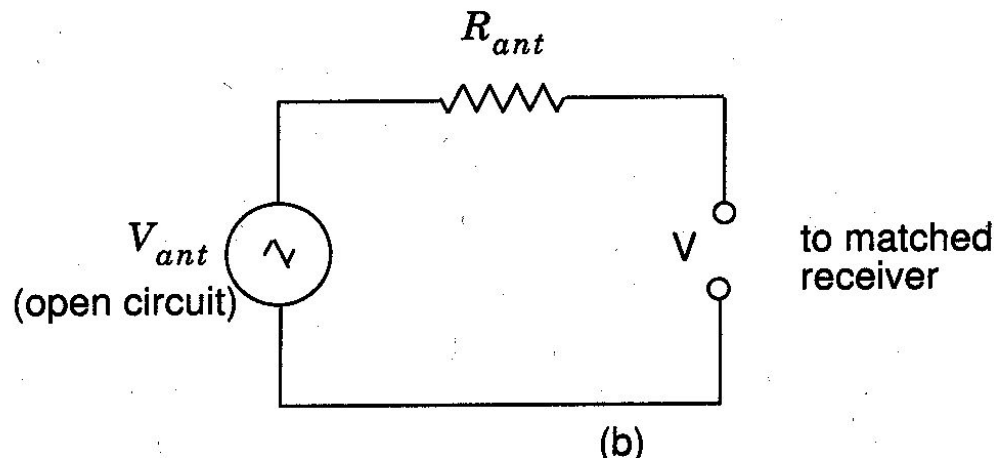
$$P_d = \frac{P_t G_t}{4\pi d^2} = \frac{EIRP}{4\pi d^2} = \frac{|E|^2}{120\pi} \text{ W/m}^2$$

- The power received at distance d is given by the power flux density times the effective aperture of the receiver antenna

$$P_r(d) = P_d A_e = \frac{|E|^2}{120\pi} A_e = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2} \text{ Watts}$$

- If the receiver antenna is modeled as a matched resistive load to the receiver, the received power is given by

$$P_r(d) = \frac{(V/2)^2}{R_{ant}} = \frac{V^2}{4R_{ant}}$$



3.4 The Three Basic Propagation Mechanisms

- Basic propagation mechanisms
 - reflection
 - diffraction
 - scattering
- **Reflection** occurs when a propagating electromagnetic wave impinges upon an object which has very large dimensions when compared to the wavelength, e.g., buildings, walls.
- **Diffraction** occurs when the radio path between the transmitter and receiver is obstructed by a surface that has sharp edges.
- **Scattering** occurs when the medium through which the wave travels consists of objects with dimensions that are small compared to the wavelength.

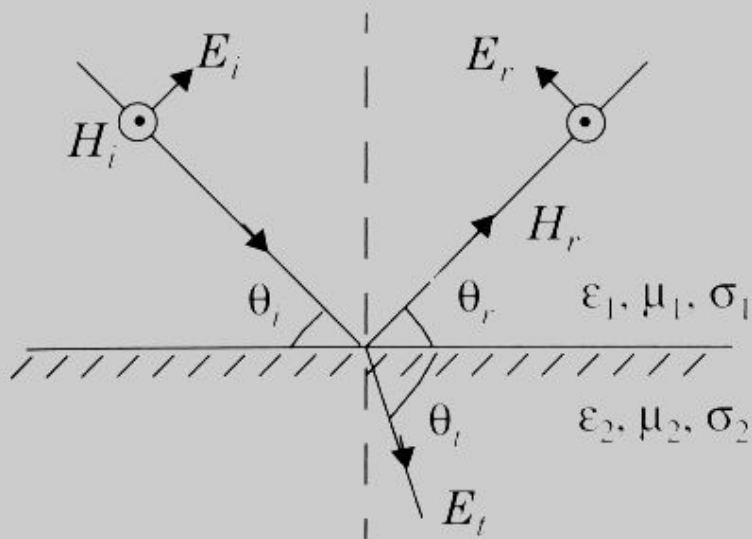
Reflection

- **Reflection** occurs when a propagating electromagnetic wave impinges upon an object which has very **large dimensions when compared to the wavelength**, e.g., buildings, walls.
- Reflection occurs when RF energy is incident upon a boundary between two materials (e.g air/ground) with different electrical characteristics
 - Permittivity μ
 - Permeability ϵ
 - Conductance σ
- Reflecting surface must be large relative to λ of RF energy
- Reflecting surface must be smooth relative to λ of RF energy
 - “specular” reflection

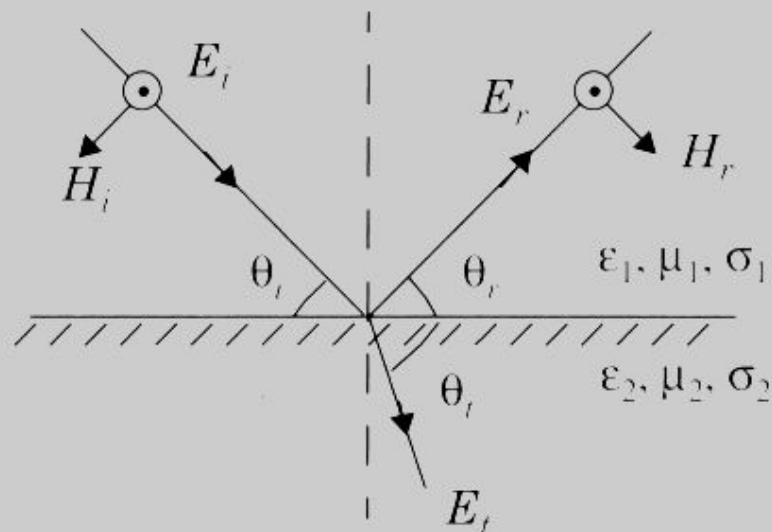
Reflection

- Perfect conductors reflect with no attenuation
 - Like light to the mirror
- Dielectrics reflect a fraction of incident energy
- Reflection coefficient of ground
 - (a) *vertical polarization* (v) or E field in the plane of incidence.
 - (b) *horizontal polarization* (h) or E field perpendicular to the incident plane

Reflection from Dielectric smooth surface



(a) E-field in the plane of incidence



(b) E-field normal to the plane of incidence

Figure 4.4 Geometry for calculating the reflection coefficients between two dielectrics.

$$\mathcal{E} = \epsilon_0 \epsilon_r - j\epsilon' \quad \text{For lossy dielectric material}$$

$$\mathcal{E} = \epsilon_0 \epsilon_r \quad \text{For perfect dielectric}$$

Reflection from perfect conductors

- E-field in the plane of incidence

$$\theta_i = \theta_r \quad \text{and} \quad E_i = E_r$$

- E-field normal to the plane of incidence

$$\theta_i = \theta_r \quad \text{and} \quad E_i = -E_r$$

Reflection

Can show that

$$\Gamma_v = \frac{E_r}{E_i} = \frac{-\eta_2 \sin \theta_t + \eta_1 \sin \theta_i}{\eta_2 \sin \theta_t + \eta_1 \sin \theta_i}$$

$$\Gamma_h = \frac{E_r}{E_i} = \frac{\eta_2 \sin \theta_i - \eta_1 \sin \theta_t}{\eta_2 \sin \theta_i + \eta_1 \sin \theta_t}$$

where $\eta_i = \sqrt{\mu_i / \epsilon_i}$, $\mu_i, \epsilon_i, \sigma_i$ are the permittivity, permeability and conductance of the media, and $\epsilon_i = \epsilon_0 \epsilon_{r_i} - j\sigma_i / (2\pi f)$. Other relationships required in deriving the results : $n_i = \sqrt{\mu_i \epsilon_i}$, $H \propto E / \eta_i$.

Reflection

Snell's law : $\theta_i = \theta_r, \frac{\sin(90 - \theta_i)}{\sin(90 - \theta_t)} = \frac{n_2}{n_1}$

Boundary conditions :

$$[E_1]_{//} = [E_2]_{//} \quad [H_1]_{//} = [H_2]_{//}$$

$$[\varepsilon_1 E_1]_{\perp} = [\varepsilon_2 E_2]_{\perp} \quad [\mu_1 H_1]_{\perp} = [\mu_2 H_2]_{\perp}$$

Reflection

If medium 1 is free space ($\varepsilon = \varepsilon_0$) and medium 2 is an dielectric, then $\Gamma_h \rightarrow -1, \Gamma_v \rightarrow -1$ regardless of ε_r if $\theta_i \rightarrow 0^\circ$. Ground may be modeled as a perfect reflector (reflection coefficient of unit magnitude) when an incident wave grazes the earth, regardless of polarization or ground dielectric properties.

Brewster Angle

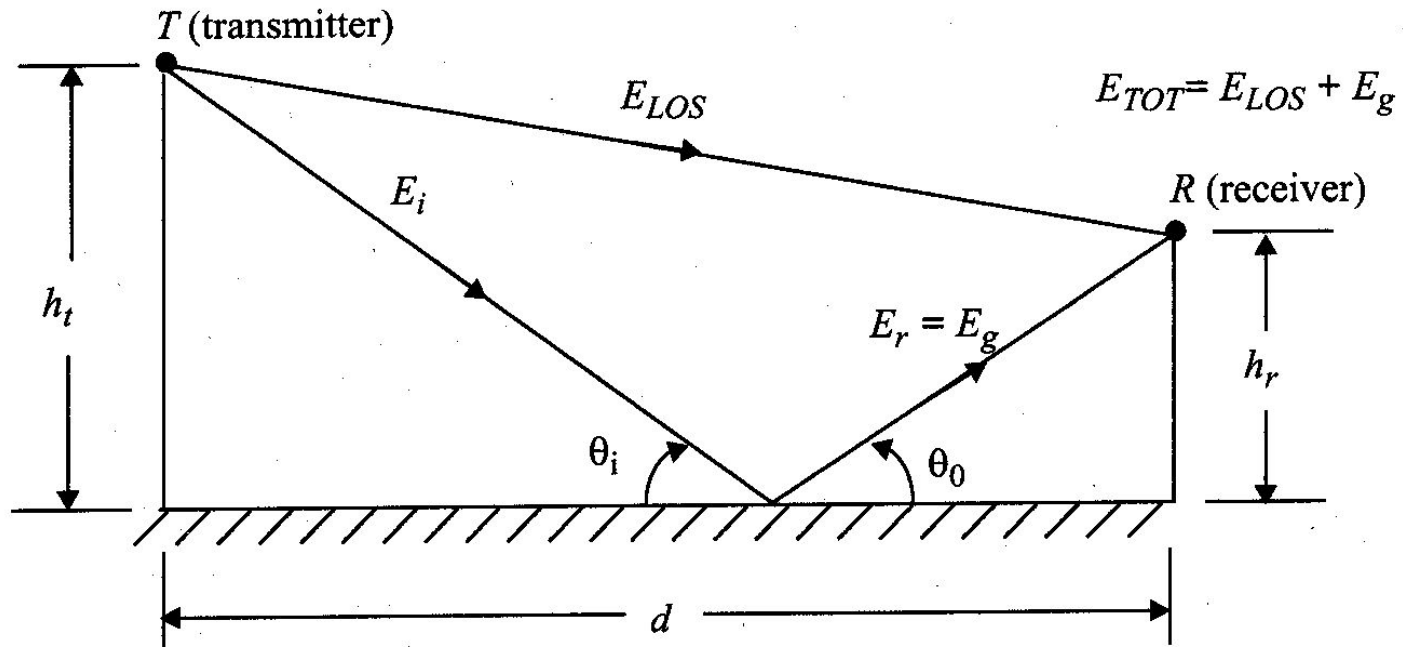
- It is the angle at which no reflection occurs in the medium of origin. It occurs when the incident angle θ_B is such that the reflection coefficient Γ_{II} is equal to zero. And

- $\sin \theta_B = \sqrt{\frac{\epsilon_1}{\epsilon_1 + \epsilon_2}}$

- It is only for vertical polarization.

- In case first medium is free space and 2nd medium
 $\epsilon_r = \epsilon_r$ then $\sin \theta_B = \sqrt{\epsilon_r - 1}$

- Ground Reflection (2-ray) Model



- E_{TOT} is the electric field that results from a **combination** of a direct line-of-sight

$$\vec{E}_{TOT} = \vec{E}_{LOS} + \vec{E}_g$$

let E_0 be $|\vec{E}|$ at reference point d_0 then

$$\vec{E}(d, t) = \left(\frac{E_0 d_0}{d} \right) \cos \left(\omega_c \left(t - \frac{d}{c} \right) \right) \quad d > d_0$$

- $\frac{E_0 d_0}{d}$ is the amplitude of the electric field at distance d
- $\omega_c = 2\pi f_c$ where f_c is the carrier frequency of the signal
- Notice at different distances d the wave is at a different phase because of the form similar to

$$\vec{E}(d, t) = A \cos(\omega_c t + \theta)$$

$$\mathbf{E}_g = \Gamma \mathbf{E}_i$$

$$\mathbf{E}_t = (1 + \Gamma) \mathbf{E}_i$$

For the direct path let $d = d'$;

for the reflected path $d = d''$ then

$$\bar{E}_{TOT}(d, t) = \left(\frac{E_0 d_0}{d'} \right) \cos \left(\omega_c \left(t - \frac{d'}{c} \right) \right) + \Gamma \left(\frac{E_0 d_0}{d''} \right) \cos \left(\omega_c \left(t - \frac{d''}{c} \right) \right)$$

- for large T–R separation : θ_i goes to 0 (angle of incidence to the ground of the reflected wave) and

$$\Gamma = -1$$

- **Phase difference** can occur depending on the phase difference between direct and reflected E fields
- The **phase difference** is θ_Δ due to **Path difference**,
- Path difference($\Delta = d'' - d'$), between $\bar{E}_{LOS}$ and $\bar{E}_g$

Method of image

From two triangles with sides d and $(h_t + h_r)$ or $(h_t - h_r)$

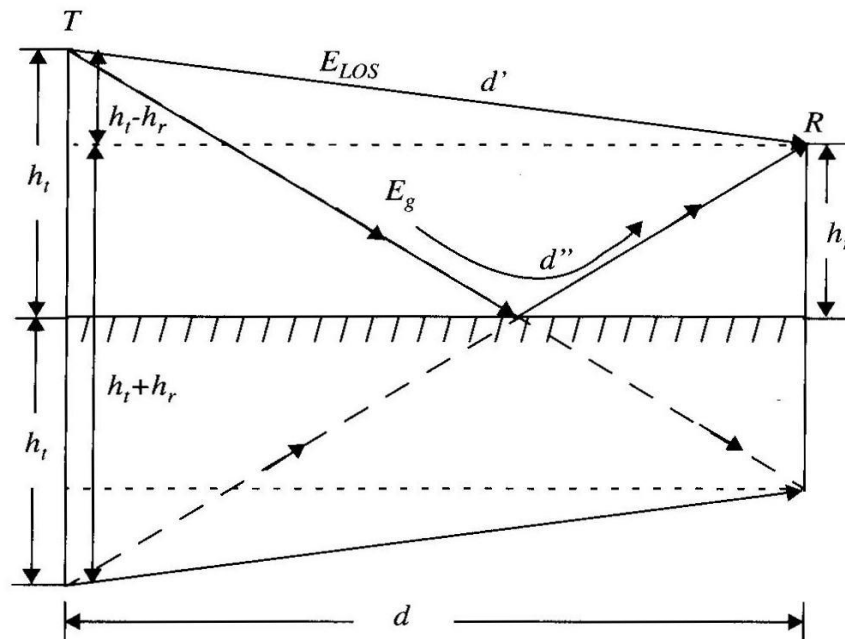


Figure 4.8 The method of images is used to find the path difference between the line-of-sight and the ground reflected paths.

$$\Delta = d'' - d' = \sqrt{(h_t + h_r)^2 + d^2} - \sqrt{(h_t - h_r)^2 + d^2}$$

- Δ can be expanded using a Taylor series expansion

$$\begin{aligned} f(x) = \sqrt{1+x} &= f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \dots \\ &= 1 + x \frac{1}{2} (1+0)^{-1/2} + \frac{x^2}{2!} \left(-\frac{1}{4} \right) (1+0)^{-3/2} + \dots \\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots \end{aligned}$$

if x is small $f(x) \approx 1 + \frac{1}{2}x$

$$\begin{aligned}
\Delta = d'' - d' &= \sqrt{(h_t + h_r)^2 + d^2} - \sqrt{(h_t - h_r)^2 + d^2} \\
&= d \sqrt{\left(\left(\frac{h_t + h_r}{d} \right)^2 + 1 \right)} - d \sqrt{\left(\left(\frac{h_t - h_r}{d} \right)^2 + 1 \right)} \\
&\approx d \left(1 + \frac{1}{2} \left(\frac{h_t + h_r}{d} \right)^2 \right) - d \left(1 + \frac{1}{2} \left(\frac{h_t - h_r}{d} \right)^2 \right) \\
&\approx \frac{1}{2d} \left((h_t + h_r)^2 - (h_t - h_r)^2 \right) \\
&\approx \frac{1}{2d} \left((h_t^2 + 2h_t h_r + h_r^2) - (h_t^2 - 2h_t h_r + h_r^2) \right) \\
&\approx \frac{2h_t h_r}{d}
\end{aligned}$$

- which works well for $d \gg (h_t + h_r)$, which means

and $\left(\frac{h_t + h_r}{d} \right)^2$ and $\left(\frac{h_t - h_r}{d} \right)^2$ are small

- the phase difference between the two arriving signals is

$$\theta_{\Delta} \text{ radians} = \left(\frac{\Delta}{\lambda} \text{ wavelengths} \right) \left(\frac{2\pi \text{ radians}}{\text{wavelength}} \right) = \frac{2\pi\Delta}{\lambda} = \frac{2\pi\Delta}{\left(\frac{c}{f_c} \right)} = \frac{\omega_c \Delta}{c}$$

$$|E_{TOT}(t)| = 2 \frac{E_0 d_0}{d} \sin \left(\frac{\theta_{\Delta}}{2} \right)$$

$$\frac{\theta_{\Delta}}{2} \approx \frac{2\pi h_r h_t}{\lambda d} < 0.3 \text{ rad}$$

$$E_{TOT}(t) \approx 2 \frac{E_0 d_0}{d} \frac{2\pi h_r h_t}{\lambda d} \approx \frac{k}{d^2} \text{ V/m}$$

- note that the magnitude is with respect to a reference of $E_0=1$ at $d_0=100$ meters, so near 100 meters the signal can be stronger than $E_0=1$
- The power received @ d is square of E.F
- $P_r(d) = P_d A_e = \frac{|E|^2}{120\pi} A_e = \frac{P_t G_t G_r \lambda^2}{4\pi^2 d^2}$
- for large distances $d \gg \sqrt{h_t h_r}$ it can be shown that

$$|\vec{E}_{TOT}(d)| \approx \frac{4\pi E_0 d_0 h_t h_r}{\lambda d^2}$$

$$P_r \approx \frac{P_t G_t G_r h_t^2 h_r^2}{d^4}$$

$$\frac{P_r}{P_t} = G_t G_r \frac{h_t^2 h_r^2}{d^4} \text{ or}$$

$$L_{dB} = 40 \log_{10} d - [20 \log_{10}(h_t h_r) + 10 \log_{10}(G_t G_r)]$$

Two-ray path loss model:

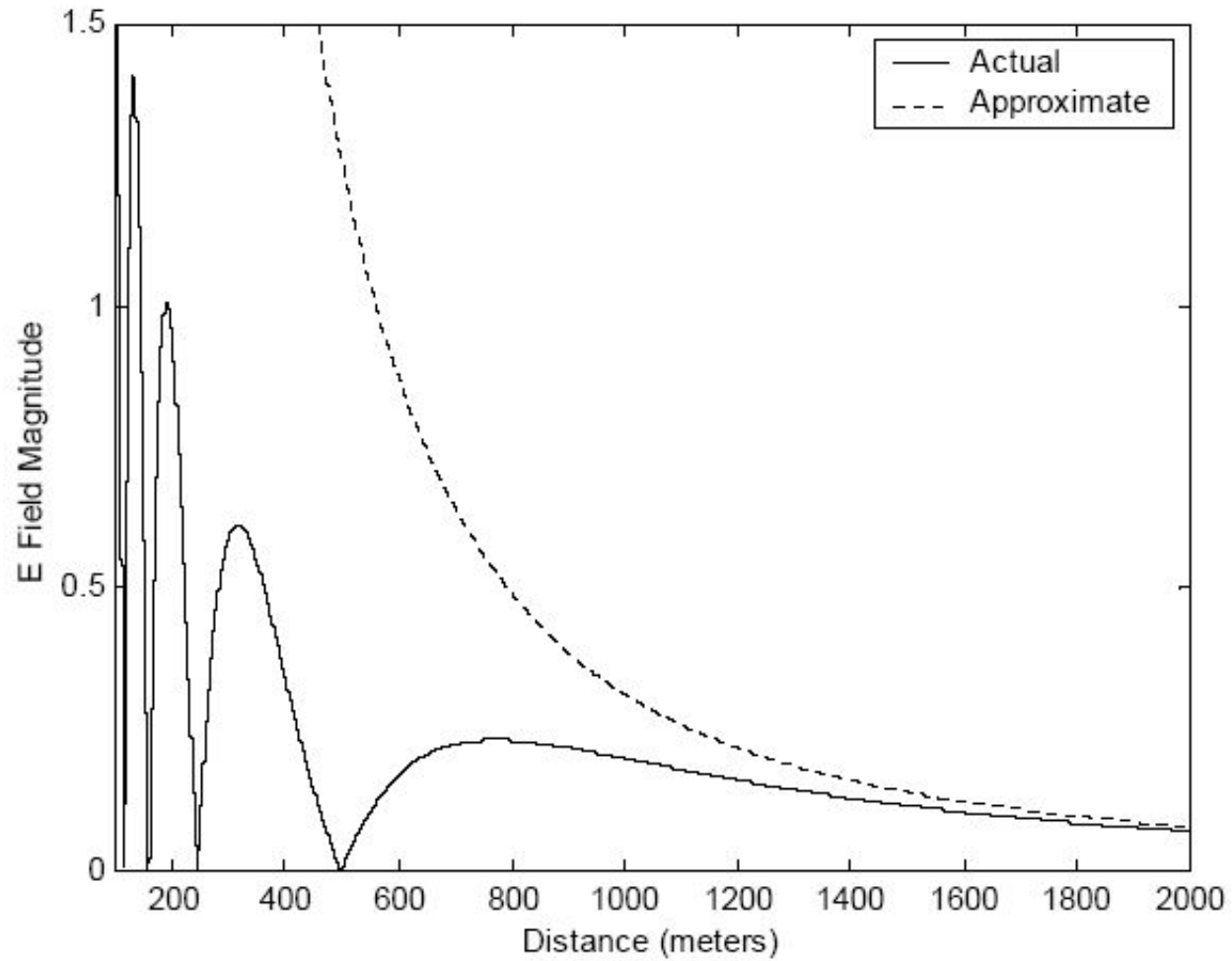
$$PL \text{ (dB)} = 40 \log d - [10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r]$$

Now d^4 instead of d^2 for free space

- $P_r \propto \frac{1}{d^4} P_t$

- $P_r(d) = P_r(d_0) \left(\frac{d_0}{d} \right)^4$

- For $d_0=100\text{meter}$, $E_0=1$, $f_c=1\text{ GHz}$, $h_t=50\text{ meters}$, $h_r=1.5\text{ meters}$, at $t=0$



The LOS is itself subject to free space loss

$$P_{LOS} = G_t G_r P_t \left(\frac{\lambda}{4\pi d'} \right)^2$$

Assuming small incidence angle θ_i ($\Gamma = -1$), then

$$\frac{P_r}{P_t} = G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 |1 - \exp(j\Delta\phi)|^2 = G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2 4 \sin^2 \left(\frac{\Delta\phi}{2} \right)$$

Diffraction

Phenomena: Radio signal can propagate around the curved surface of the earth, beyond the horizon and behind obstructions.

Radio path between transmitter and receiver obstructed by surface with sharp irregular edges.

Although the received field strength decreases rapidly as a receiver moves deeper into the obstructed (shadowed) region, the diffraction field still exists and often has sufficient strength to produce a useful signal.

The field strength of a diffracted wave in the shadowed region is the vector sum of the electric field components of all the secondary wavelets in the space around the obstacles.

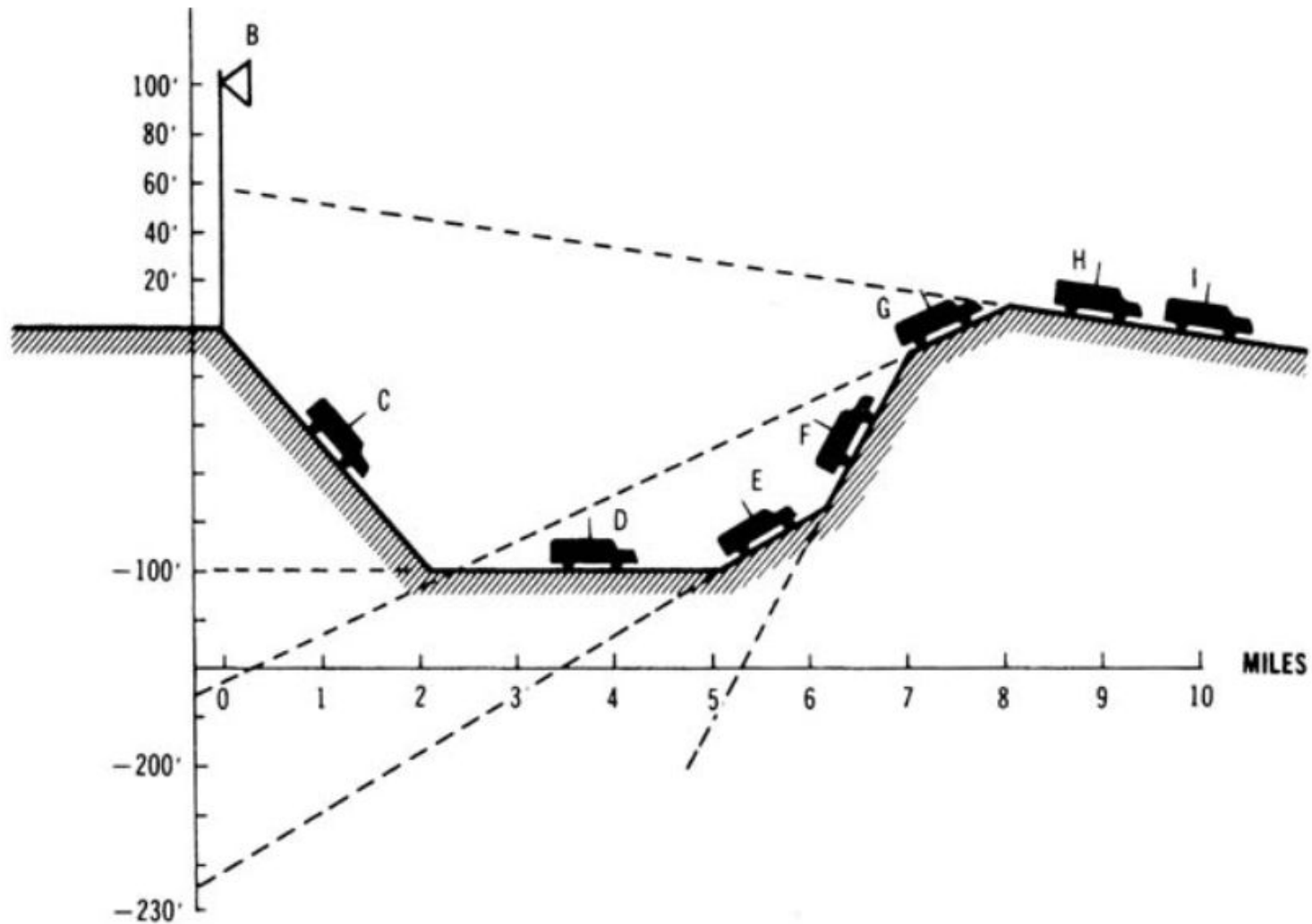
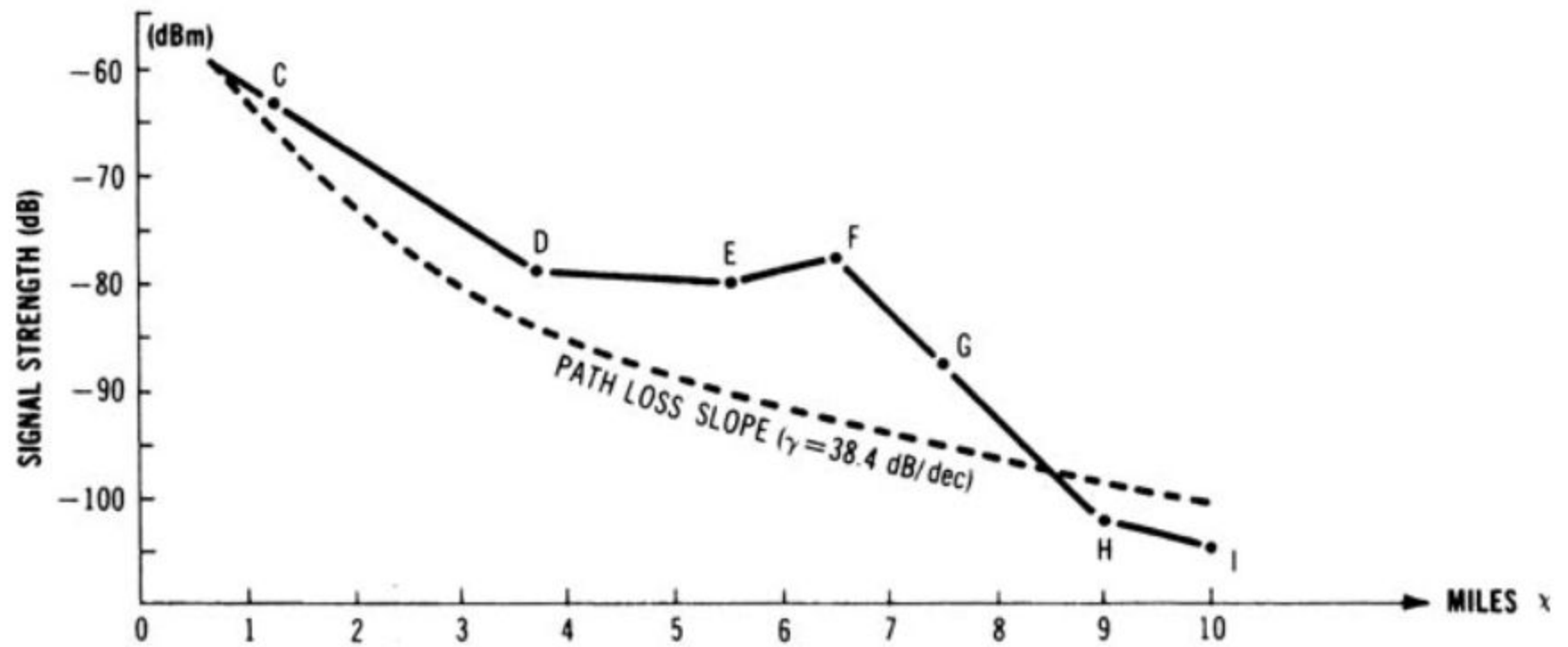


Illustration of the terrain effect on the effective antenna gain at each position. (a) Hilly terrain contour



Knife-edge Diffraction Model

It is essential to estimate the signal attenuation caused by diffraction of radio waves over hills and buildings in predicting the field strength in the given service area.

In practice, prediction for diffraction loss is a process of theoretical approximation modified by necessary empirical corrections.

The simplest case: shadowing is caused by a single object such as a hill or mountain.

Diffraction

RF energy can propagate:

- around the curved surface of the Earth
- beyond the line-of-sight horizon
- Behind obstructions

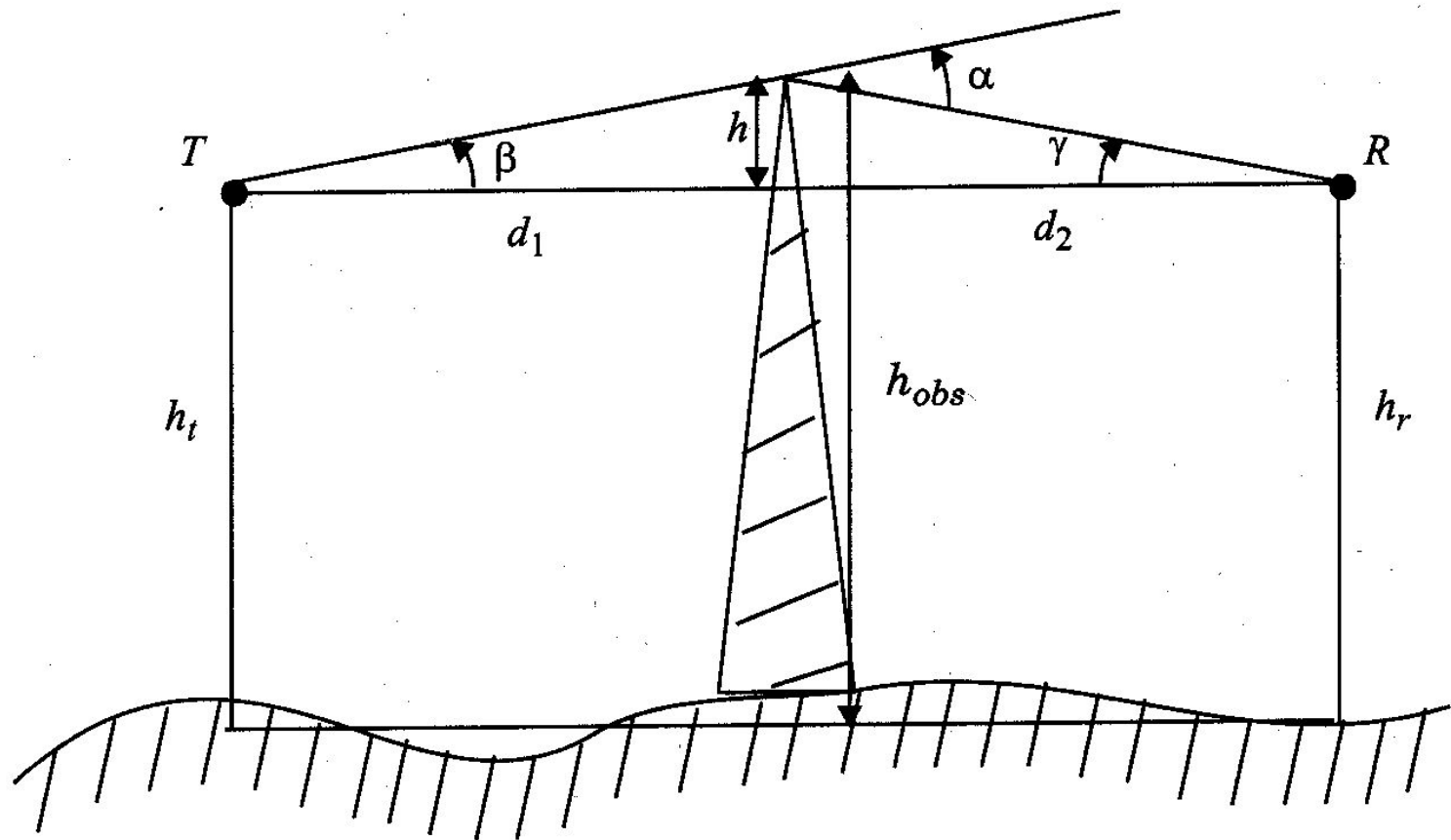
Although EM field strength decays rapidly as Rx moves deeper into “shadowed” or obstructed (OBS) region

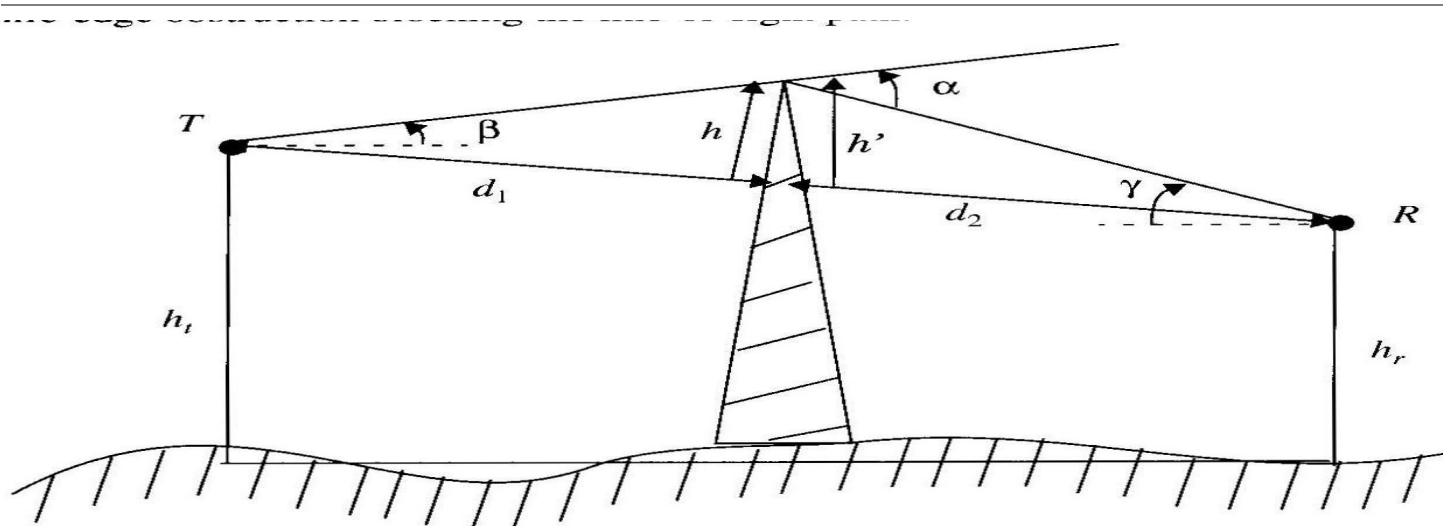
The diffraction field often has sufficient strength to produce a useful signal

Huygen's principle says ,all points on a wavefront can be considered as point sources for the production of secondary wavelets, and these wavelets combine to produce a new wavefront in the direction of propagation.

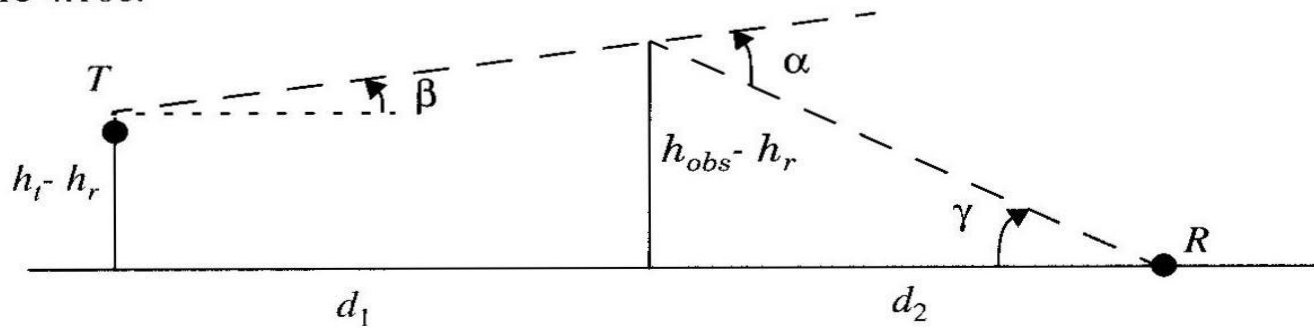
Diffraction caused by propagation of secondary wavelets into a shadowed region.

- Diffraction



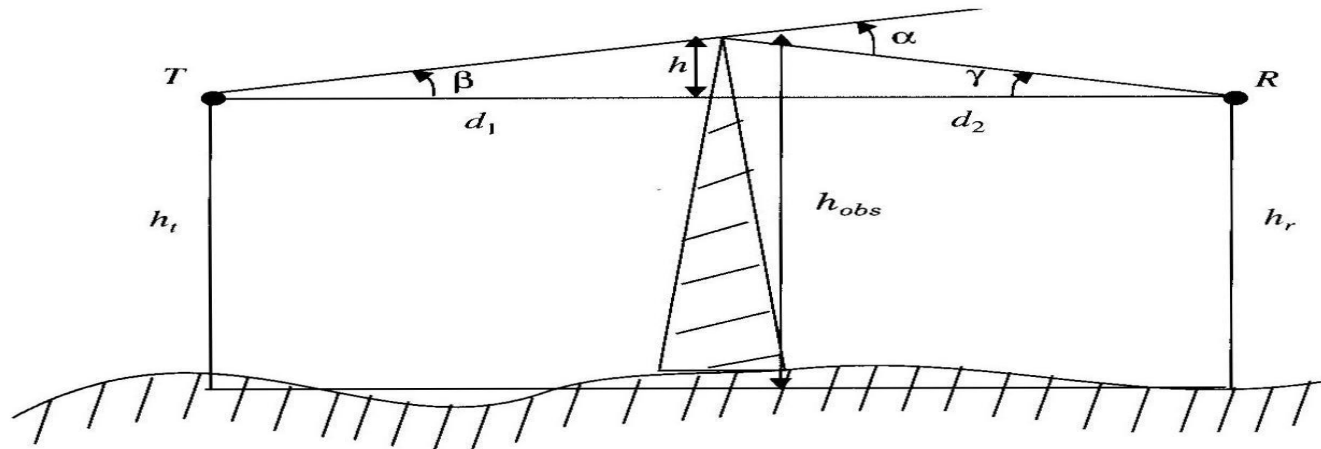


(b) Knife-edge diffraction geometry when the transmitter and receiver are not at the same height. Note that if α and β are small and $h \ll d_1$ and d_2 , then h and h' are virtually identical and the geometry may be redrawn as shown in Figure 4.10c.

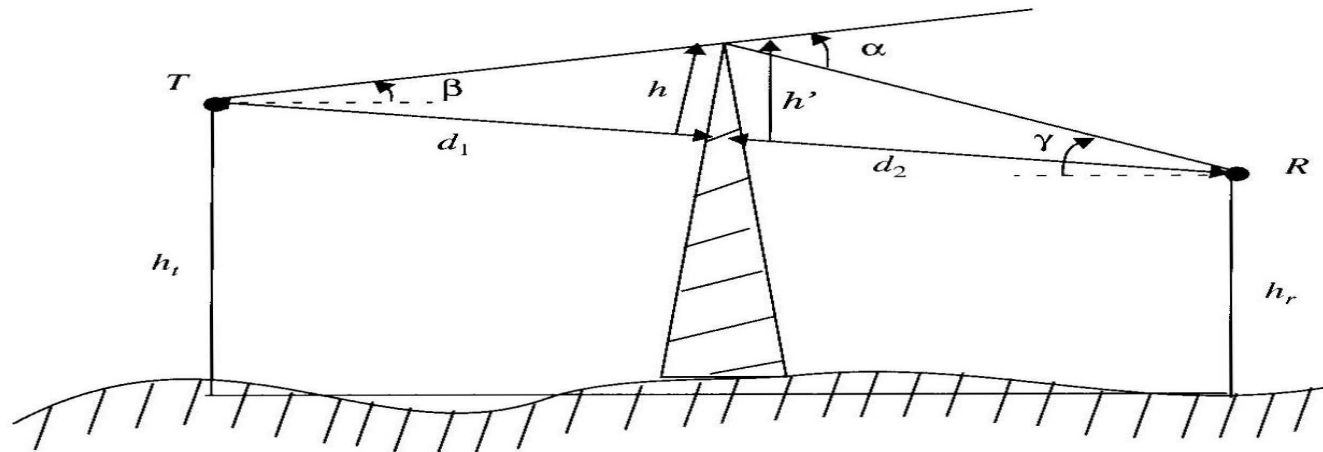


(c) Equivalent knife-edge geometry where the smallest height (in this case h_r) is subtracted from all other heights.

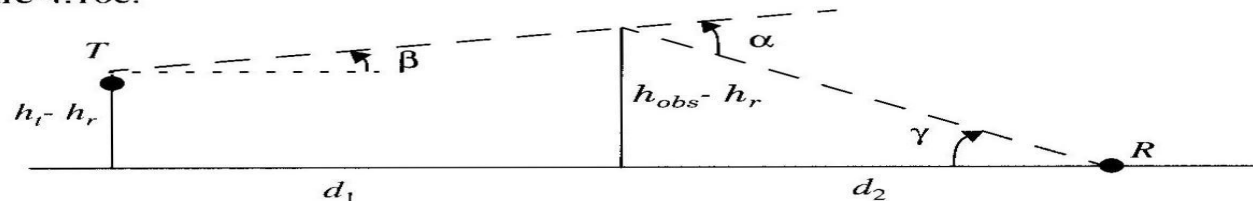
Figure 4.10 Diagrams of knife-edge geometry.



(a) Knife-edge diffraction geometry. The point T denotes the transmitter and R denotes the receiver, with an infinite knife-edge obstruction blocking the line-of-sight path.



(b) Knife-edge diffraction geometry when the transmitter and receiver are not at the same height. Note that if α and β are small and $h \ll d_1$ and d_2 , then h and h' are virtually identical and the geometry may be redrawn as shown in Figure 4.10c.



(c) Equivalent knife-edge geometry where the smallest height (in this case h_r) is subtracted from all other heights.

Figure 4.10 Diagrams of knife-edge geometry.

Phase difference between a direct LOS and diffracted path is a function of height and position of the obstruction, as well as transmitter and receiver location.

Concept of diffraction loss as a function of the path difference around an obstruction is defined using **Fresnel Zones**.

Fresnel zones represents successive regions where secondary waves have a path length from Tx to RX which are $n\lambda/2$ greater than(>) total path length of LOS path.

The difference between the direct path and diffracted path, call **excess path length** (Δ), assume $h \ll d_1, d_2$ and $h \gg \lambda$.

$$\Delta \approx \frac{h^2}{2} \frac{(d_1 + d_2)}{d_1 d_2}$$

The corresponding phase difference

$$\phi = \frac{2\pi\Delta}{\lambda} = \frac{2\pi}{\lambda} \frac{h^2}{2} \frac{(d_1 + d_2)}{d_1 d_2} \quad \& \quad \phi = \frac{\pi}{2} v^2$$

Fresnel-Kirchoff diffraction parameter

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}} = \alpha \sqrt{\frac{2d_1 d_2}{\lambda(d_1 + d_2)}}$$

Diffraction

Can show that path difference

$$\Delta d = \frac{h^2}{2} \frac{(d_1 + d_2)}{d_1 d_2} \text{ or } \Delta \phi = \frac{2\pi \cdot \Delta d}{\lambda} = \frac{\pi}{2} \frac{2h^2 (d_1 + d_2)}{\lambda d_1 d_2}$$

To take care of antenna having same or different height, consider only α and assuming that $h \ll d_1, d_2$ and $h \gg \lambda$, then

$$\alpha = \beta + \gamma \approx \tan \beta + \tan \gamma = h \left(\frac{d_1 + d_2}{d_1 d_2} \right)$$

Concentric circle on the plane represents the loci of the origins of secondary wavelets which propagate to the Rx such that the total path length increases by $\lambda/2$ for successive circle.. These circle are called as **Fresnel's Zone**.

Radius of **nth** fresnel zone circle is denoted by r_n .

$$r_n = \sqrt{\frac{n\lambda d_1 d_2}{d_1 + d_2}}$$

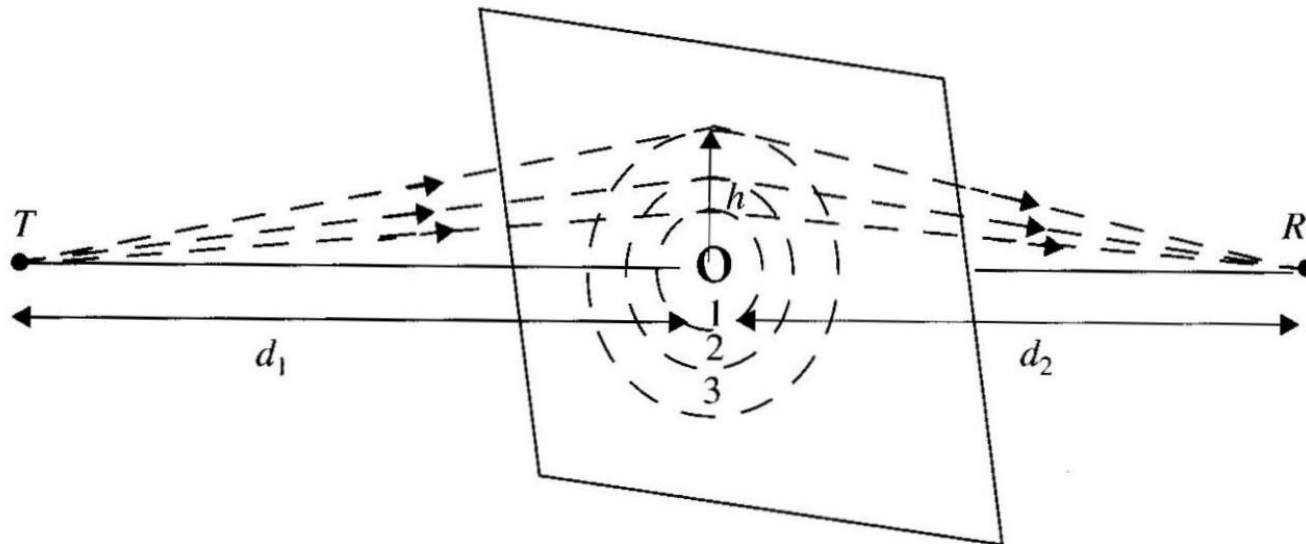


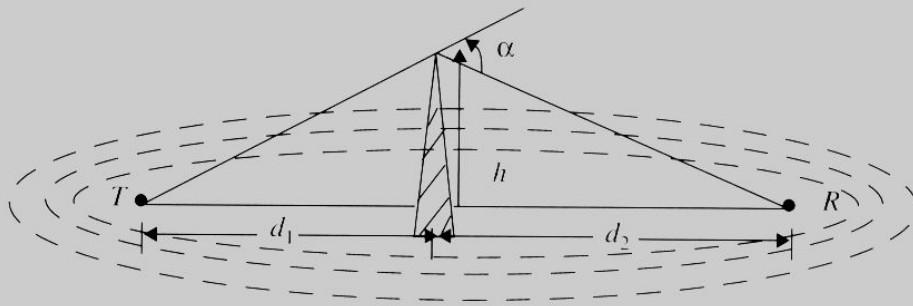
Figure 4.11 Concentric circles which define the boundaries of successive Fresnel zones.

Knife Edge Diffraction Model

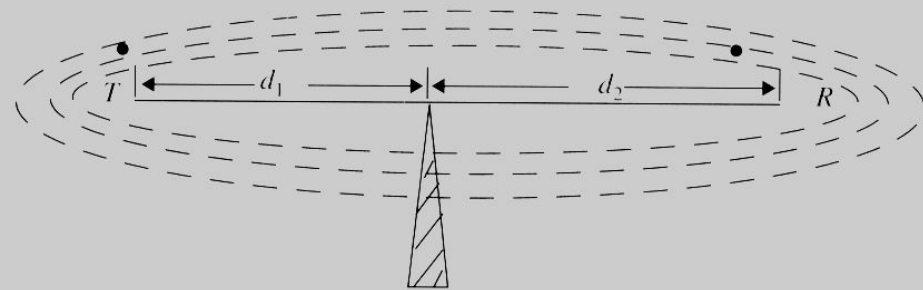
For estimating the signal attenuation caused by diffraction of radio waves over hills and buildings.

When shadowing is caused by a single object like hills or mountain, the attenuation caused by diffraction can be estimated by treating the obstructing as a diffraction knife edge.

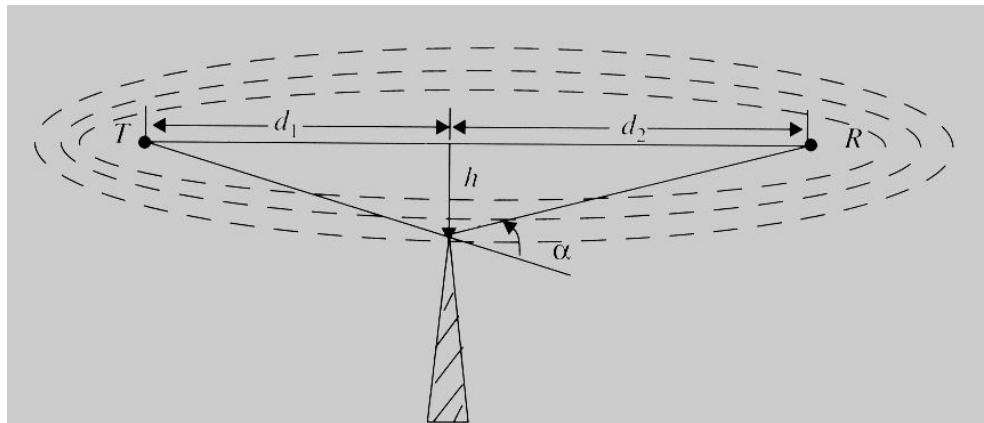
Fresnel diffraction geometry



(a) α and v are positive, since h is positive



(b) α and v are equal to zero, since h is equal to zero



(c) α and v are negative, since h is negative

The wavefront on top of an obstruction generates secondary (weaker) waves.

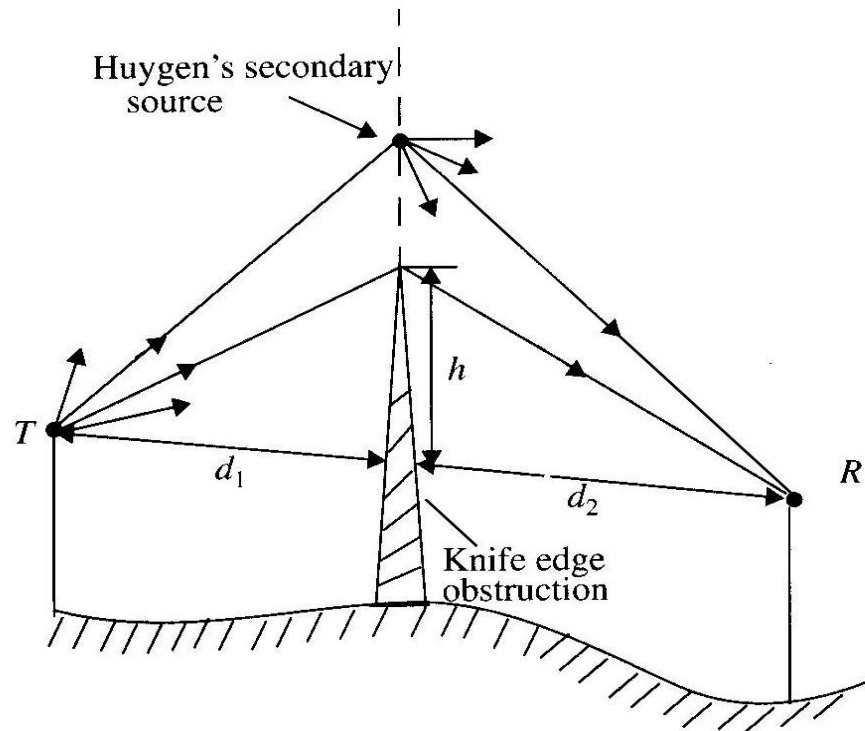


Figure 4.13 Illustration of knife-edge diffraction geometry. The receiver R is located in the shadow region.

Field strength at point R is the vector sum of the fields due to all of secondary's Huygen's source in the plane above the knife edge.

The electric field strength (E_d) of a knife edge diffracted edge is given by:

$$\frac{E_d}{E_0} = F(v) = \frac{(1+j)}{2} \int_v^{\infty} \exp((-j\pi t^2)/2) dt$$

Where,

E_0 is the free space field strength &

$F(v)$ is the complex Fresnel integral is the function of Fresnel-Kirchoff's diffraction parameter(v)

The diffraction gain due to the presence of a knife edge, as compared the free space E-field

$$G_d(\text{dB}) = 20\log|F(v)|$$

Diffraction gain

$$\begin{aligned} G_d(\text{dB}) &= 0 & v \leq -1 \\ G_d(\text{dB}) &= 20\log(0.5 - 0.62v) & -1 \leq v \leq 0 \\ G_d(\text{dB}) &= 20\log(0.5 \exp(-0.95v)) & 0 \leq v \leq 1 \\ G_d(\text{dB}) &= 20\log\left(0.4 - \sqrt{0.1184 - (0.38 - 0.1v)^2}\right) & 1 \leq v \leq 2.4 \\ G_d(\text{dB}) &= 20\log\left(\frac{0.225}{v}\right) & v > 2.4 \end{aligned}$$

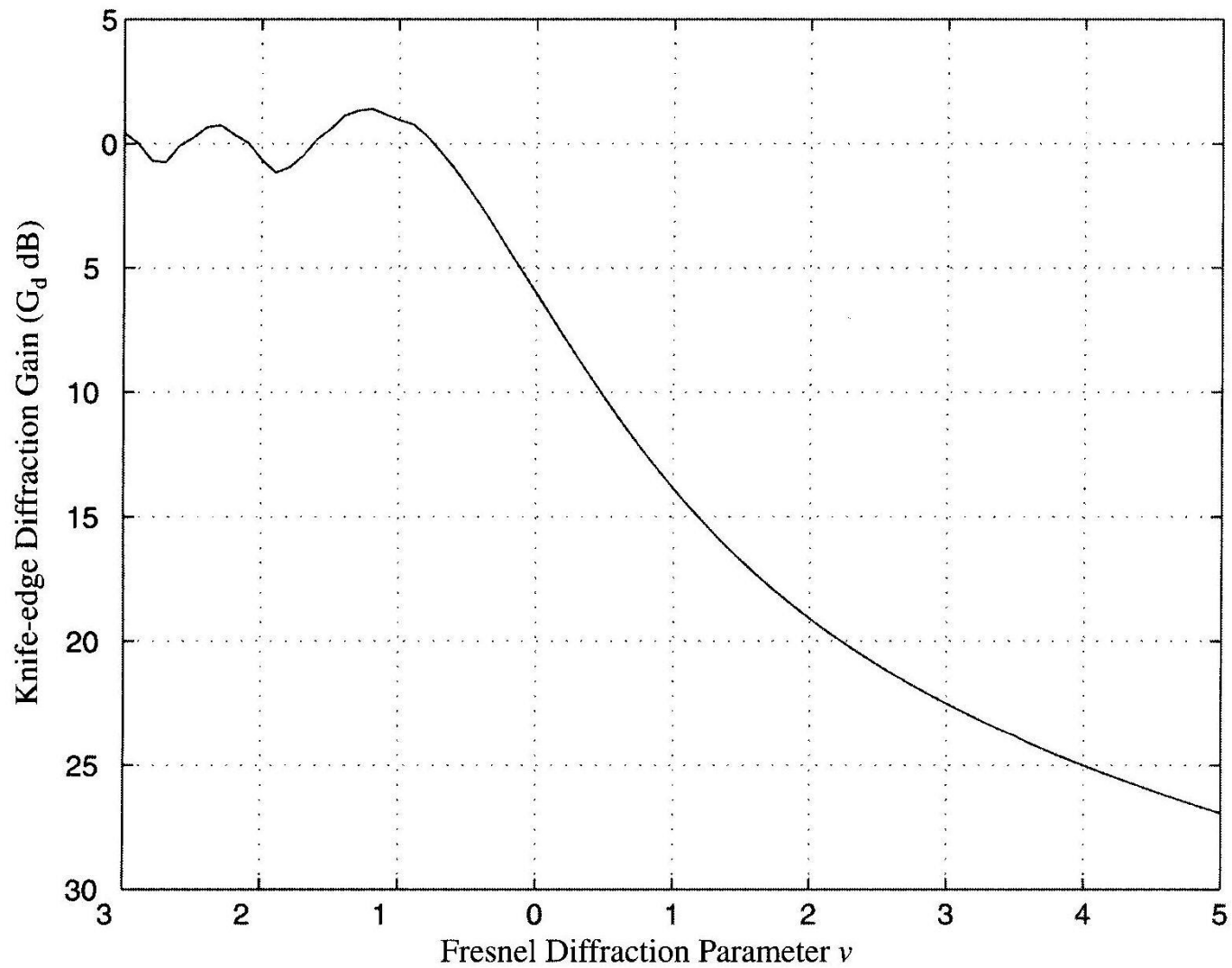


Figure 4.14 Knife-edge diffraction gain as a function of Fresnel diffraction parameter v .

Example 4.7

Compute the diffraction loss for the three cases shown in Figure 4.12. Assume $\lambda = 1/3$ m, $d_1 = 1$ km, $d_2 = 1$ km, and (a) $h = 25$ m, (b) $h = 0$, (c) $h = -25$ m. Compare your answers using values from Figure 4.14, as well as the approximate solution given by Equation (4.61.a)–(4.61.e). For each of these cases, identify the Fresnel zone within which the tip of the obstruction lies.

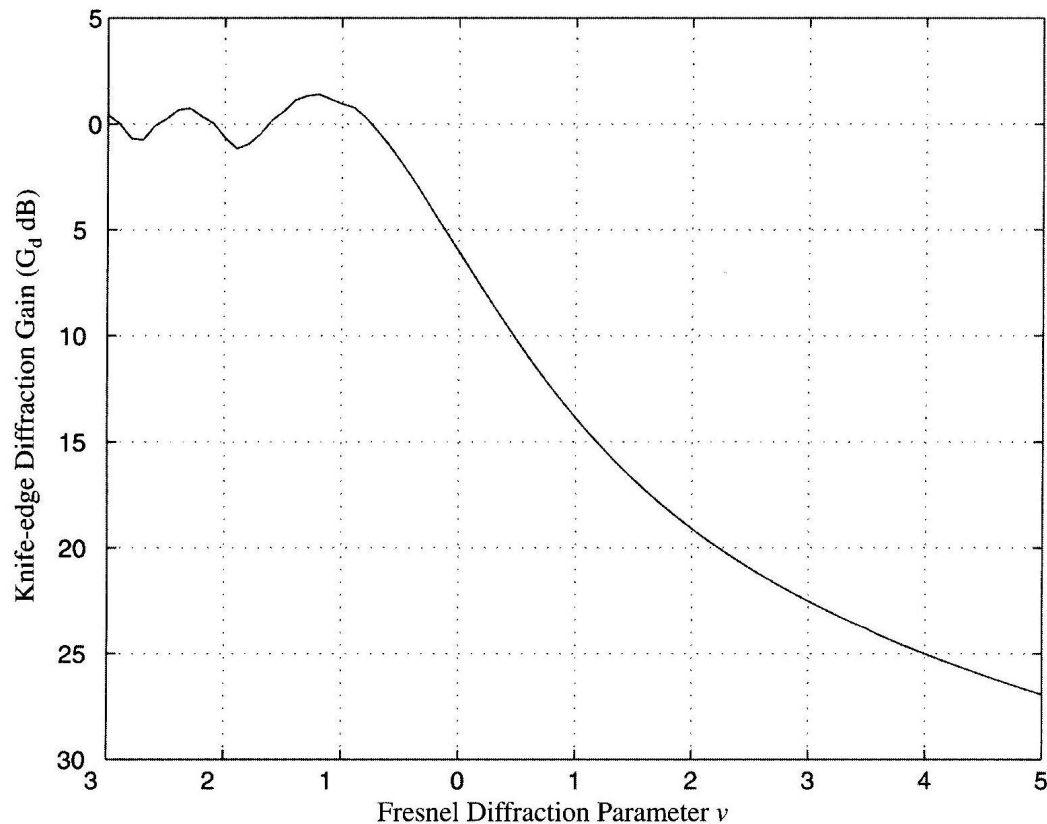


Figure 4.14 Knife-edge diffraction gain as a function of Fresnel diffraction parameter v .

Given:

$$\lambda = 1/3 \text{ m}$$

$$d_1 = 1 \text{ km}$$

$$d_2 = 1 \text{ km}$$

(a) $h = 25 \text{ m}$

Using Equation (4.56), the Fresnel diffraction parameter is obtained as

$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}} = 25 \sqrt{\frac{2(1000 + 1000)}{(1/3) \times 1000 \times 1000}} = 2.74.$$

From Figure 4.14, the diffraction loss is obtained as 22 dB.

Using the numerical approximation in Equation (4.61.e), the diffraction loss is equal to 21.7 dB.

The path length difference between the direct and diffracted rays is given by Equation (4.54) as

$$\Delta = \frac{h^2(d_1 + d_2)}{2 d_1 d_2} = \frac{25^2(1000 + 1000)}{2 \times 1000 \times 1000} = 0.625 \text{ m}.$$

To find the Fresnel zone in which the tip of the obstruction lies, we need to compute n which satisfies the relation $\Delta = n\lambda/2$. For $\lambda = 1/3 \text{ m}$, and $\Delta = 0.625 \text{ m}$, we obtain

$$n = \frac{2\Delta}{\lambda} = \frac{2 \times 0.625}{0.3333} = 3.75.$$

Therefore, the tip of the obstruction completely blocks the first three Fresnel zones.

(b) $h = 0$ m

Therefore, the Fresnel diffraction parameter $v = 0$.

From Figure 4.14, the diffraction loss is obtained as 6 dB.

Using the numerical approximation in Equation (4.61.b), the diffraction loss is equal to 6 dB.

For this case, since $h = 0$, we have $\Delta = 0$, and the tip of the obstruction lies in the middle of the first Fresnel zone.

(c) $h = -25$ m

Using Equation (4.56), the Fresnel diffraction parameter is obtained as -2.74 .

From Figure 4.14, the diffraction loss is approximately equal to 1 dB.

Using the numerical approximation in Equation (4.61.a), the diffraction loss is equal to 0 dB.

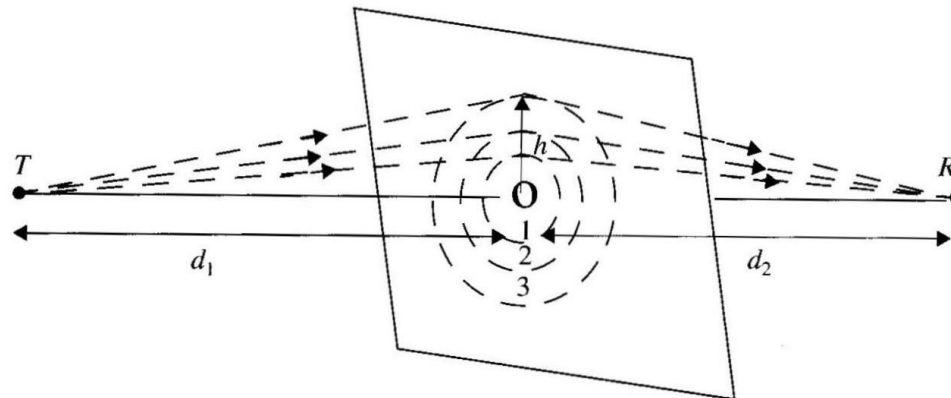
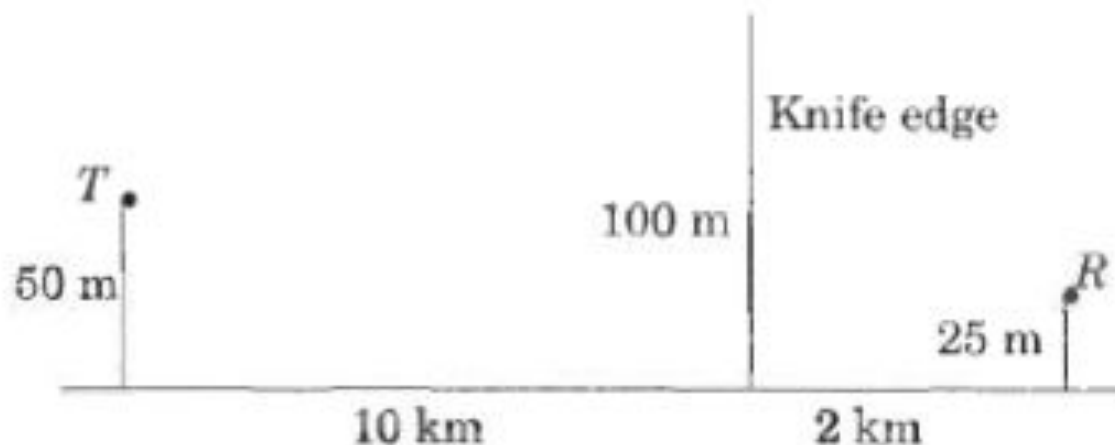


Figure 4.11 Concentric circles which define the boundaries of successive Fresnel zones.

Example 4.8

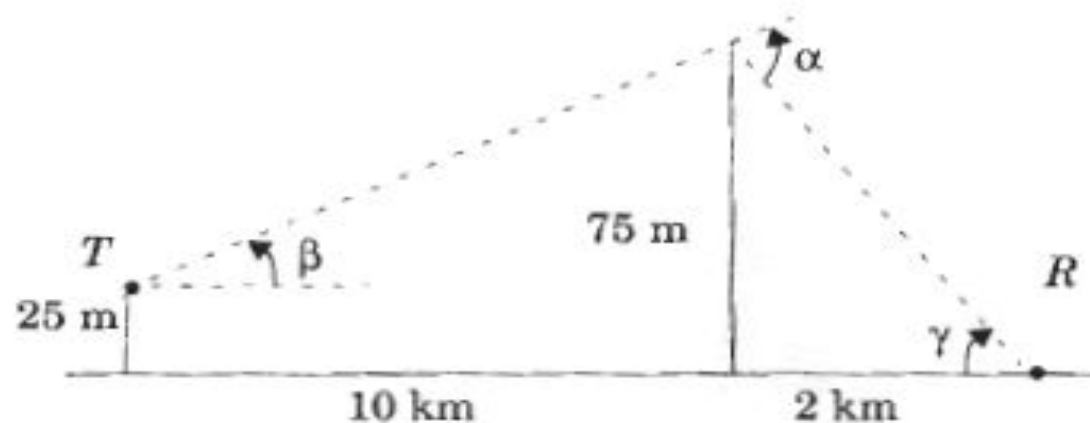
Given the following geometry, determine (a) the loss due to knife-edge diffraction, and (b) the height of the obstacle required to induce 6 dB diffraction loss. Assume $f = 900$ MHz.



Solution

(a) The wavelength $\lambda = \frac{c}{f} = \frac{3 \times 10^8}{900 \times 10^6} = \frac{1}{3} \text{ m}$.

Redraw the geometry by subtracting the height of the smallest structure.



$$\beta = \tan^{-1}\left(\frac{75 - 25}{10000}\right) = 0.2865^\circ$$

$$\gamma = \tan^{-1}\left(\frac{75}{2000}\right) = 2.15^\circ$$

and

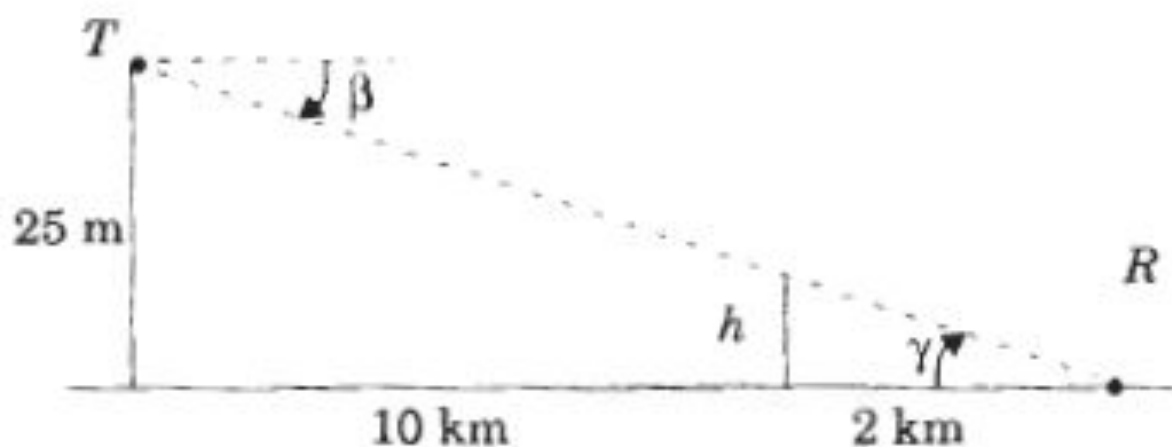
$$\alpha = \beta + \gamma = 2.434^\circ = 0.0424 \text{ rad}$$

Then using Equation (4.56)

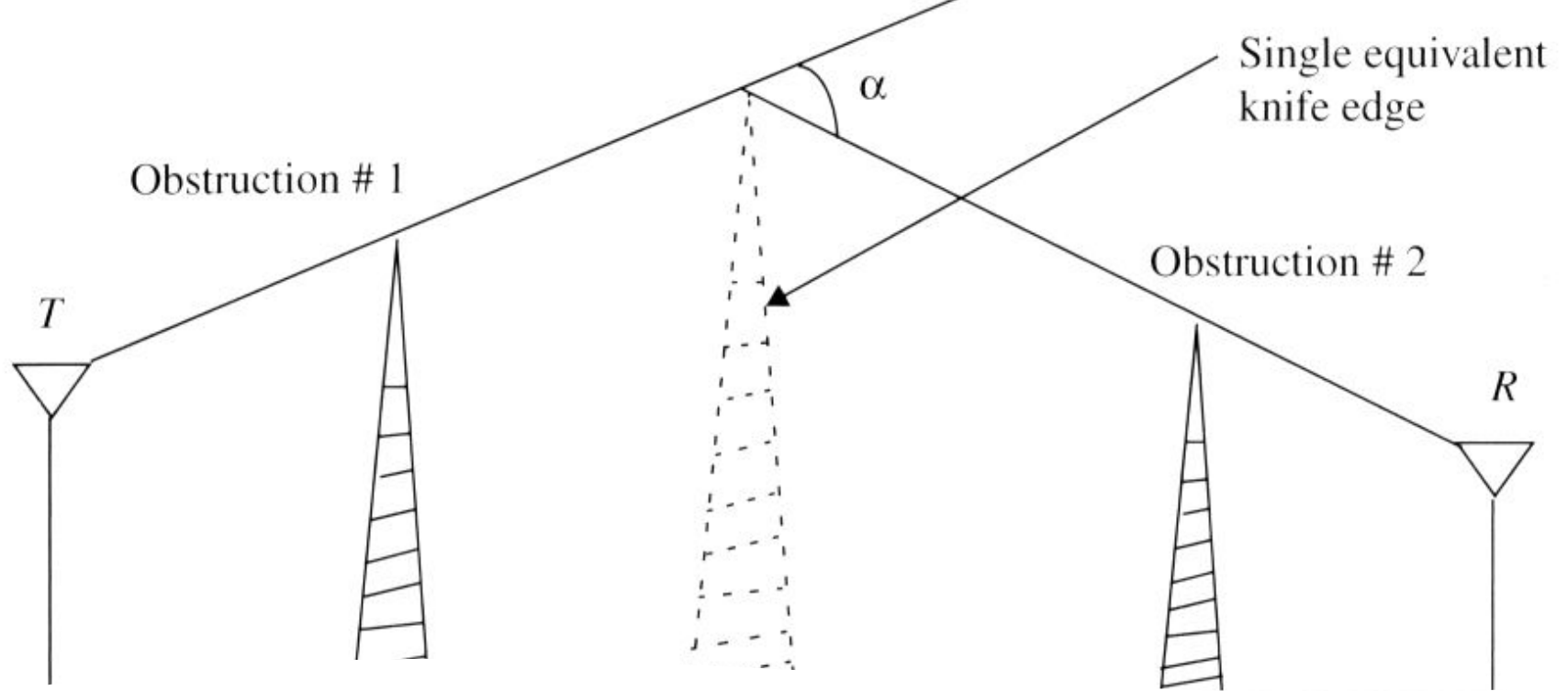
$$v = 0.0424 \sqrt{\frac{2 \times 10000 \times 2000}{(1/3) \times (10000 + 2000)}} = 4.24.$$

From Figure 4.14 or (4.61.e), the diffraction loss is 25.5 dB.

- (b) For 6 dB diffraction loss, $v = 0$. The obstruction height h may be found using similar triangles ($\beta = \gamma$), as shown below.



It follows that $\frac{h}{2000} = \frac{25}{12000}$, thus $h = 4.16$ m.

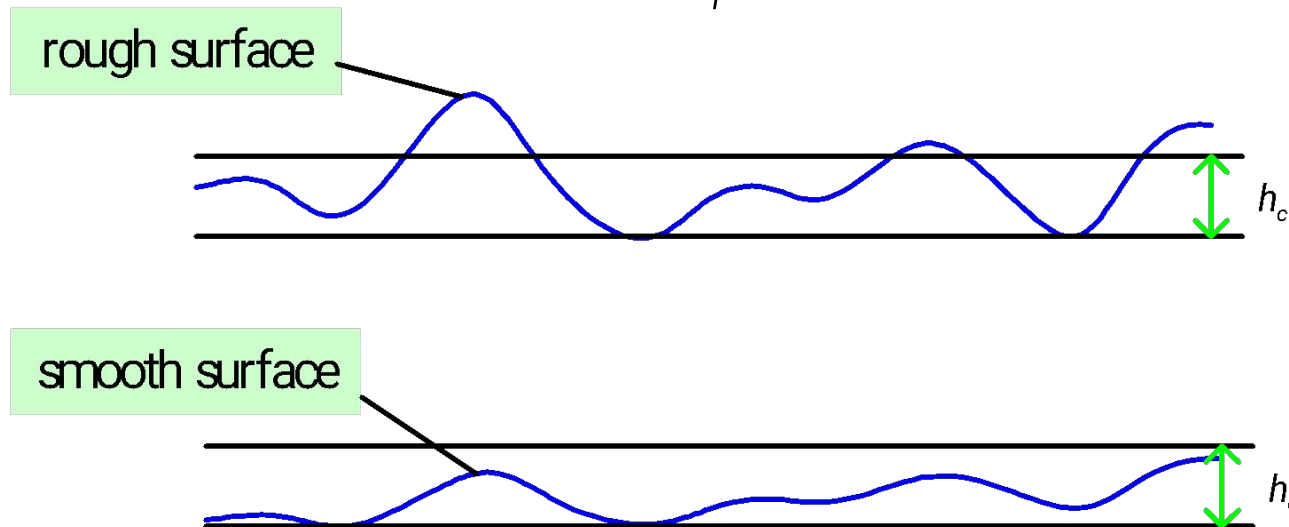


Multiple knife-edge diffraction

- Multiple Knife-edge diffraction : For the presence of two knife edges, replace it by an equivalent knife edge.

- The actual received signal is often stronger than what is predicted by reflection and diffraction
- **Scattering**
 - when a radio wave impinges on a rough surface, the reflected energy is spread out, e.g., trees, lamp posts.
- Surface roughness is test using Rayleigh criterion which defines a critical height h_c for a given angle of incidence θ_i

$$h_c = \frac{\lambda}{8 \sin \theta_i}$$



Scattering



Rough surfaces

Lamp posts and trees, scatter all directions

Critical height for bumps is $f(\lambda, \text{incident angle})$, 4.62

Smooth if its minimum to maximum protuberance h is less than critical height.

Scattering loss factor modeled with Gaussian distribution, 4.63, 4.64.



Nearby metal objects (street signs, etc.)

Usually modeled statistically

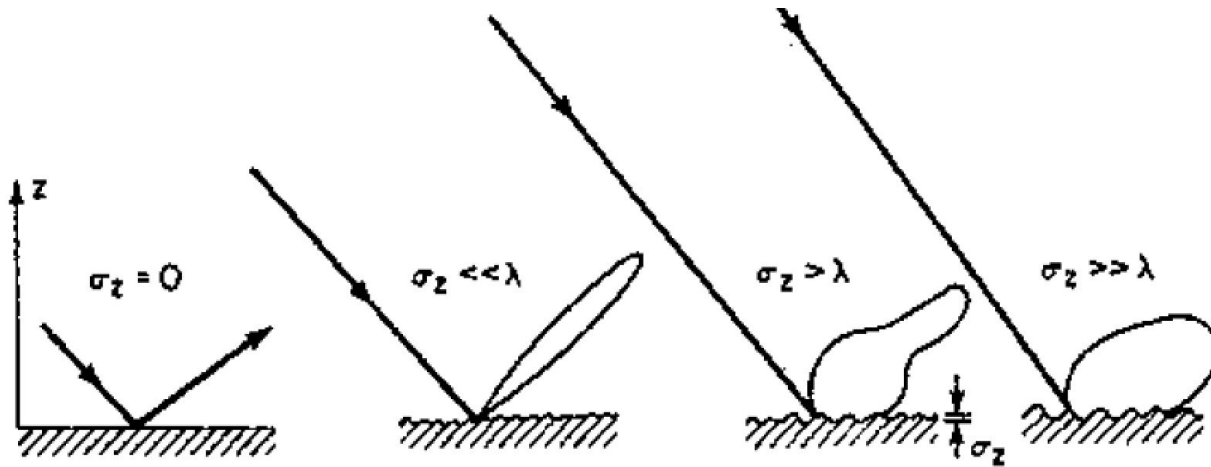


Large distant objects

Analytical model: Radar Cross Section (RCS)

Bistatic radar equation, 4.66

- It is therefore expected that the received signal is stronger than predicted from reflection and diffraction models alone.

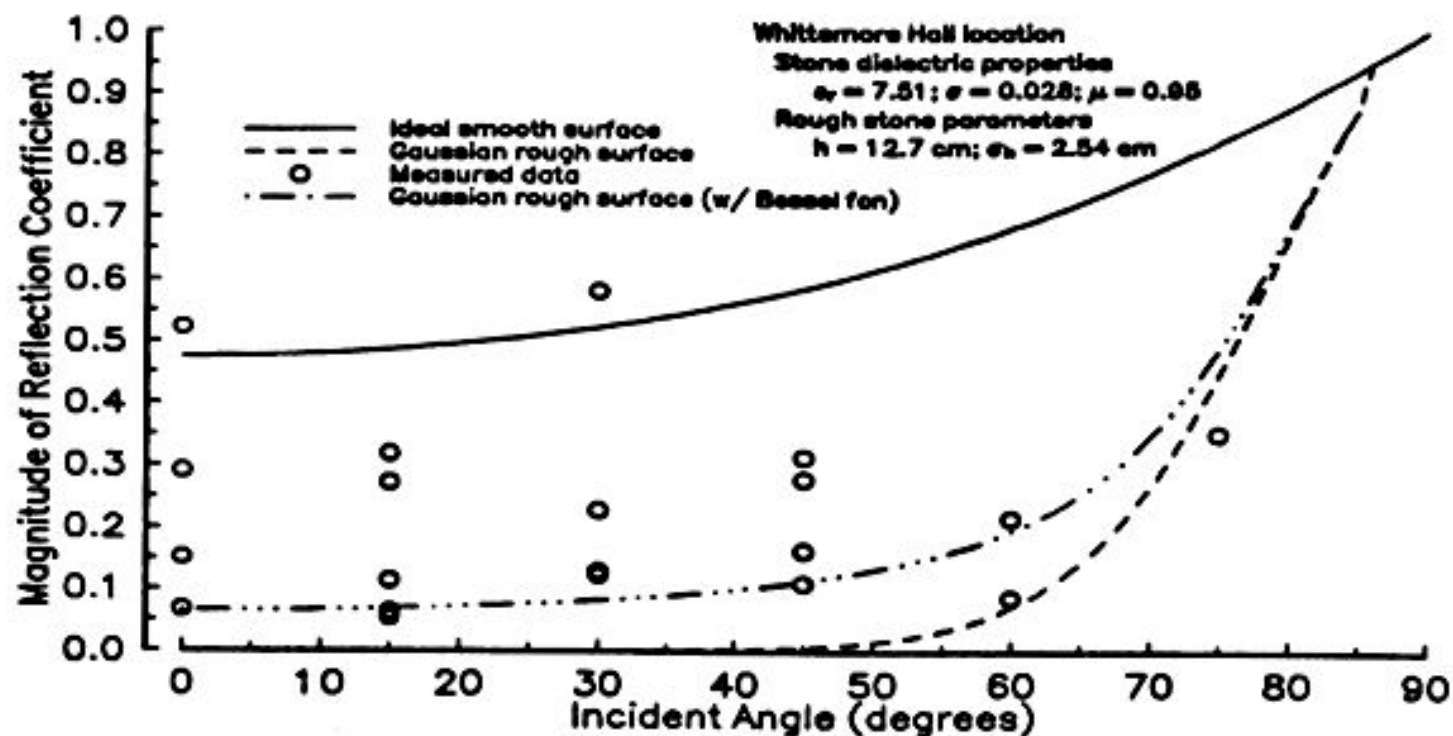


- For rough surface, the flat surface reflection coefficient needs to be multiplied by a **scattering loss factor**

$$\rho_s = \exp \left[\left(\frac{\pi \sigma_h \sin \theta_i}{\lambda} \right)^2 \right]$$

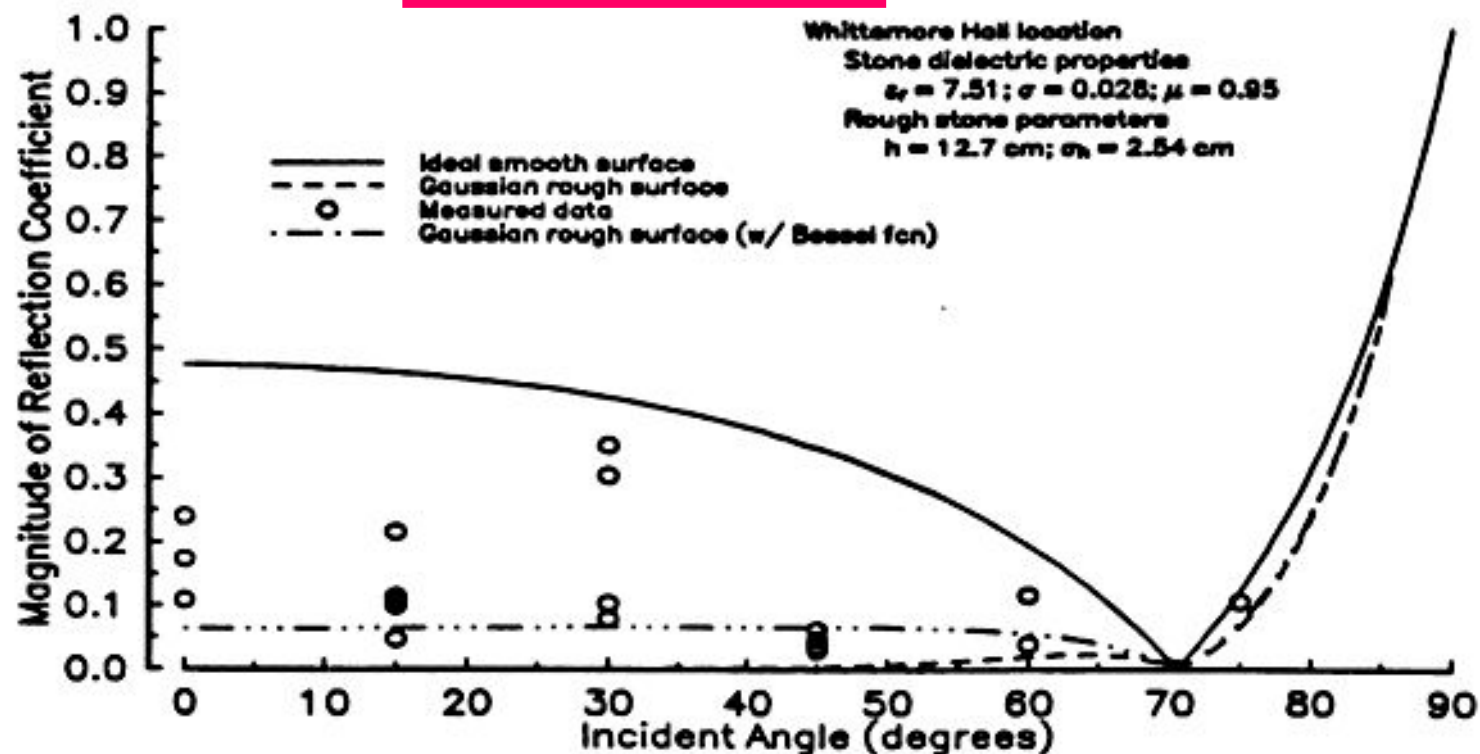
σ_h is the standard deviation of the surface height.

Measured Reflection Coefficients of Rough Stone Wall $f = 4 \text{ GHz}$ $\longleftrightarrow$ Perpendicular Polarization (vertical antenna polarization)



(a) E-field in the plane of incidence (parallel polarization).

Measured Reflection Coefficients of Rough Stone Wall
 $f = 4 \text{ GHz}$ $\longleftrightarrow$ Parallel Polarization (horizontal antenna polarization)



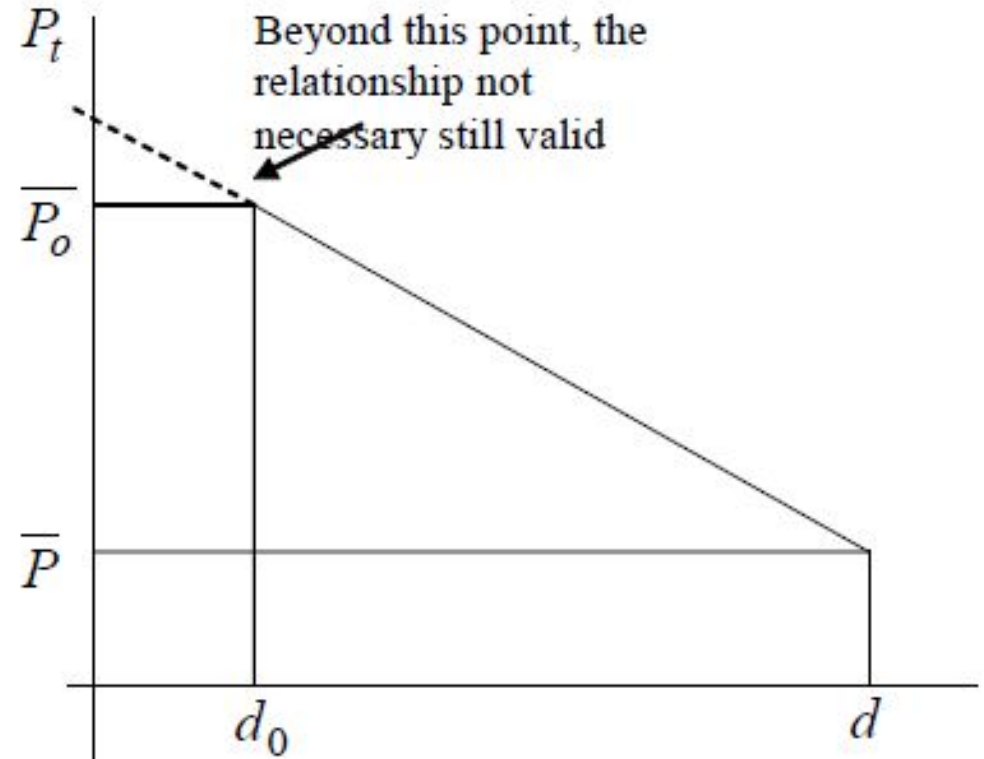
(b) E-field normal to plane of incidence (perpendicular polarization).

Propagation Models

- Large scale models predict behavior averaged over **distances $\gg \lambda$**
 - Function of distance & significant environmental features, roughly frequency independent
 - Breaks down as distance decreases
 - Useful for modeling the range of a radio system and rough capacity planning,
 - Experimental rather than the theoretical for previous three models
 - **Path loss models**, Outdoor models, Indoor models

- Small scale (fading) models describe **signal variability on a scale of λ**
 - Multipath effects (phase cancellation) dominate, path attenuation considered constant
 - Frequency and bandwidth dependent
 - Focus is on modeling “Fading”: rapid change in signal over a short distance or length of time.

Log Distance Path Loss Models



Log Distance Path Loss Models

After considering all these effects, log-distance path loss model is given as

$$\bar{P}_r \propto \left(\frac{1}{d}\right)^n$$

where n is the *path loss exponent*. For Free-space propagation model, $n=2$, and for two-ray model $n=4$.

Log Distance Path Loss Models

At a reference point d_0 with received power $\overline{P}_0$, can show that

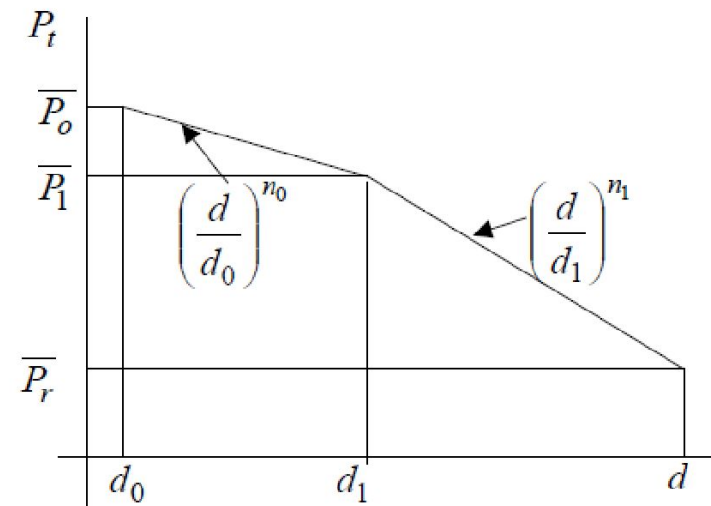
$$\frac{\overline{P}_r}{\overline{P}_0} = \left(\frac{d_0}{d} \right)^n = \left(\frac{d}{d_0} \right)^{-n}$$

The path loss from the reference point

$$\begin{aligned} L_{path}(d) &= \overline{P}_0(\text{in dB}) - \overline{P}_r(\text{in dB}) \\ &= 10n \cdot \log_{10} \left(\frac{d}{d_0} \right) \end{aligned}$$

Log Distance Path Loss Models

Need to be careful if you need to extrapolate the curve beyond $d < d_0$. It could follow another relationship.



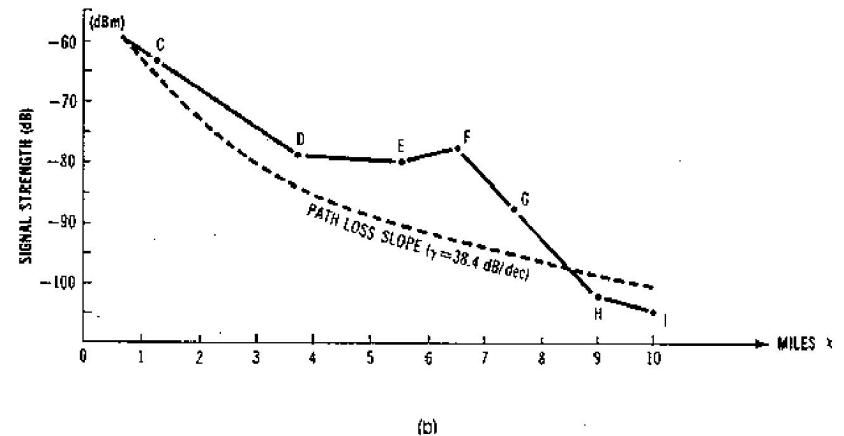
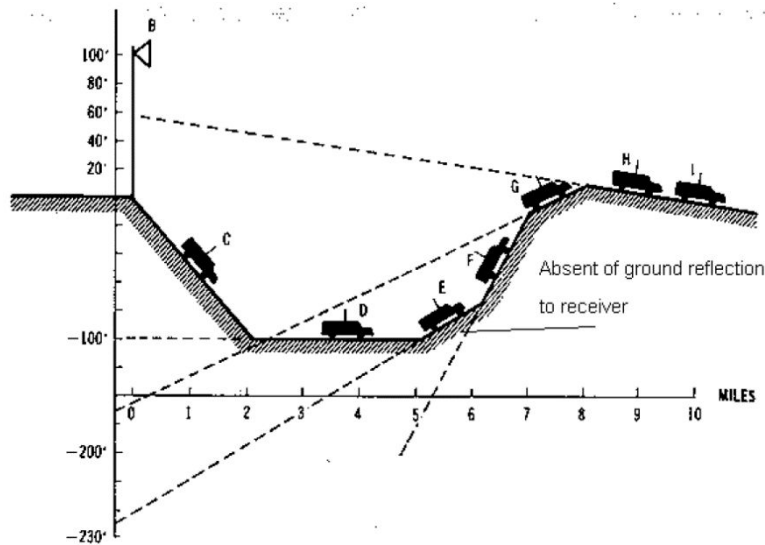
Typical large-scale path loss

Table 4.2 Path Loss Exponents for Different Environments

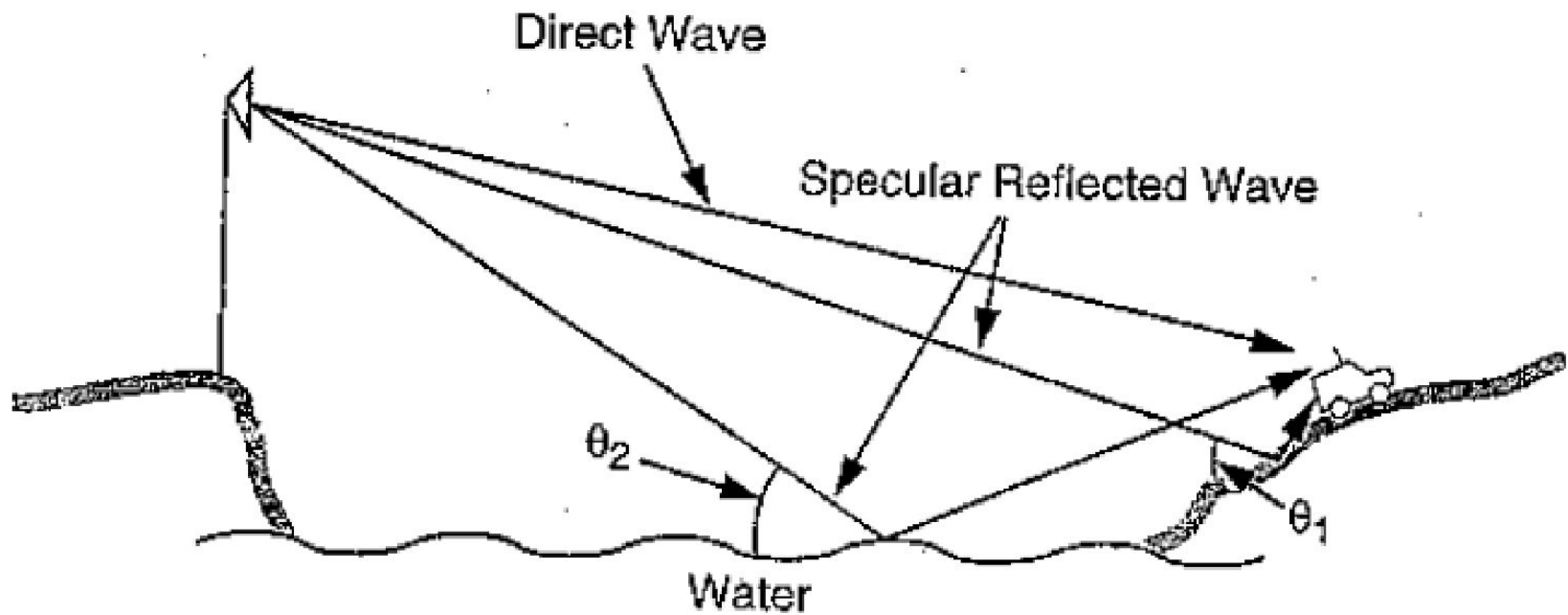
Environment	Path Loss Exponent, n
Free space	2
Urban area cellular radio	2.7 to 3.5
Shadowed urban cellular radio	3 to 5
In building line-of-sight	1.6 to 1.8
Obstructed in building	4 to 6
Obstructed in factories	2 to 3

For Example

- Sometime different values are used for n depending on the distance from the transmitter.

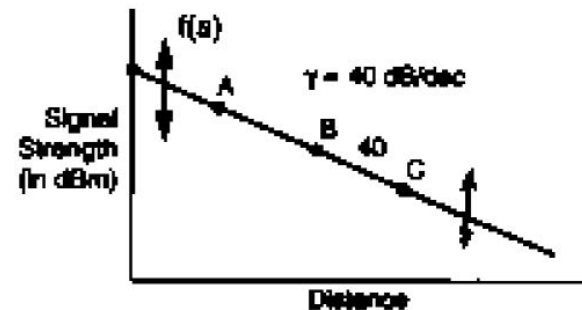
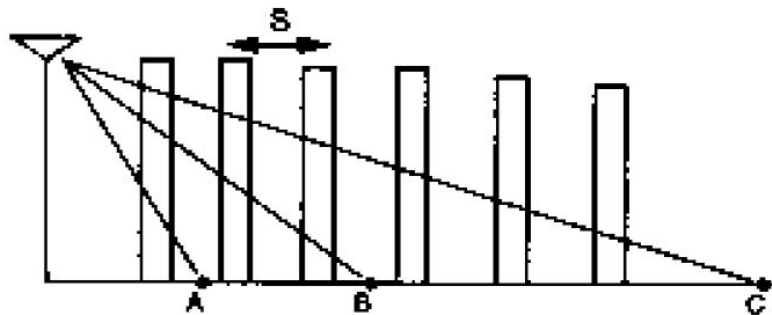
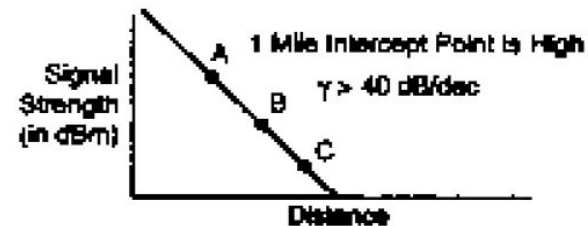
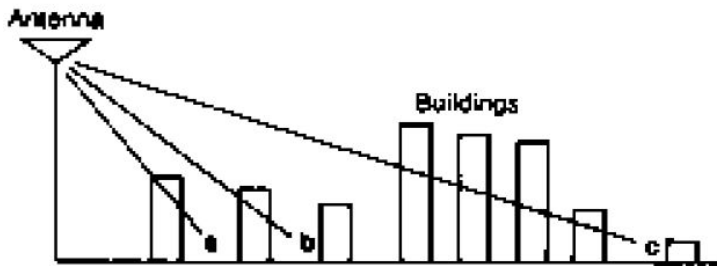
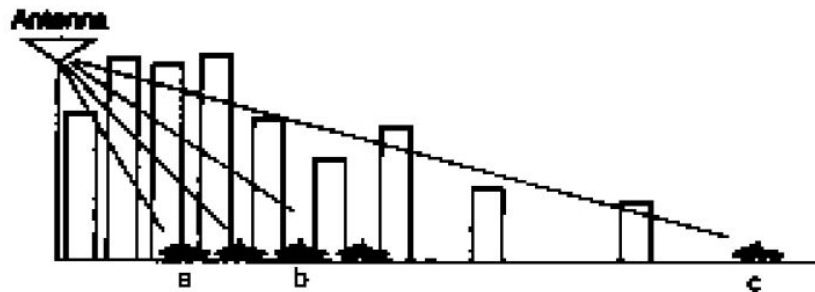


For Example



For Example

n does not directly reflect the strength of the received power



3.5 Practical Link Budget Design using path Loss Models

- Radio propagation models combine
 - analytical method (logic and theory,)
 - empirical method (practical observation and experience.)
- Log-distance Path Loss Model
 - average received signal power decreases logarithmically with distance
- The average path loss

$$\overline{PL}(d) \propto \left(\frac{d}{d_0} \right)^n$$

or

$$\overline{PL}(d)(dB) = \overline{PL}(d_0) + 10n \log \left(\frac{d}{d_0} \right)$$

- **Log-normal Shadowing**

- Surrounding environmental clutter may be different at two different locations having the same T-R separation.
- Measurements have shown that at any value d , the path loss $PL(d)$ at a particular location is random and distributed normally (normal in dB)

$$PL(d) = \overline{PL}(d) + X_{\sigma} = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_{\sigma}$$

and

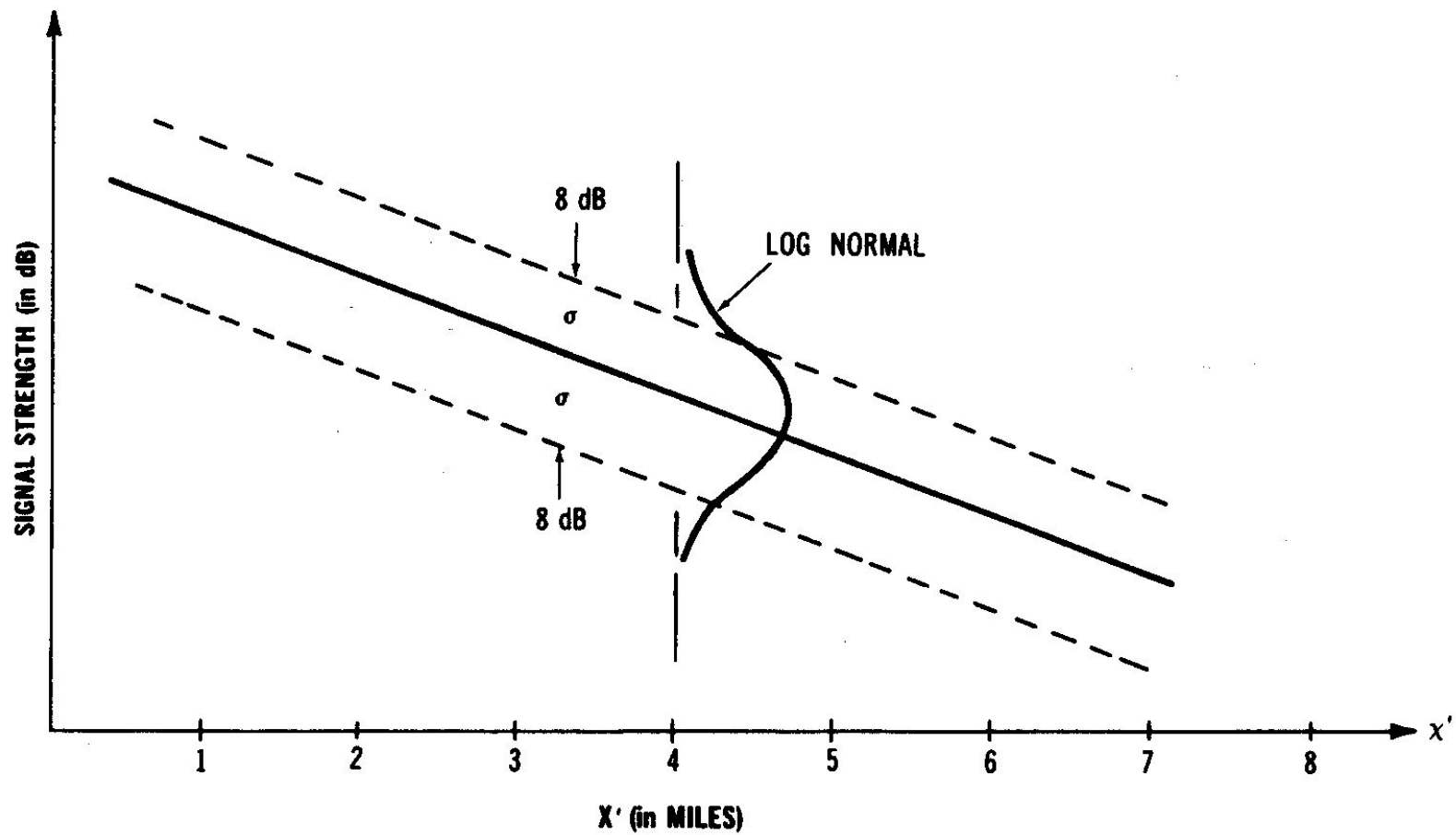
$$P_r(d) = P_t(d) - PL(d)$$

X_{σ} : **zero-mean Gaussian distributed random variable** (in dB) with **standard deviation** σ

- The probability that the received signal level will exceed a certain value γ can be calculated from

$$\Pr[P_r(d) > \gamma] = Q\left(\frac{\gamma - \overline{P_r(d)}}{\sigma}\right)$$

where $\overline{P_r(d)} = P_t(d) - \overline{PL}(d)$



Log-Normal Shadowing Model

- Shadowing occurs when objects block LOS between transmitter and receiver
- A simple statistical model can account for unpredictable “shadowing”
 - $PL(d)(dB) = PL(d) + X_0$,
 - Add a 0-mean Gaussian RV to Log-Distance PL
 - Variance σ is usually from 3 to 12.
 - Reason for Gaussian

Measured large-scale path loss

- Determine n and σ by mean and variance
- Equ. 4.70
- Equ. 4.72
- Basic of Gaussian distribution

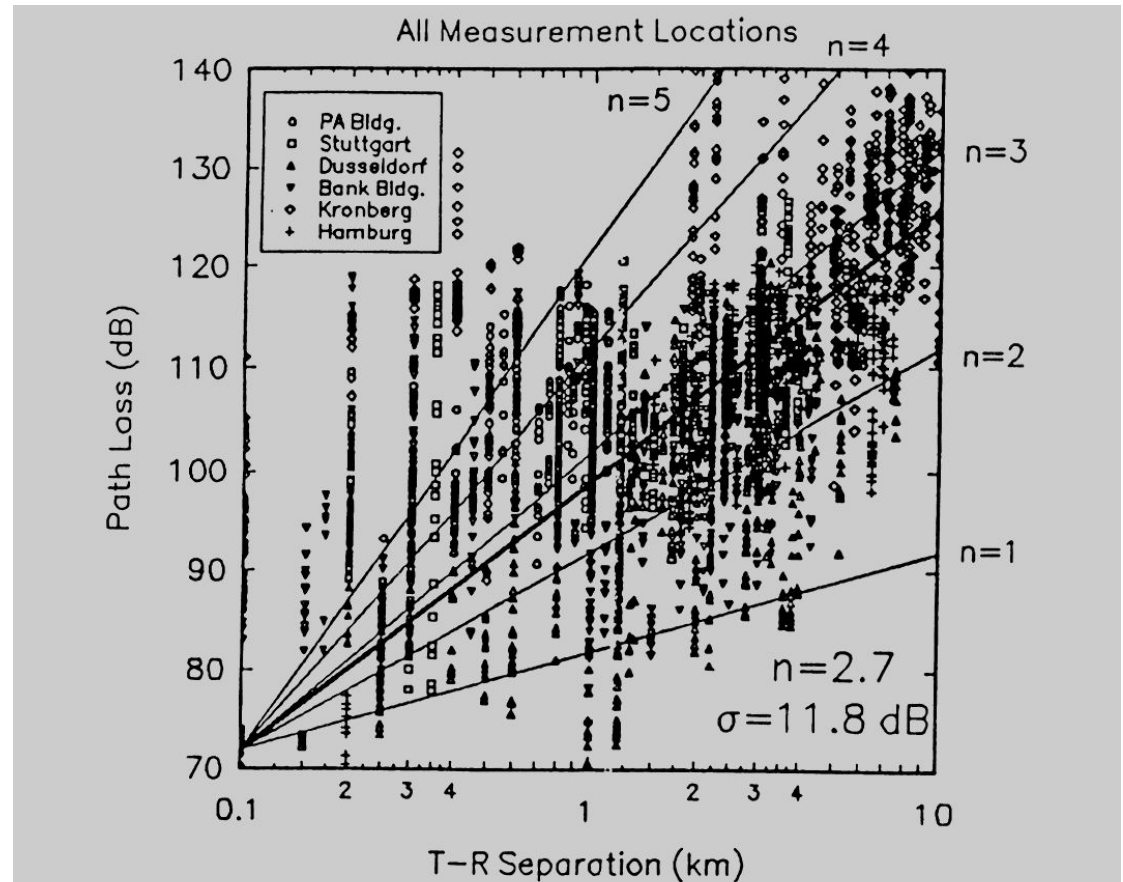
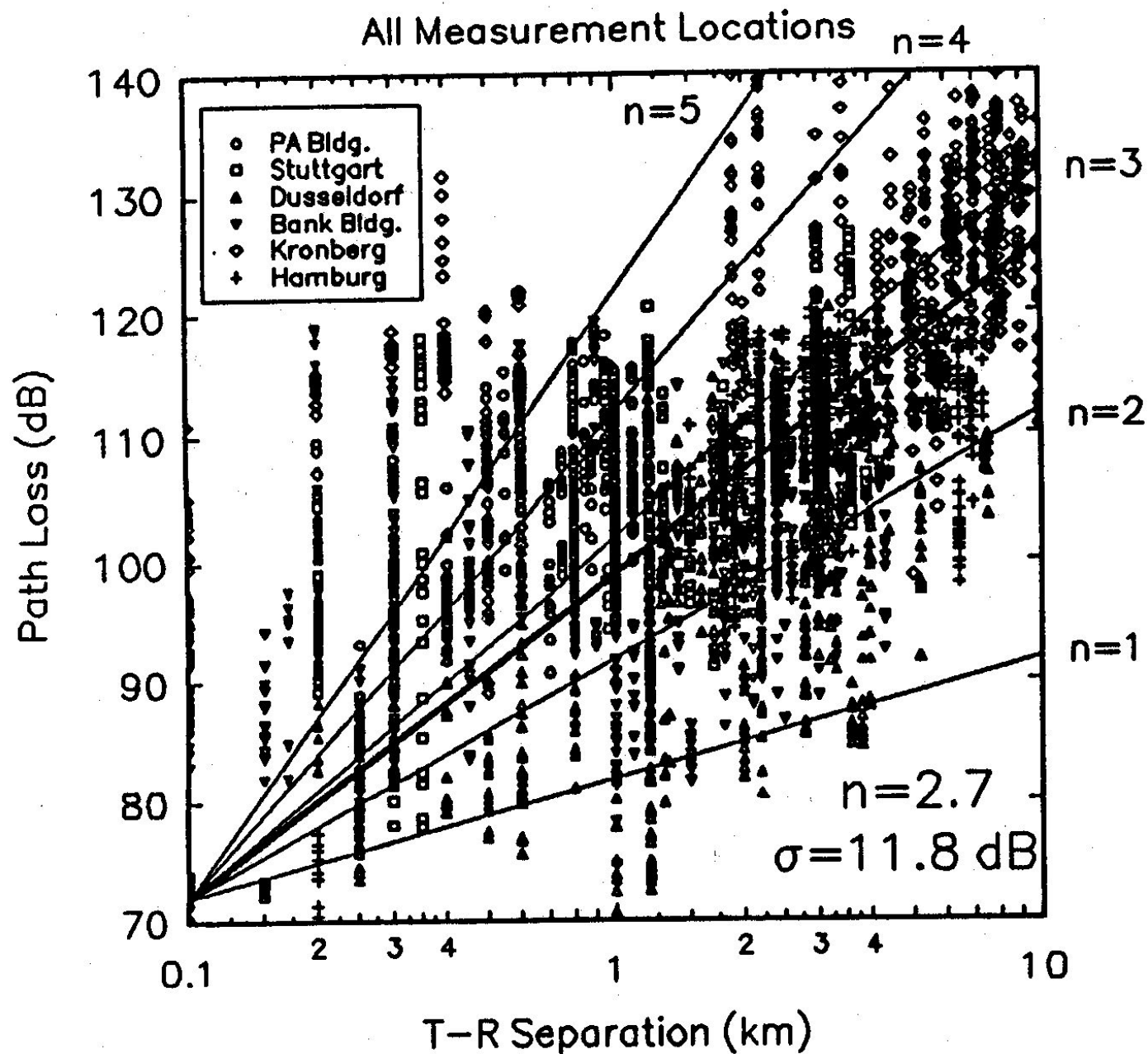


Figure 4.17 Scatter plot of measured data and corresponding MMSE path loss model for many cities in Germany. For this data, $n = 2.7$ and $\sigma = 11.8$ dB [from [Sei91] © IEEE].



Area versus Distance coverage model with shadowing model

- Percentage for SNR larger than a threshold
- Equ. 4.79
- Exam. 4.9

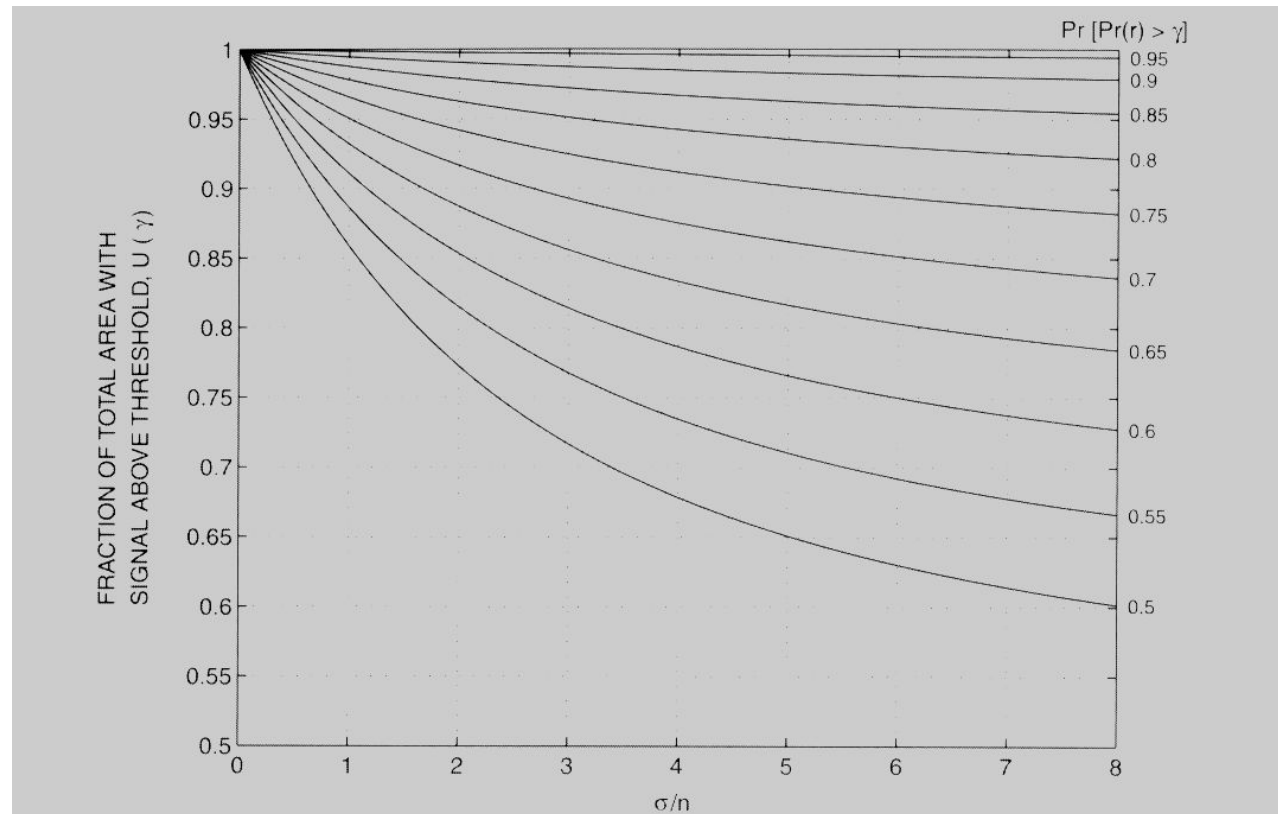


Figure 4.18 Family of curves relating fraction of total area with signal above threshold, $U(\gamma)$ as a function of probability of signal above threshold on the cell boundary.

Popular Land Mobile Services Models

- Original Okumura-Hata Model
- COST 231 Extension
- ITU Extension
- Okumura-Hata-Davidson
- MSAM LMS package

Available large-scale models:

- **Friis free space propagation model:** to model propagation path loss
- **Log distance path loss model:** incorporates both path loss and *shadowing effect*
- **Two ray ground reflection model**, also called as plane-earth loss model
- **Single knife-edge diffraction model** - incorporates diffraction loss
- **Hata - Okumura Model** - an empirical model

Okumura Model

$$L_{50}(\text{dB}) = L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{AREA}$$

Where,

- L_{50} is the 50th percentile (i.e., median) value of propagation path loss,
 - L_F is the free space propagation loss,
 - $A_{mu}(f, d)$ is the median attenuation relative to free space,
 - $G(h_{te})$ is the base station antenna height gain factor,
 - $G(h_{re})$ is the mobile antenna height gain factor, and
 - G_{AREA} is the gain due to the type of environment.
-
- Note that the antenna height gains are strictly a function of height and have nothing to do with antenna patterns.

Okumura Model

- The major disadvantage with the model is its low response to rapid changes in terrain, therefore the model is fairly good in urban areas, but not as good in rural areas.
- Common standard deviations between predicted and measured path loss values are around 10 to 14 dB.
- G(h_{re}):
$$G(h_{te}) = 20 \log \left(\frac{h_{te}}{200} \right) \quad 1000\text{m} > h_{te} > 30\text{m}$$

$$G(h_{re}) = 10 \log \left(\frac{h_{re}}{3} \right) \quad h_{re} \leq 3\text{m}$$

$$G(h_{re}) = 20 \log \left(\frac{h_{re}}{3} \right) \quad 10\text{m} > h_{re} > 3\text{m}$$

Okumura and Hata's model

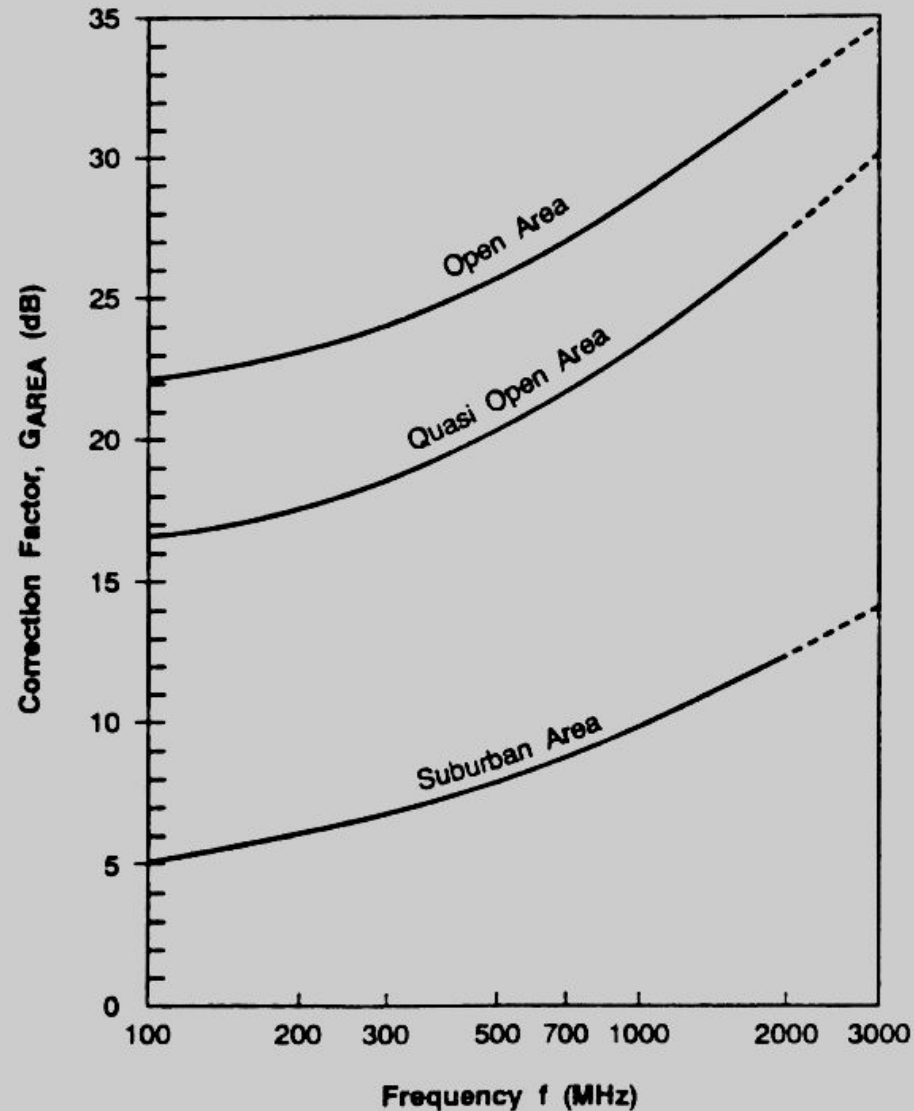
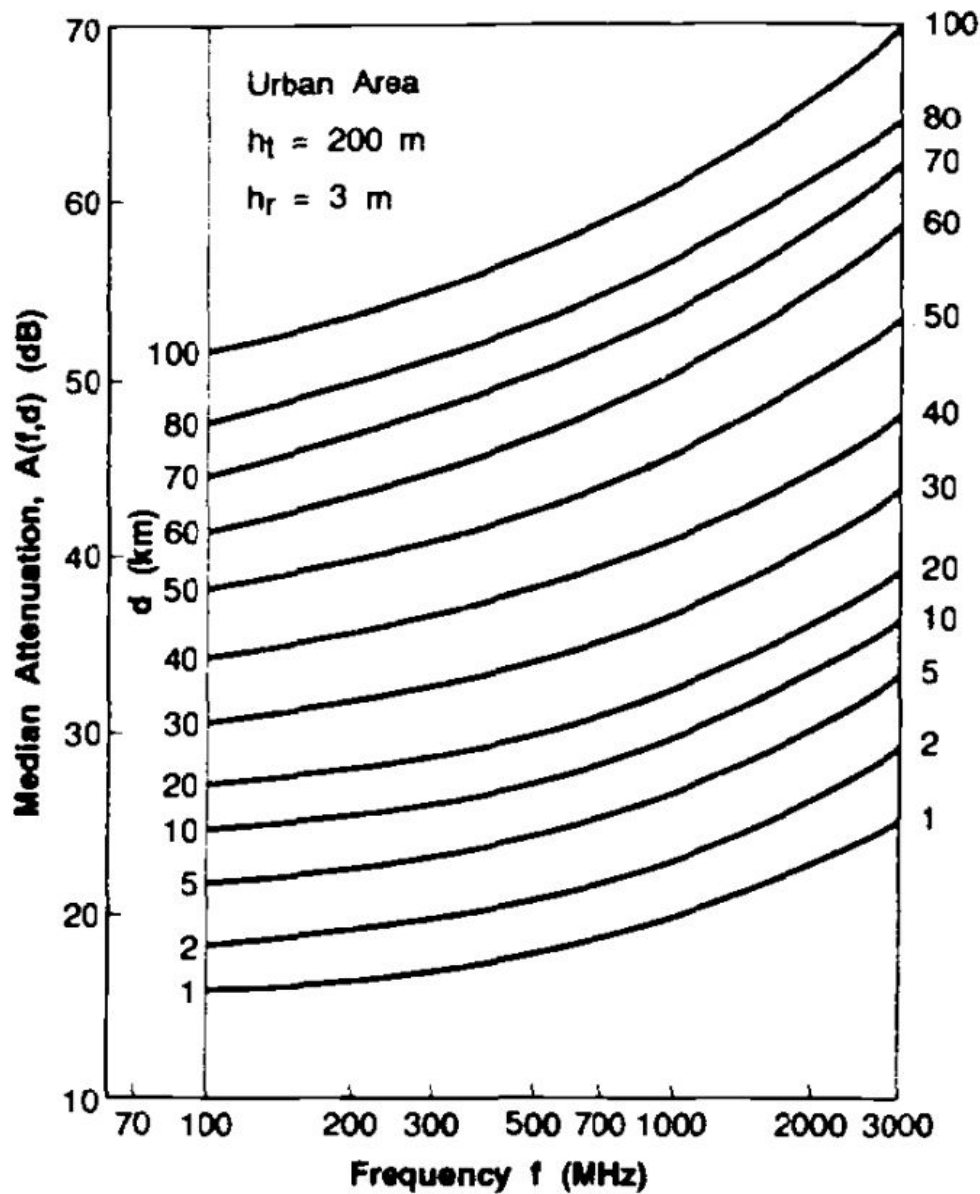


Figure 4.24 Correction factor, G_{AREA} , for different types of terrain [from [Oku68] © IEEE].



Median attenuation relative to free space ($A_{mu}(f,d)$), over a quasi-smooth terrain [From [Oku68] © IEEE).

Find the **median path loss** using Okumura's model for $d = 50\text{km}$ $h_{te} = 100\text{ m}$, $h_{re} = 10\text{ m}$ in a **sub-urban** environment. If the base station transmitter radiates an EIRP of 1 kW at a carrier frequency of 900 MHz, find the **power at the receiver** (assume a unity gain receiving antenna).

The free space path loss L_F can be calculated

$$L_F = 10\log\left[\frac{\lambda^2}{(4\pi)^2 d^2}\right] = 10\log\left[\frac{(3 \times 10^8 / 900 \times 10^6)^2}{(4\pi)^2 \times (50 \times 10^3)^2}\right] = 125.5\text{ dB}.$$

From the Okumura curves

$$A_{mu}(900\text{ MHz}(50\text{ km})) = 43\text{ dB}$$

and

$$G_{ARFA} = 9\text{ dB}.$$

$$G(h_{te}) = 20\log\left(\frac{h_{te}}{200}\right) = 20\log\left(\frac{100}{200}\right) = -6 \text{ dB}.$$

$$G(h_{re}) = 20\log\left(\frac{h_{re}}{3}\right) = 20\log\left(\frac{10}{3}\right) = 10.46 \text{ dB}.$$

The total mean path loss is

$$\begin{aligned} L_{50}(\text{dB}) &= L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{AREA} \\ &= 125.5 \text{ dB} + 43 \text{ dB} - (-6) \text{ dB} - 10.46 \text{ dB} - 9 \text{ dB} \\ &= 155.04 \text{ dB}. \end{aligned}$$

Therefore, the median received power is

$$\begin{aligned} P_r(d) &= EIRP(\text{dBm}) - L_{50}(\text{dB}) + G_r(\text{dB}) \\ &= 60 \text{ dBm} - 155.04 \text{ dB} + 0 \text{ dB} = -95.04 \text{ dBm}. \end{aligned}$$

The Hata Model

Standard formula for median path loss in urban areas is given by

$$L_{50}(\text{urban}) = 69.55 + 26.16 \log_{10} f - 13.82 \log_{10} h_1 - a(h_2) \\ + (44.9 - 6.55 \log_{10} h_1) \log_{10} d$$

For a small to medium sized city, the correction factor is given by

$$a(h_2) = (1.1 \log_{10} f - 0.7) h_2 - (1.56 \log_{10} f - 0.8) \text{ dB}$$

and for a large city, is give by

$$a(h_2) = 8.29(\log_{10} 1.54 h_2)^2 - 1.1 \text{ dB} \quad \text{for } f \leq 200 \text{ MHz} \\ = 3.2(\log_{10} 11.75 h_2)^2 - 4.97 \text{ dB} \quad \text{for } f \geq 400 \text{ MHz}$$

To obtain the path loss in decibels in a suburban are the formula is modified as

$$L_{50} = L_{50}(\text{urban}) - 2 [\log_{10}(f/28)]^2 - 5.4$$

And for the path loss in decibels in open areas the formula is modified as

$$L_{50} = L_{50}(\text{urban}) - 4.78 (\log_{10} f)^2 + 18.33 (\log_{10} f) - 40.98$$

Hata Model

- Empirical formulation of the graphical data in the Okamura model. Valid 150MHz to 1500MHz, Used for cellular systems
- The following classification was used by Hata:

■Urban area $L_{dB} = A + B \log d - E$

■Suburban area $L_{dB} = A + B \log d - C$

■Open area $L_{dB} = A + B \log d - D$

$$A = 69.55 + 26.16 \log f - 13.82 h_b$$

$$B = 44.9 - 6.55 \log h_b$$

$$C = 2(\log(f / 28))^2 + 5.4$$

$$D = 4.78 \log(f / 28)^2 + 18.33 \log f + 40.94$$

$$E = 3.2(\log(11.75 h_m))^2 - 4.97 \quad \text{for large cities, } f \geq 300 \text{MHz}$$

$$E = 8.29(\log(1.54 h_m))^2 - 1.1 \quad \text{for large cities, } f < 300 \text{MHz}$$

$$E = (1.11 \log f - 0.7) h_m - (1.56 \log f - 0.8) \quad \text{for medium to small cities}$$

Modified Cost 231

(European Cooperative for Scientific and Technical Research)

Mean Loss

P_t : transmitter power in watts

f : 1.5 – 2.0 GHz

h_1 : 30 to 200 m

h_2 : 1 to 10 m

d : 1 to 100 Km

$$L_{50}(\text{dB}) = 46.33 + 33.9 \log_{10} f - 13.82 \log_{10} h_1 - a(h_2) + (44.9 - 6.55 \log_{10} h_1) (\log_{10} d)^b + C_m$$

**Where C_m = 0 dB, for small/medium cities
= 3 dB, for large cities**

Field Strength

$$E \text{ in dB(dB mV/m)} = 63.1 + 10 \log_{10} P_t - 13.9 \log_{10} f + 13.82 \log_{10} h_1 + a(h_2) - (44.9 - 6.55 \log_{10} h_1)(\log_{10} d)^b - C_m$$

b as defined earlier

PCS Extension of Hata Model

- **COST-231 Hata Model, European standard**
- Higher frequencies: up to 2GHz
- Smaller cell sizes
- Lower antenna heights

$$L_{dB} = F + B \log d - E + G$$

$$F = 46.3 + 33.9 \log f - 13.82 \log h_b \quad f > 1500 \text{ MHz}$$

$$G = \begin{cases} 3 & \text{Metropolitan centers} \\ 0 & \text{Medium sized city and suburban areas} \end{cases}$$

ITS MODELS

- **ITM (Irregular Terrain Model) Based on Longley-Rice**
 - **Point-To-Point Mode**
 - **Point-To-Area Mode**
- **IF-77 (Johnson-Gierhart) For Aeronautical Mobile Applications used in ITU-R P.528**
- **FAA updated this program in 2010**
- **ITS HF Propagation Analysis Package**
 - **VOACAP, VOAAREA and S/I VOACAP Gives Point-To-Point, Area Coverage Signal-To-Interference Ratio for VOA**
 - **REC533 Point-To-Point Prediction of HF Broadcast, ITU-R P.533-5**
 - **RECAREA: Area Coverage of HF Broadcast**
 - **ICEPAC, ICEAREA and S/I ICEPAC: Ionospheric Communications enhanced Profile Analysis and Circuit Prediction Program**

Irregular Terrain Model Dimensions of Variability

- **On fixed paths hourly median signal levels have been measured and record for periods of years and median signal levels and standard deviations determined. Results in time variability. Looking at the data one might be able to say for 95% of the time the attenuation did not exceed 36 dB.**
- **Looking at a second similar path with all parameters constant, we would find the statistics are different and the long term statistics have changed. We find that there is a path-to-path variability. We call this location variability. Now we would perhaps say there will be 70% of locations the attenuation does not exceed 36 dB for at least 95% of the time.**
- **If we rerun the experiments in different but like appearing situations we again observe changes in the statistics. These are referred to as situational variability. Now we would say in 90% of like situations, there will be at least 70% of locations the attenuation does not exceed 36 dB for at least 95% of the time.**
- **Prediction of Tropospheric Radio Transmission Loss over Irregular Terrain – Longley & Rice - Ref ESSA TR ERL 79-ITS 67**
- **Use of the ITS ITM in the Area Prediction Mode NTIA 82-100**

ITU-R RECOMMENDATIONS

- ITU-R P. 1144 **Guide** to the Application of the Propagation Methods of Radiocommunication Study Group 3
- ITU-R P. 452 Prediction Procedure for the Evaluation of **Microwave Interference** Between Stations on the Surface of the Earth at Frequencies Above About 0.7 GHz
- ITU-R P. 530 Propagation Data and Prediction Methods Required for the Design of **Terrestrial Line-of-Sight** Systems
- ITU-R P. 533 **HF Propagation** Prediction Method
- ITU-R P. 1411-3: Propagation Data and Prediction Methods for the Planning of **Short-Range** Outdoor Radiocommunication Systems and Radio Local Area Networks in the Frequency Range 300 MHz to 100 GHz
- ITU-R P. 1546-2: Method for **Point-to-Area** Predictions for Terrestrial Services in the Frequency Range 30 to 3000 MHz.
- ITU-R P.1812 A **path-specific** propagation prediction method for point-to-area terrestrial services in the VHF and UHF bands
- ITU-R P. 526 Propagation by **Diffraction**
- ITU-R P. 676 Attenuation by **atmospheric gases**
- ITU-R P.840 Attenuation due to **clouds and fog**

SMALL SCALE FADING

for

Multipath effects

Large-Scale Propagation

• Large-scale propagation refers to signal behavior over **long distances** and is influenced by **path loss** and **shadowing** due to obstacles like buildings, terrain, and foliage.

- **Path Loss:** Signal attenuation as it travels farther from the transmitter.
- **Shadowing (Slow Fading):**
Caused by large obstacles blocking the signal path, leading to gradual signal strength variations over tens to hundreds of meters.
- **Models Used:** Free-space model, two-ray ground reflection, log-distance path loss model, etc.

Small-Scale Propagation

• Small-scale propagation deals with rapid signal fluctuations over short distances (centimeters to meters) or short time intervals.

- **Caused By:** Multipath propagation, Doppler shifts due to motion.
- **Types:**
 - **Flat Fading:** All frequencies fade equally.
 - **Frequency-Selective Fading:** Different frequencies fade differently due to delay spread.
 - **Fast Fading:** Rapid changes within a symbol duration.
 - **Slow Fading:** Changes slower than the symbol duration.

Multipath Propagation

Multipath propagation occurs when transmitted signals reach the receiver via **multiple paths** due to reflection, diffraction, and scattering.

- **Effects:**
 - **Inter-Symbol Interference (ISI):** Overlapping symbols due to delayed paths.
 - **Constructive/Destructive Interference:** Signal strength may increase or decrease depending on phase alignment.

Mitigation Techniques: Equalization, diversity schemes, adaptive modulation

Topics:

- Parameters of Mobile Multipath Channels
- Small – Scale Fading:
 - Fading Effects due to Multipath Time Delay
 - Fading Effects due to Doppler Spread
- Rayleigh and Ricean Distributions

Major Categories of Fading :

Large Scale Fading :

This is the *loss* that propagation models try to account for mostly dependant on the distance from the transmitter to the receiver also known as *Large Scale Path Loss, Log-Normal Fading or Shadowing*

Small Scale Fading :

Could be 20-30 dB over a fraction of a wavelength. It is Caused by the superposition or cancellation of multipath propagation signals, the speed of the transmitter or receiver or the bandwidth of the transmitted signal. It is also known as *Multipath Fading or Rayleigh Fading*

Small Scale Fading:

The type of fading experienced by a signal propagating through a channel can be determined by the nature of the transmitted signal with respect to the characteristics of the channel.

Factors influencing small scale fading:

- Multipath propagation.
- Speed of the mobile.
- Speed of the surrounding objects.
- Transmission bandwidth of the signal.

4.1 Small-Scale Multipath Propagation

- The three most important effects
 - Rapid changes in signal strength over a small travel distance or time interval
 - Random frequency modulation due to varying Doppler shifts on different multipath signals
 - Time dispersion caused by multipath propagation delays
- Factors influencing small-scale fading
 - **Multipath propagation**: reflection objects and scatters
 - **Speed of the mobile**: Doppler shifts
 - **Speed of surrounding objects**
 - **Transmission bandwidth of the signal**
 - The received signal will be distorted if the transmission bandwidth is greater than the bandwidth of the multipath channel.
 - **Coherent bandwidth**: bandwidth of the multipath channel.

Parameters of Mobile Multipath Channels:

In order to compare different multipath channels we need parameters which quantify the multipath channel, they are:

1. Delay spread
2. Coherence bandwidth
3. Doppler spread
4. Coherence time

T_s : reciprocal bandwidth (symbol period)

B_s : bandwidth of the transmitted signal

B_c : coherent bandwidth

σ_τ : rms delay spread

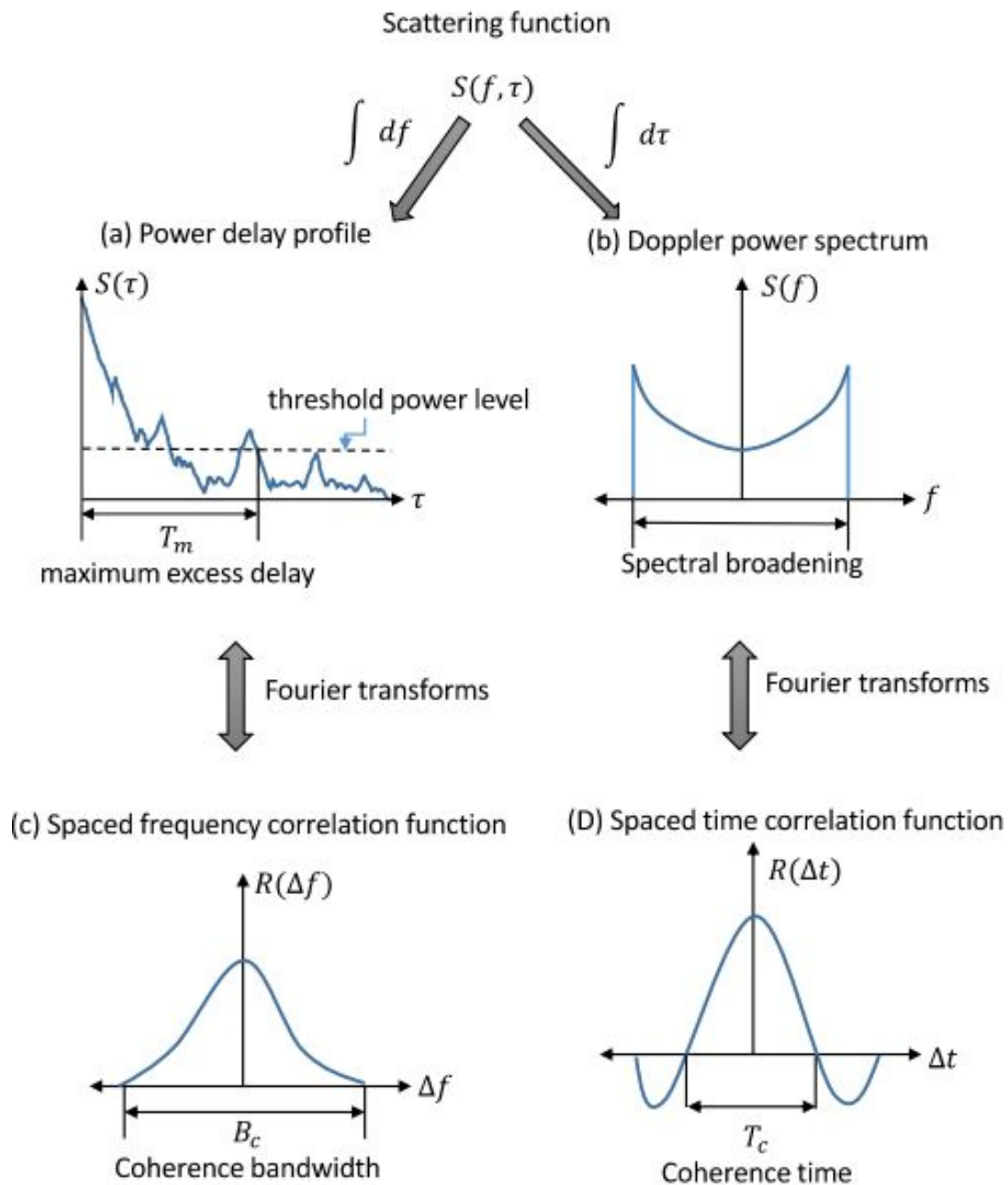
The scattering function combines information about **Doppler spread and the path delays.**

Consider the typical U-shaped Jakes Doppler power spectral density,

- the maximum Doppler shift f_m and
- an exponential delay spread with mean τ_{rms}
- $P_{local-mean}$ is the received power level averaged over an area of tens or hundreds of meters.

$$S(f, \tau) = \frac{P_{local-mean}}{4\pi f_m \sqrt{\left(1 - \frac{(f-f_c)^2}{f_m^2}\right)}} \frac{1}{\tau_{rms}} \exp\left[-\frac{\tau}{\tau_{rms}}\right]$$

Relation between
scattering function,
power delay profile,
Doppler power
spectrum,
spaced frequency
correlation function
and spaced time
correlation function



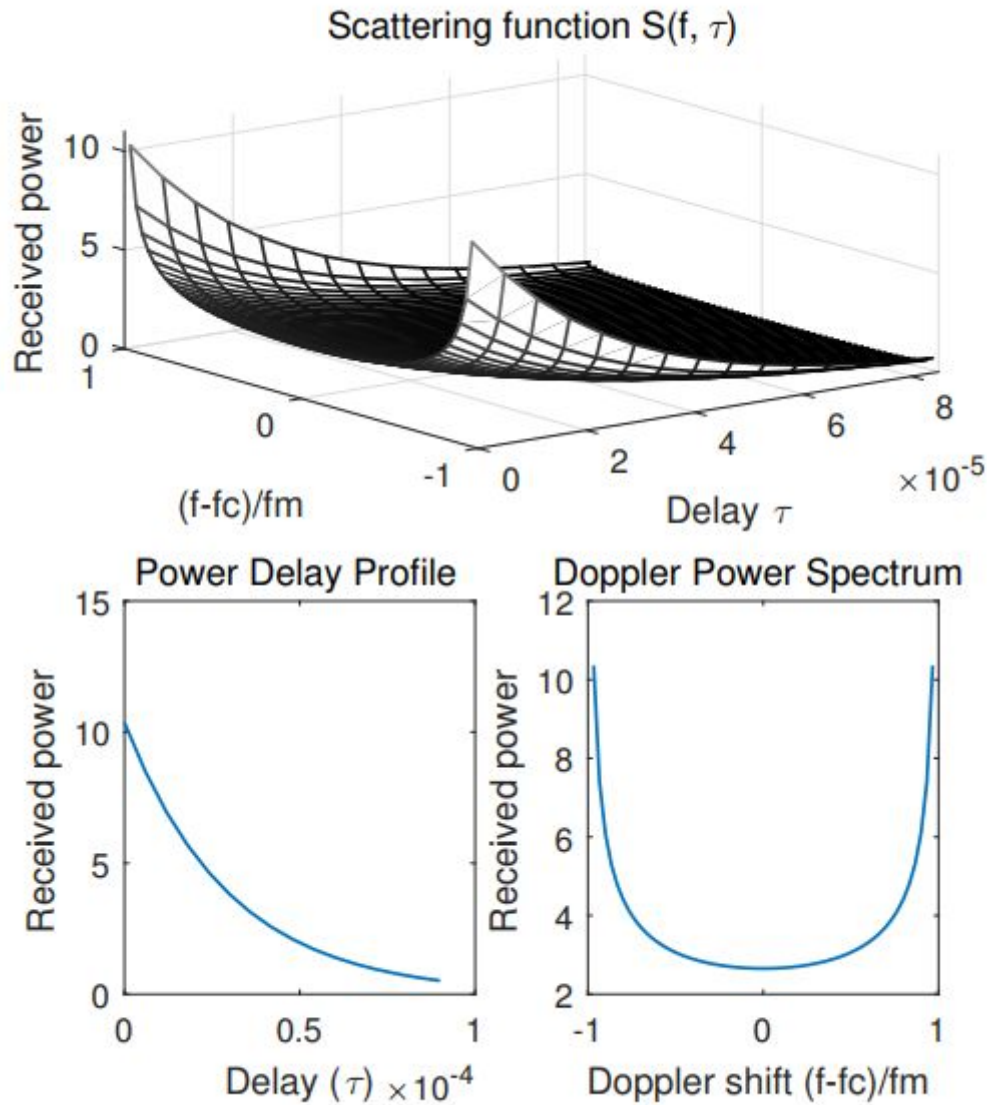


Fig: Scattering function, power delay profile and Doppler power spectrum

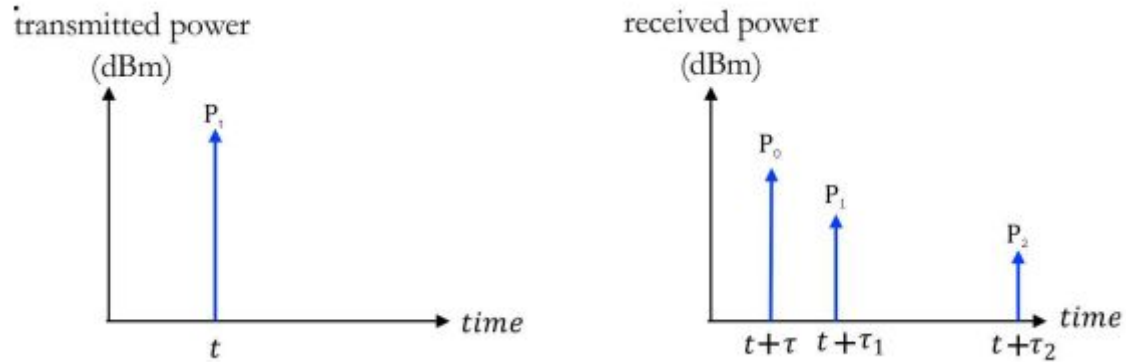


Fig: A typical discrete power delay profile plot for a multipath channel with 3 paths

$$\tau_{rms} = \sqrt{\frac{\int_{-\infty}^{\infty} S(\tau)(\tau - \bar{\tau})^2 d\tau}{\int_{-\infty}^{\infty} S(\tau) d\tau}}$$

where, the mean delay $\bar{\tau}$ is given by

$$\bar{\tau} = \sqrt{\frac{\int_{-\infty}^{\infty} S(\tau)\tau d\tau}{\int_{-\infty}^{\infty} S(\tau) d\tau}}$$

For discrete PDP (as in Figure 11.4), the RMS delay spread (τ_{rms}) can be calculated as

$$\tau_{rms} = \sqrt{\frac{\sum_i p_i (\tau_i - \bar{\tau})^2}{\sum_i p_i}}$$

where, p_i is the power of the i^{th} path, τ_i is the delay of the i^{th} path and the mean delay $\bar{\tau}$ is given by

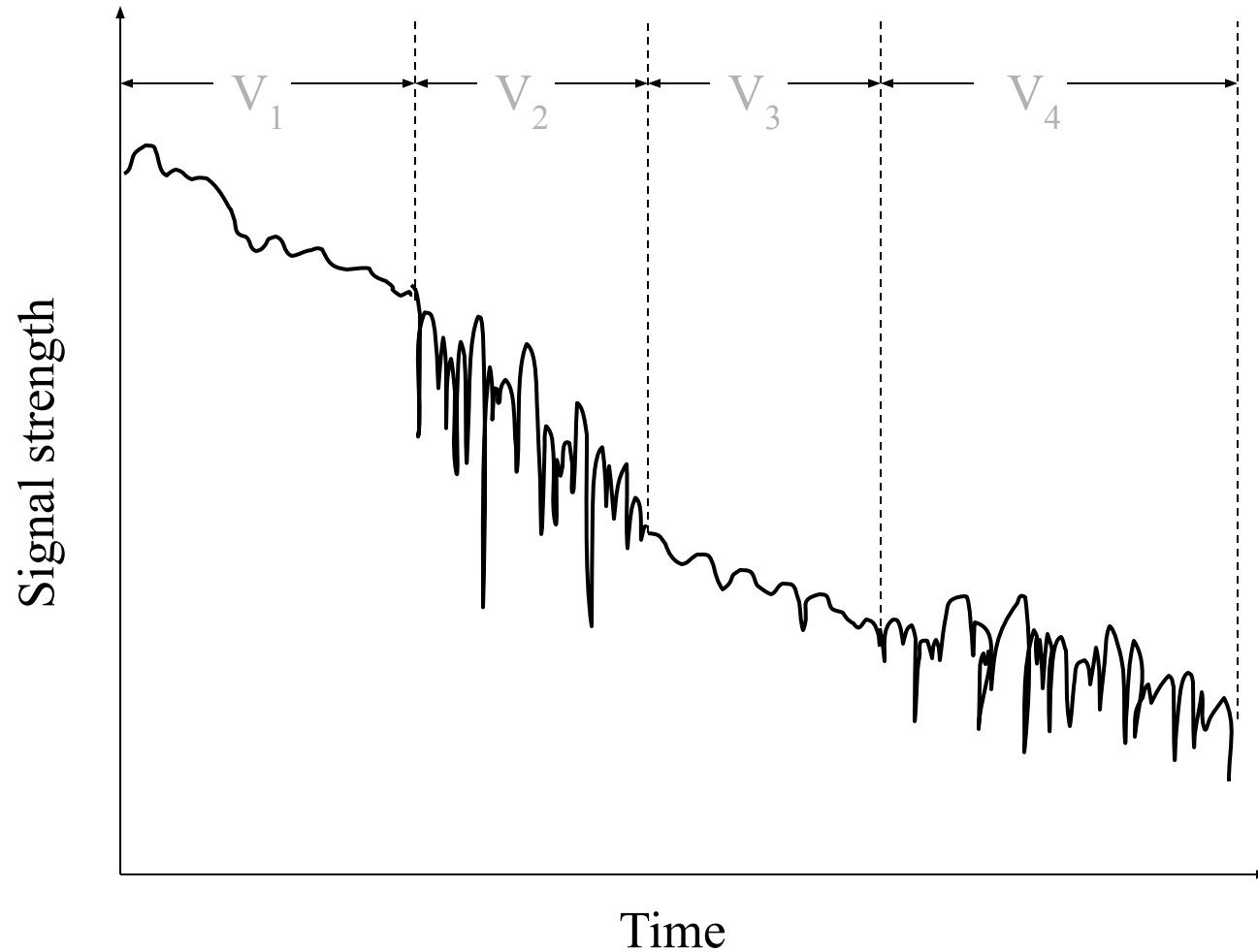
$$\bar{\tau} = \sqrt{\frac{\sum_i p_i \tau_i}{\sum_i p_i}}$$

The ratio of RMS delay spread (τ_{rms}) and symbol time duration (τ_{sym}) quantifies the strength **of intersymbol interference (ISI)**.

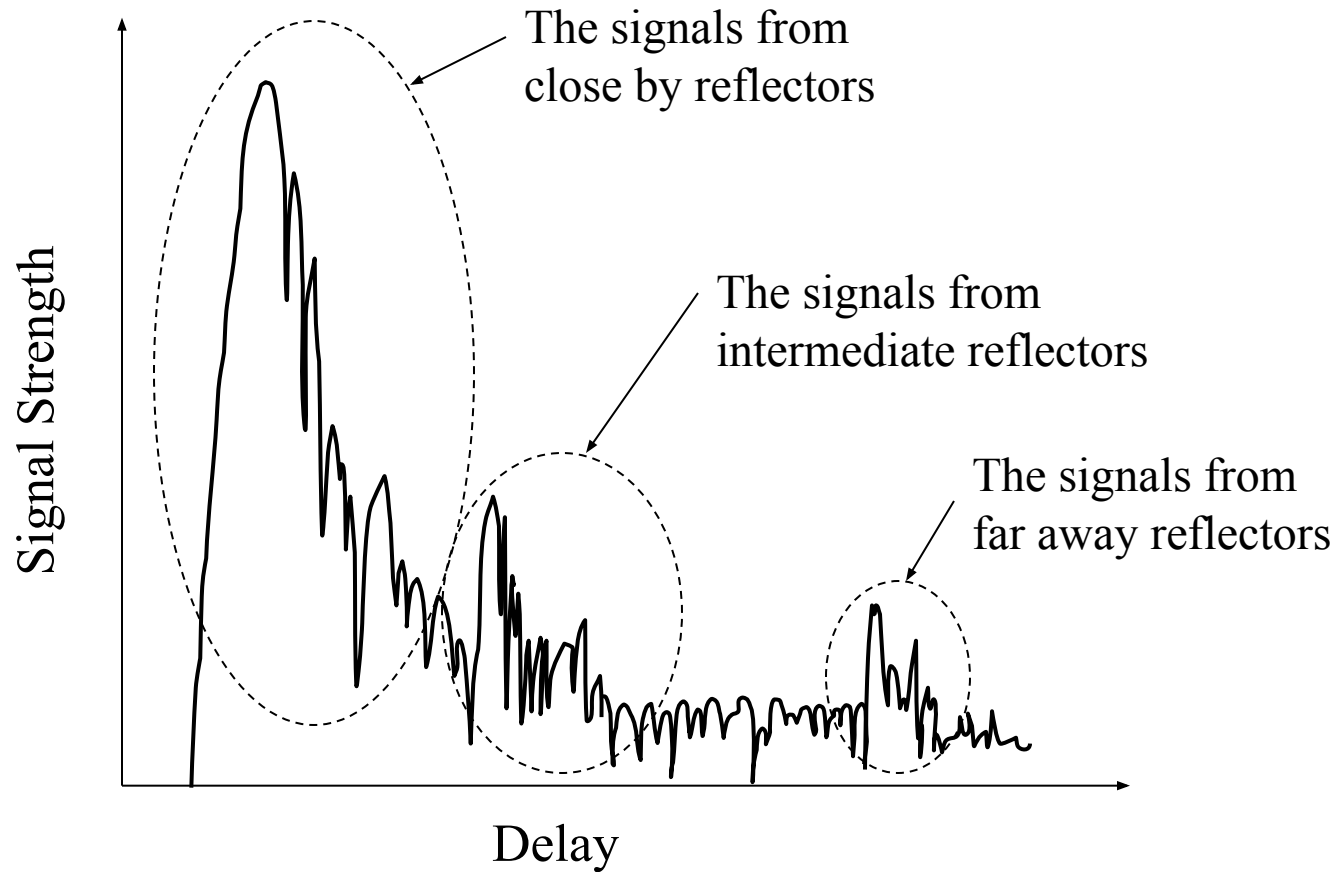
Delay Spread

- When a signal propagates from a transmitter to a receiver, signal suffers one or more reflections.
- This forces signal to follow different paths.
- Each path has different path length, so the time of arrival for each path is different.
- This effect which spreads out the signal is called “**Delay Spread**”.

Moving Speed Effect



Delay Spread



Delay Spread : refers to the time dispersion of a signal caused by multipath propagation.

- When a signal is transmitted, it may reach the receiver via multiple paths (reflections, scattering, diffraction).
- Each path has a different length, so the signal components arrive at slightly different times.
- This causes the received signal to spread out in time, leading to **inter-symbol interference (ISI)** in digital systems.

Key Metric:

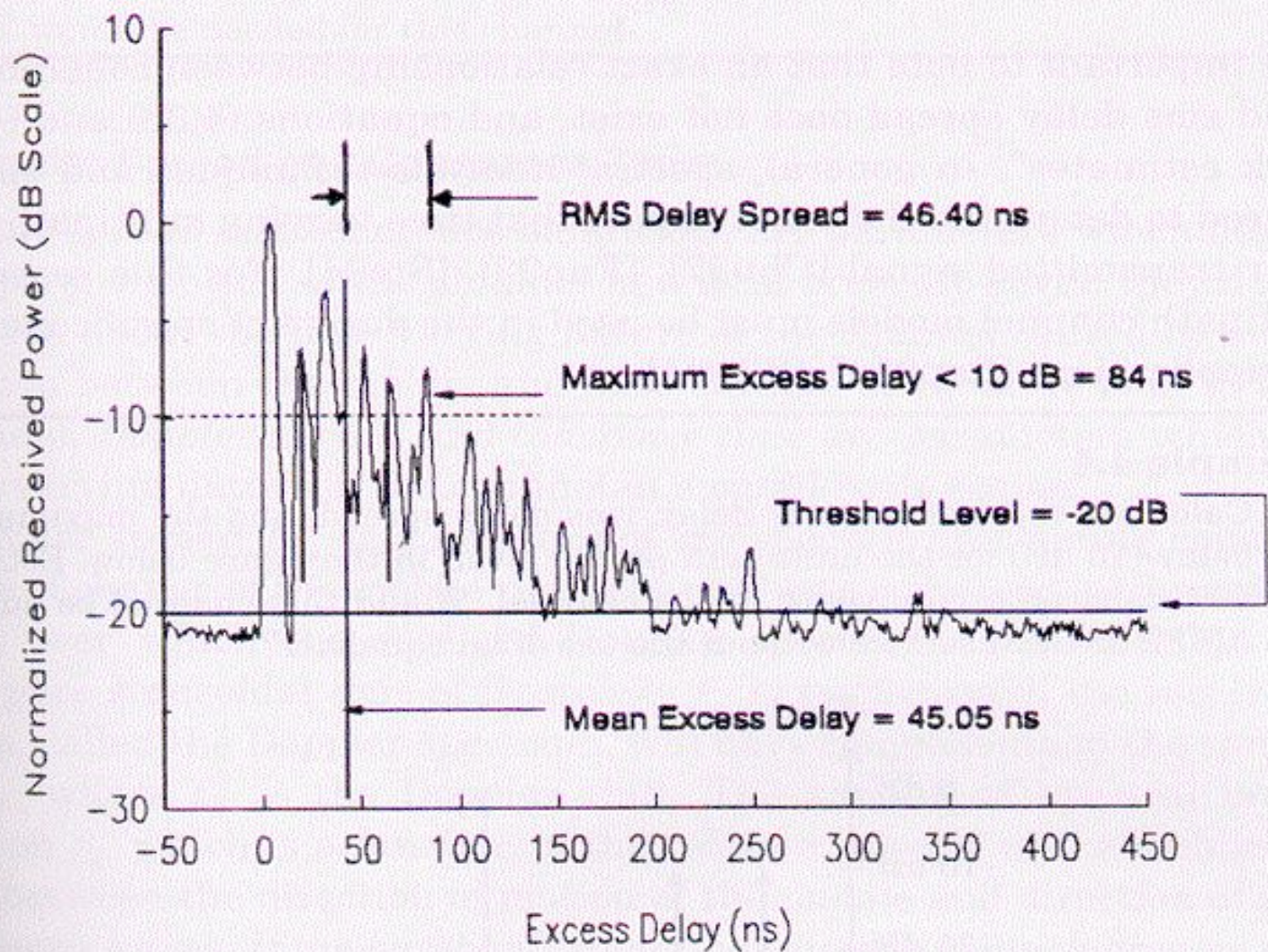
- **RMS Delay Spread**: Measures the average time difference between the earliest and latest arriving signal components.
- A larger delay spread means more distortion and a narrower coherence bandwidth, affecting frequency-selective fading.

Delay Spread:

- Mean excess delay
- RMS delay spread
- Excess delay spread

Mean excess delay is the first moment of the power delay profile and is defined by the equation

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_h P(\tau_k) \tau_k}{\sum_h P(\tau_k)}$$



RMS delay spread is the square root of the second central moment of the power delay profile and is defined by the equation:

$$\sigma_{\tau} = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}$$

where

$$\overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2} = \frac{\sum_h P(\tau_k) \tau_k^2}{\sum_h P(\tau_k)}$$

Maximum excess delay is defined as the $\tau_x - \tau_0$, where τ_0 is the first arriving signal and τ_x is the maximum delay at which a multipath component is within X dB of the strongest arriving multipath signal.

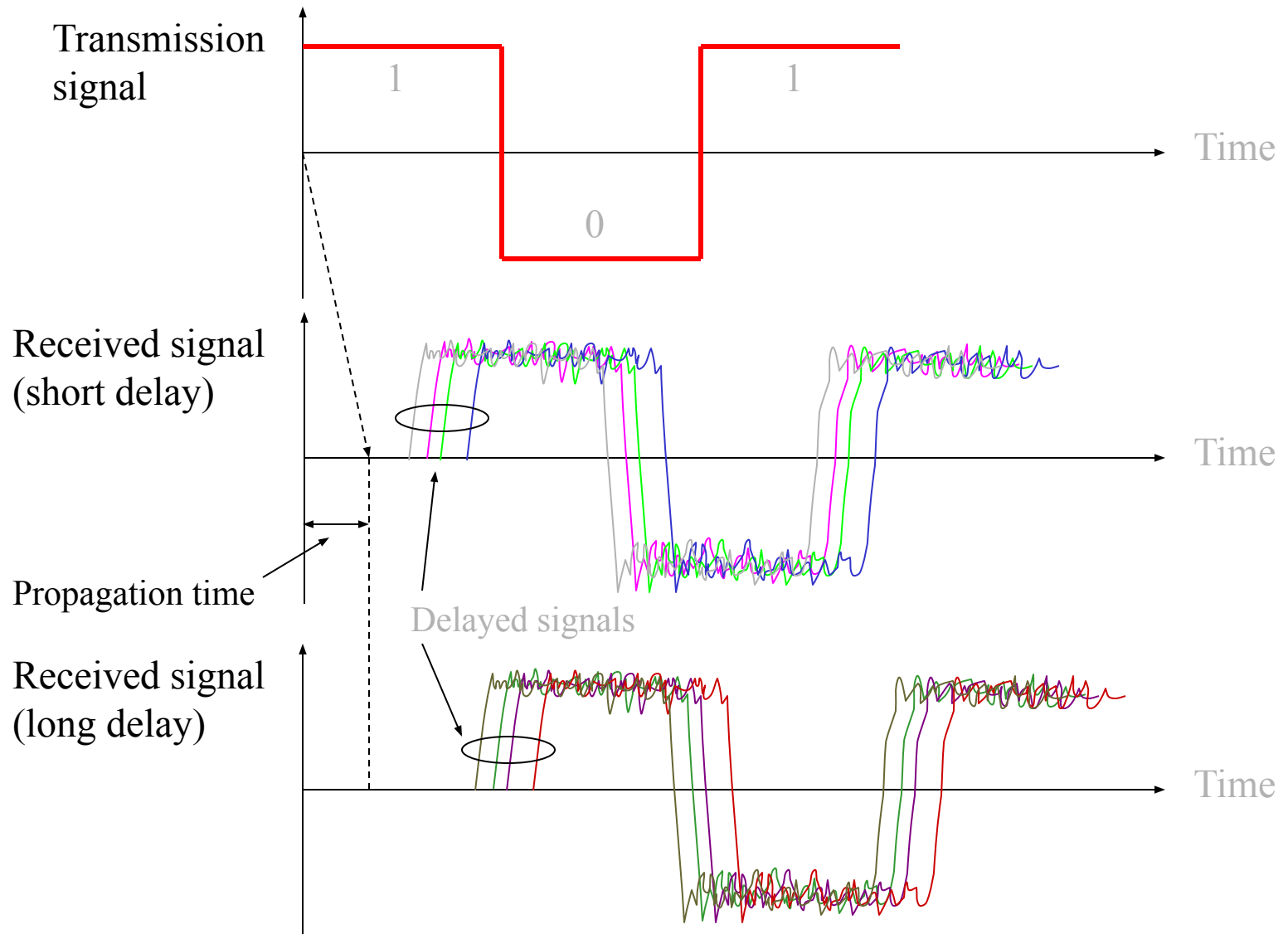
Intersymbol Interference (ISI)

- Caused by time delayed multipath signals
- Has impact on burst error rate of channel
- Second multipath is delayed and is received during next symbol
- For low bit-error-rate (BER)

$$R < \frac{1}{2\tau_d}$$

- R (digital transmission rate) limited by delay spread τ_d .

Intersymbol Interference (ISI)



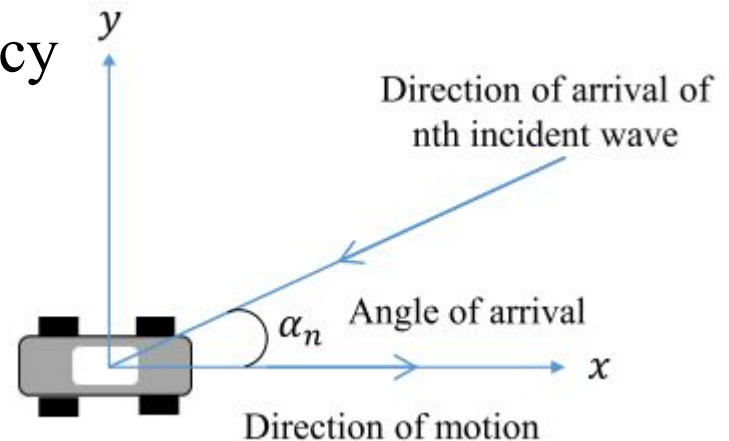
Doppler effect describes the frequency shift of a signal due to relative motion between the transmitter and receiver.

- If the transmitter or receiver is moving, the frequency of the received signal changes.
 - Moving toward each other $\rightarrow$ frequency increases.
 - Moving away $\rightarrow$ frequency decreases.

- This causes **Doppler spread/ Frequency dispersion**, which is the range of frequency shifts due to multipath components arriving at different angles and speeds.

Impact:

- Doppler spread affects the **coherence time** of the channel.
- A high Doppler spread means the channel changes rapidly, leading to fast fading



If Doppler effect exists, it makes the **impulse response of the channel time-variant** and the principle of superposition of incident waves is a linear phenomenon. Therefore, a mobile communication system can be modeled as ***a linear time variant system.***

In multipath propagation, channels are broadly classified as

- frequency-selective channels
- frequency non-selective channels (also called frequency-flat channels). [groups of scattered waves are not resolvable at the receiver, that is, the propagation delay differences of the signal components arriving at the receiver are negligible compared to the symbol interval, then the channel has only one propagation path]

• Doppler Shift

- A mobile moves at a constant velocity v , along a path segment having length d between points X and Y .
- Path length difference

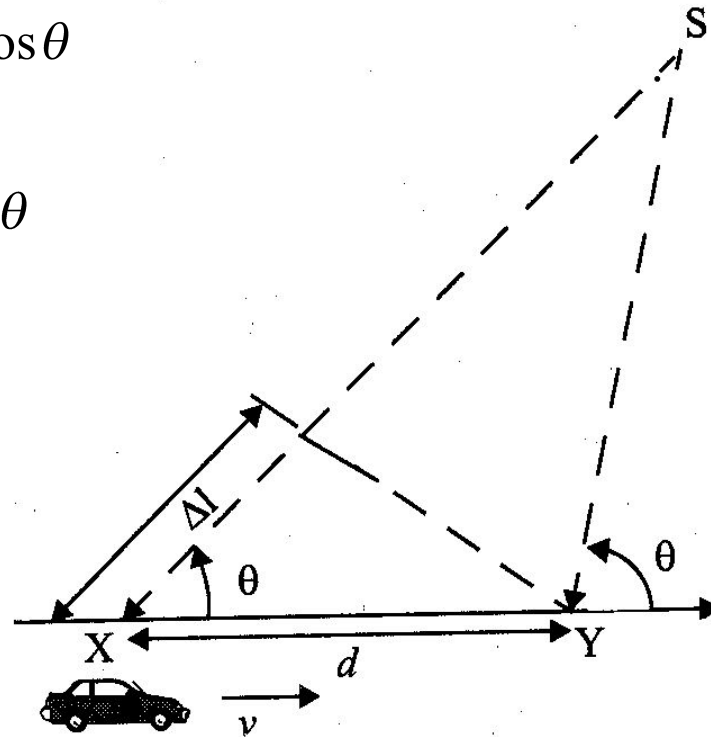
$$\Delta l = d \cos \theta = v \Delta t \cos \theta$$

- Phase change

$$\Delta \phi = \frac{2\pi \Delta l}{\lambda} = \frac{2\pi v \Delta t}{\lambda} \cos \theta$$

- Doppler shift

$$f_d = \frac{1}{2\pi} \cdot \frac{\Delta \phi}{\Delta t} = \frac{v}{\lambda} \cos \theta$$



Doppler Shift

- Doppler Effect: When a wave source and a receiver are moving towards each other, the frequency of the received signal will not be the same as the source.
 - When they are moving toward each other, the frequency of the received signal is higher than the source.
 - When they are opposing each other, the frequency decreases.

Thus, the frequency of the received signal is

$$f_R = f_C - f_D$$

where f_C is the frequency of source carrier,

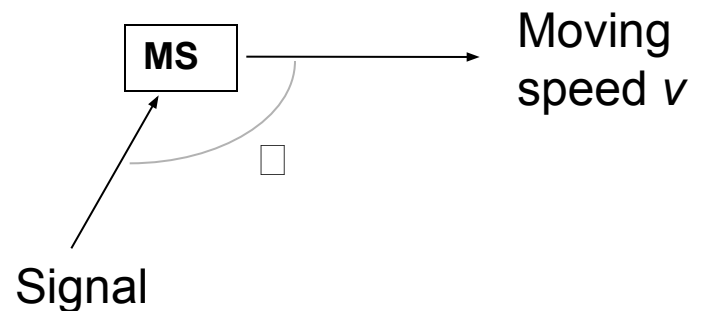
f_D is the Doppler frequency.

- Doppler Shift in frequency:

$$f_D = \frac{v}{\lambda} \cos \theta$$

where v is the moving speed,

λ is the wavelength of carrier.



Q.1 Consider a transmitter which radiates a sinusoidal carrier frequency of **1850 MHz**. For a vehicle moving **60 mph**, compute the received carrier frequency if the mobile is moving (a) directly towards the transmitter, (b) directly away from the transmitter, (c) in a direction which is perpendicular to the direction of arrival of the transmitted signal.

Solution: Given: Carrier frequency $f = 1850 \text{ MHz}$

Therefore, wavelength $= c/f_c = 0.162 \text{ m}$

Vehicle speed $v = 60 \text{ mph} = 26.82 \text{ m/s}$

(a) The vehicle is moving directly towards the transmitter.

The Doppler shift in this case is positive and the received frequency

$$\begin{aligned} \text{is given by equation } f &= f_c + f_d = 1850 \times 10^6 + \frac{26.82}{0.162} = \\ &= 1850.00016 \text{ MHz} \end{aligned}$$

(b) The vehicle is moving directly away from the transmitter. The Doppler shift in this case is negative and hence the received frequency is given by

$$f = f_c - f_d$$

$$f = 1850 * 10^6 - (26.82/0.162) = 1849.999834\text{MHz}$$

(c) The vehicle is moving perpendicular to the angle of arrival of the transmit-ted signal.

In this case, $\theta = 90^\circ$ $\cos\theta = 0$ and there is no Doppler shift.

The received signal frequency is the same as the transmitted frequency of 1850 MHz.

Cochannel Interference

- Cells having the same frequency interfere with each other.
- r_d is the desired signal
- r_u is the interfering undesired signal
- β is the protection ratio for which
 $r_d \leq \beta r_u$ (so that the signals interfere the least)
- If $P(r_d \leq \beta r_u)$ is the probability that $r_d \leq \beta r_u$,
Cochannel probability $P_{co} = P(r_d \leq \beta r_u)$

4.4.2 Coherent Bandwidth

- **Coherent bandwidth**, B_c , is a statistic measure of the range of frequencies over which the channel can be considered to be “flat”.
- Two sinusoids with frequency separation greater than B_c are affected quite differently by the channel.
- Coherent bandwidth is defined as the bandwidth over which the frequency correlation function is above 0.9, then

the coherent bandwidth is approximately

$$B_c \approx \frac{1}{50\sigma_\tau}$$

- If the frequency correlation function is above 0.5

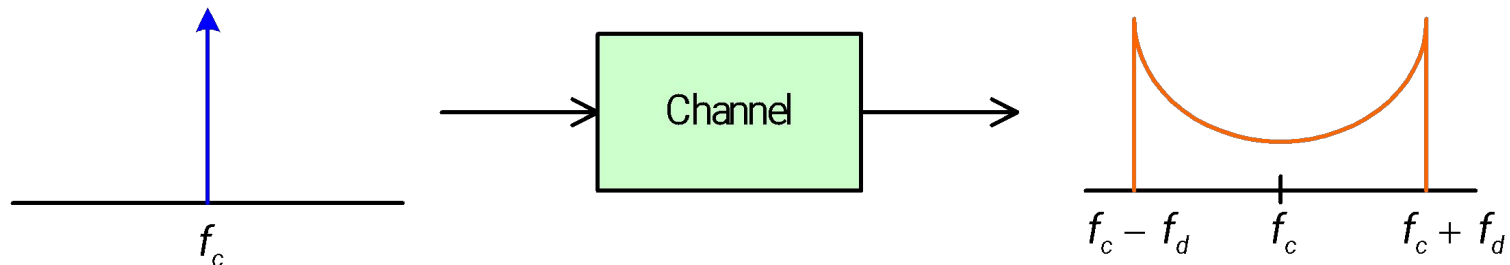
$$B_c \approx \frac{1}{5\sigma_\tau}$$

Coherence Bandwidth

- Coherence bandwidth B_c :
 - Represents correlation between 2 fading signal envelopes at frequencies f_1 and f_2 .
 - Is a function of delay spread.
 - Two frequencies that are larger than coherence bandwidth fade independently.
 - Concept useful in diversity reception
 - Multiple copies of same message are sent using different frequencies.

4.4.3 Doppler Spread and Coherent Time

- Doppler spread and coherent time are parameters which describe the time varying nature of the channel in a small-scale region.
- When a pure sinusoidal tone of f_c is transmitted, the received signal spectrum, called the Doppler spectrum, will have components in the range $f_c - f_d$ and $f_c + f_d$, where f_d is the Doppler shift.



- f_d is a function of the relative velocity of the mobile, and the angle between the direction of motion of the mobile and direction of arrival of the scattered waves

- **Coherent time T_c** is the time domain dual of Doppler spread.
- Coherent time is used to characterize the time varying nature of the frequency dispersiveness of the channel in the time domain.

$$T_c \approx \frac{1}{f_m}$$

f_m : maximum Doppler shift given by $f_m = v / \lambda$

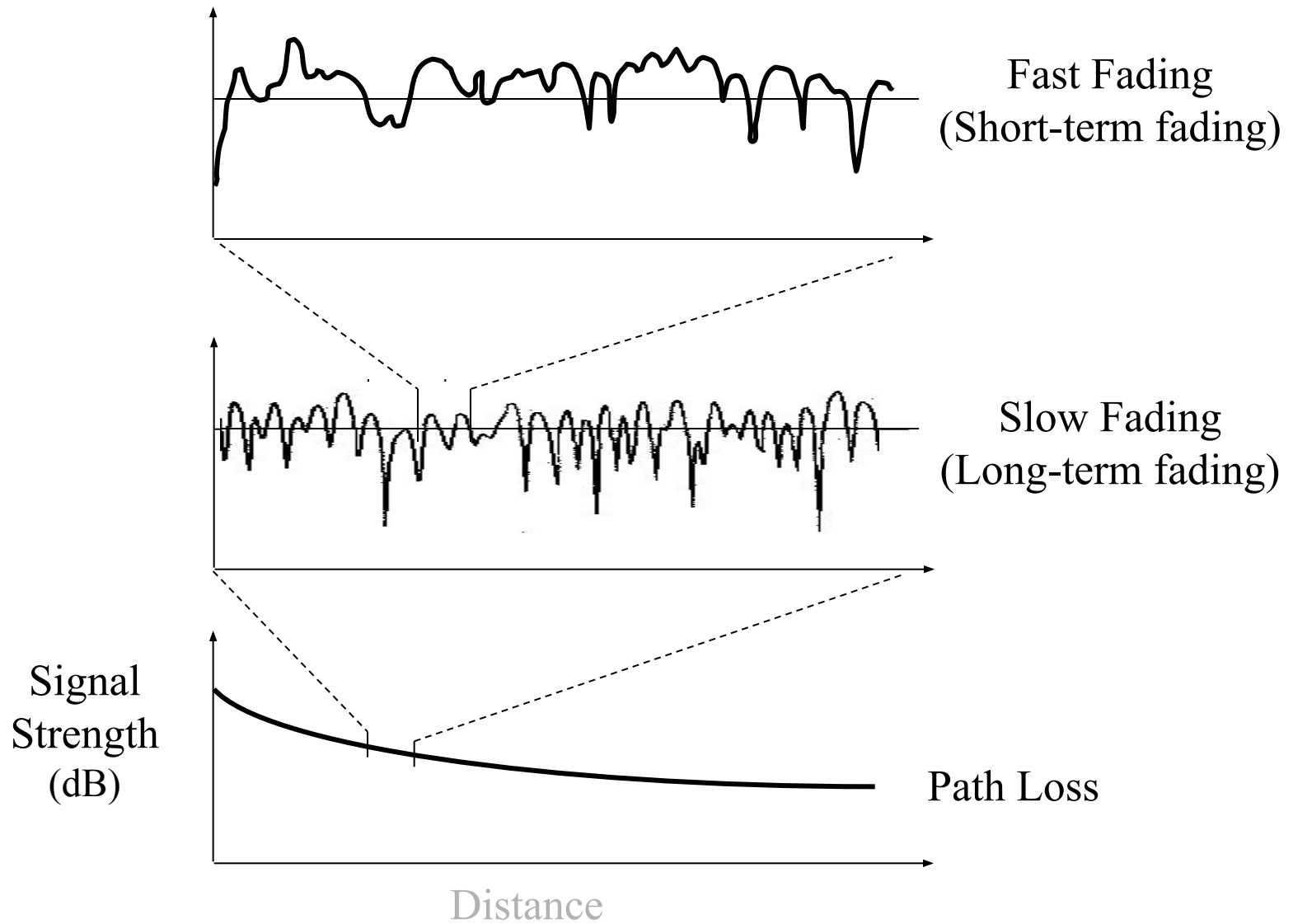
v : speed of the mobile λ : speed of the light

- Two signals arriving with a time separation greater than T_c are affected differently by the channel
- A statistic measure of the time duration over which the channel impulse response is essentially invariant.
- Coherent time is defined as the :

time over which the time co-releation function is above 0.5, then

$$T_c \approx \frac{9}{16\pi f_m}$$

Fading



4.4 Types of Small-Scale Fading

- Multipath delay spread leads to *time dispersion* and *frequency selective fading*.
- Doppler spread leads to *frequency dispersion* and *time selective fading*.
- Multipath delay spread and Doppler spread are independent of one another.

Small-Scale Fading

(Based on multipath time delay spread)

Flat Fading

1. BW of signal < BW of channel
2. Delay spread < Symbol period

Frequency Selective Fading

1. BW of signal > BW of channel
2. Delay spread > Symbol period

Small-Scale Fading

(Based on Doppler spread)

Fast Fading

1. High Doppler spread
2. Coherence time < Symbol period
3. Channel variations faster than baseband signal variations

Slow Fading

1. Low Doppler spread
2. Coherence time > Symbol period
3. Channel variations slower than baseband signal variations

T_s : reciprocal bandwidth (symbol period)

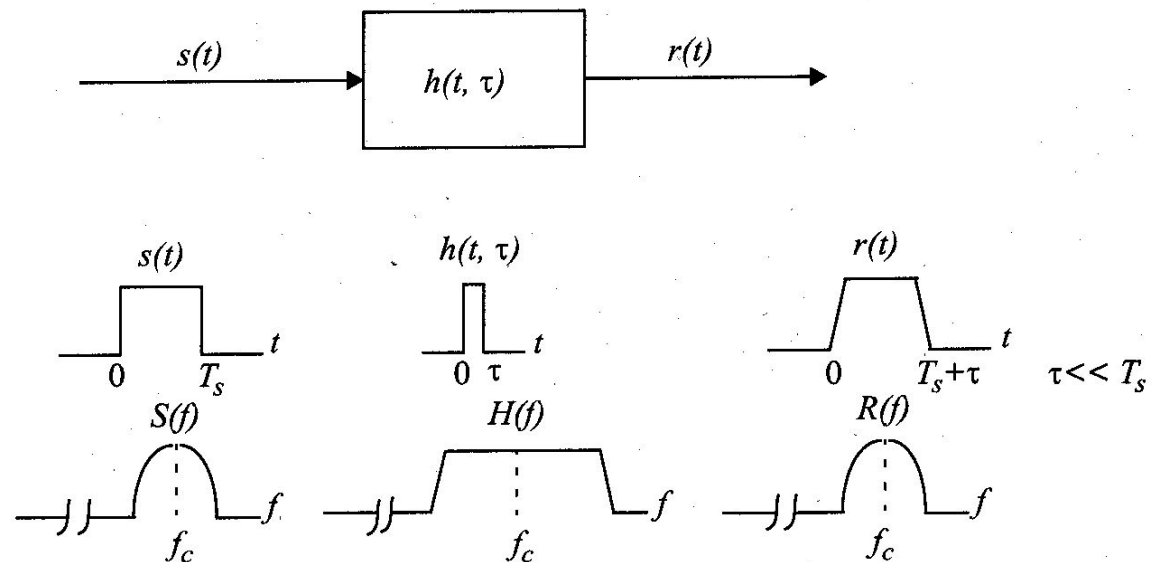
B_s : bandwidth of the transmitted signal

B_c : coherent bandwidth

σ_τ : rms delay spread

4.5.1 Flat Fading

- If the channel has a constant gain and linear phase response over a bandwidth which is greater than the bandwidth of the transmitted signal, the received signal will undergo **flat fading**.
- The received signal strength changes with time due to fluctuations in the gain of the channel caused by multipath.
- The received signal varies in gain but the spectrum of the transmission is preserved.



- Flat fading channel is also called **amplitude varying channel**.
- Also called **narrow band channel**: bandwidth of the applied signal is *narrow* as compared to the channel bandwidth.
- Time varying statistics: Rayleigh flat fading.
- A signal undergoes flat fading if

and
$$B_s \ll B_c$$

$$T_s \gg \sigma_\tau$$

T_s : reciprocal bandwidth (symbol period)

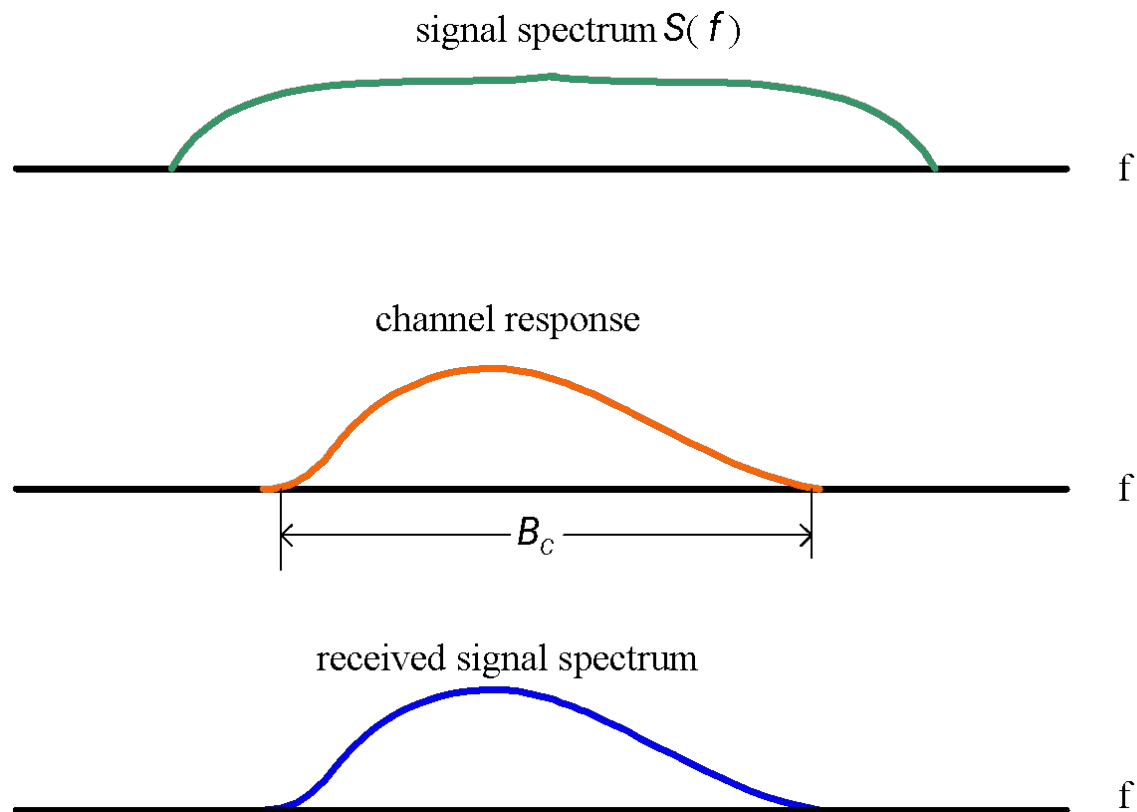
B_s : bandwidth of the transmitted signal

B_c : coherent bandwidth

σ_τ : rms delay spread

4.5.1 Frequency Selective Fading

- If the channel possesses a constant-gain and linear phase response over a bandwidth that is smaller than the bandwidth of transmitted signal, then the channel creates frequency selective fading.



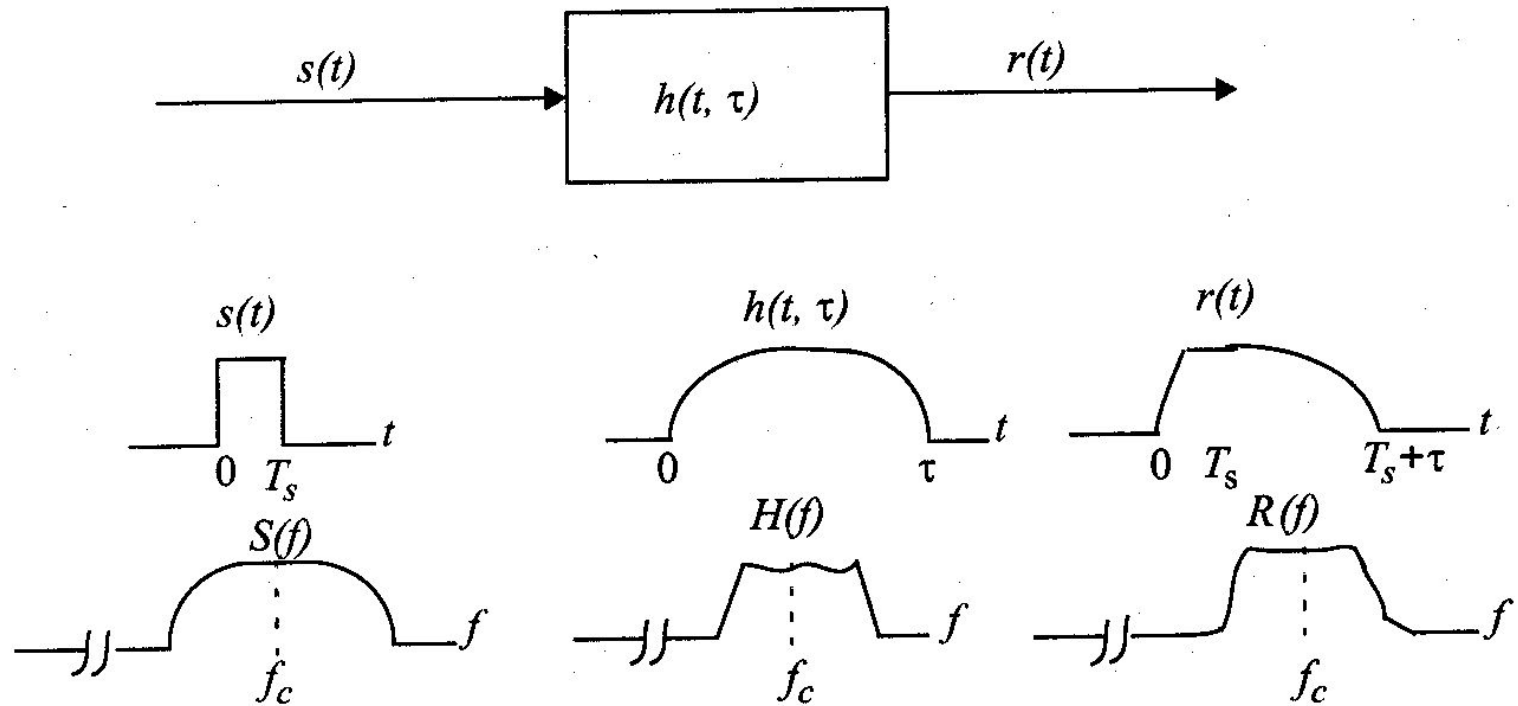
- Frequency selective fading is due to time dispersion of the transmitted symbols within the channel.
 - Induces intersymbol interference
- Frequency selective fading channels are much more difficult to model than flat fading channels.
- Statistic impulse response model
 - 2-ray Rayleigh fading model
 - computer generated
 - measured impulse response
- For frequency selective fading

and

$$B_s > B_c$$

$$T_s > \sigma_\tau$$

- Frequency selective fading channel characteristic



4.5.2 Fading Effects Due to Doppler Spread

- **Fast Fading:** The channel impulse response changes rapidly within the symbol duration.
 - The coherent time of the channel is smaller than the symbol period of the transmitted signal.
 - Cause frequency dispersion due to Doppler spreading.
- A signal undergoes fast fading if

$$T_s > T_c$$

and

$$B_s < B_d$$

- **Slow Fading**: The channel impulse response changes at a rate much slower than the transmitted baseband signal $s(t)$.
 - The Doppler spread of the channel is much less than the bandwidth of the baseband signal.
- A signal undergoes slow fading if

$$T_s \ll T_c$$

and

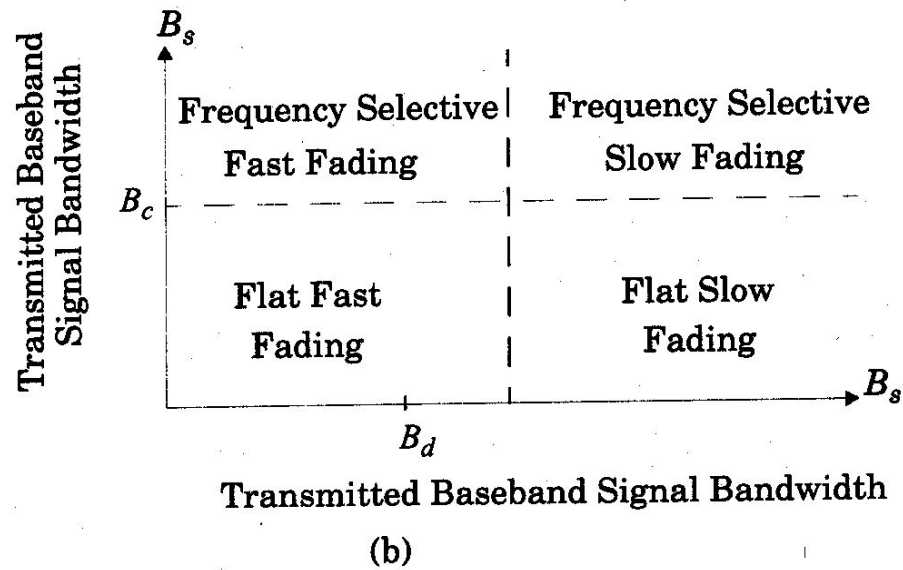
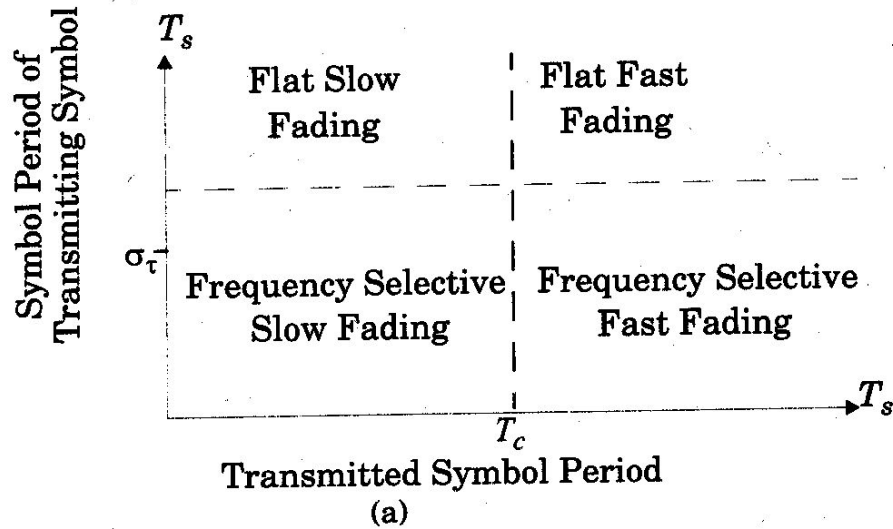
$$B_s \gg B_D$$

Slow Fading

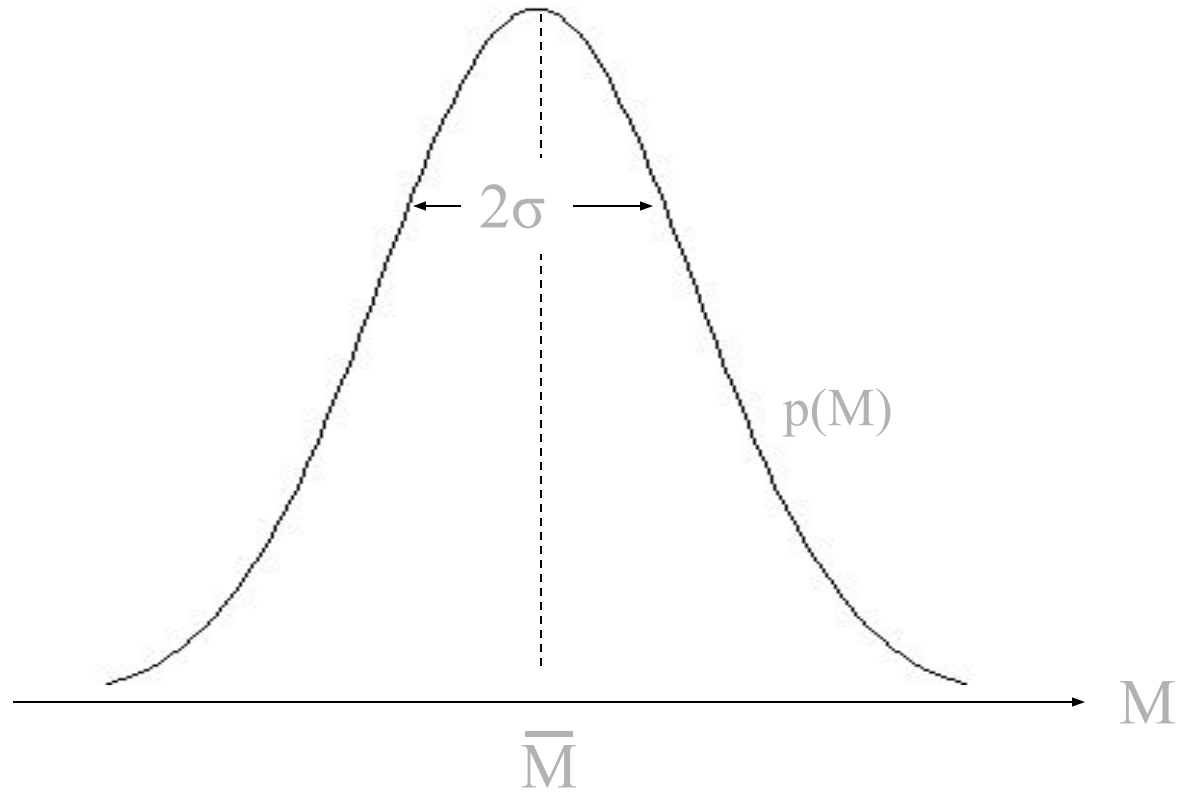
- The long-term variation in the mean level is known as slow fading (shadowing or log-normal fading). This fading caused by shadowing.
- Log-normal distribution:
 - The pdf of the received signal level is given in decibels by

$$p(M) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(M-\bar{M})^2}{2\sigma^2}},$$

where M is the true received signal level m in decibels, i.e., $10\log_{10}m$,
 $\bar{M}$ is the area average signal level, i.e., the mean of M ,
 σ is the standard deviation in decibels



Log-normal Distribution



The pdf of the received signal level

Fast Fading

- The signal from the transmitter may be reflected from objects such as hills, buildings, or vehicles.
 - When MS far from BS, the envelope distribution of received signal is Rayleigh distribution. The pdf is

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2}{2\sigma^2}}, \quad r > 0$$

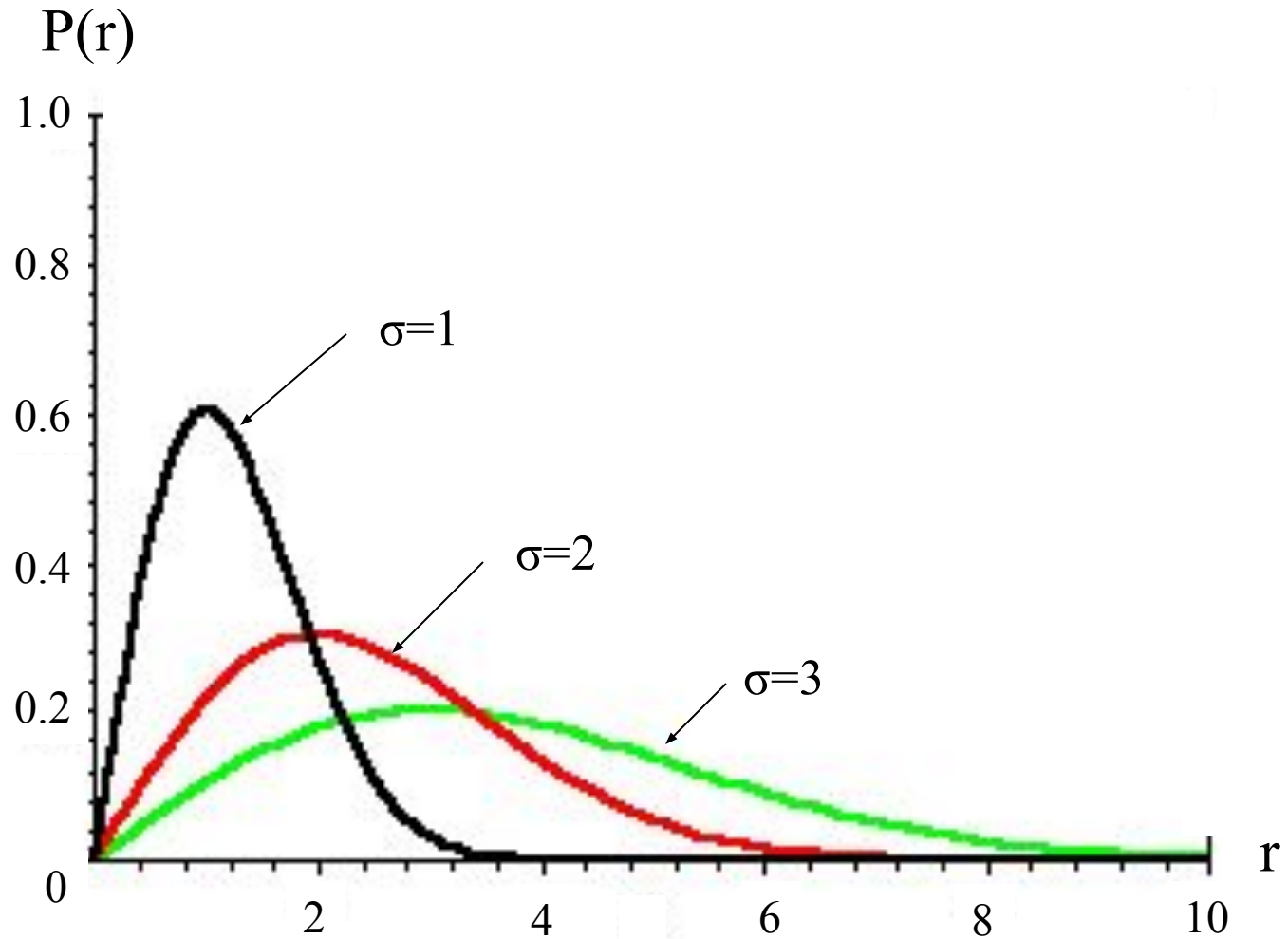
where σ is the standard deviation.

- Middle value r_m of envelope signal within sample range to be satisfied by

$$P(r \leq r_m) = 0.5.$$

- We have $r_m = 1.777 \square$

Rayleigh Distribution



The pdf of the envelope variation

Fast Fading (Continued)

- When MS far from BS, the envelope distribution of received signal is Rician distribution. The pdf is

$$p(r) = \frac{r}{\sigma^2} e^{-\frac{r^2 + \alpha^2}{2\sigma^2}} I_0\left(\frac{r\alpha}{\sigma}\right), \quad r \geq 0$$

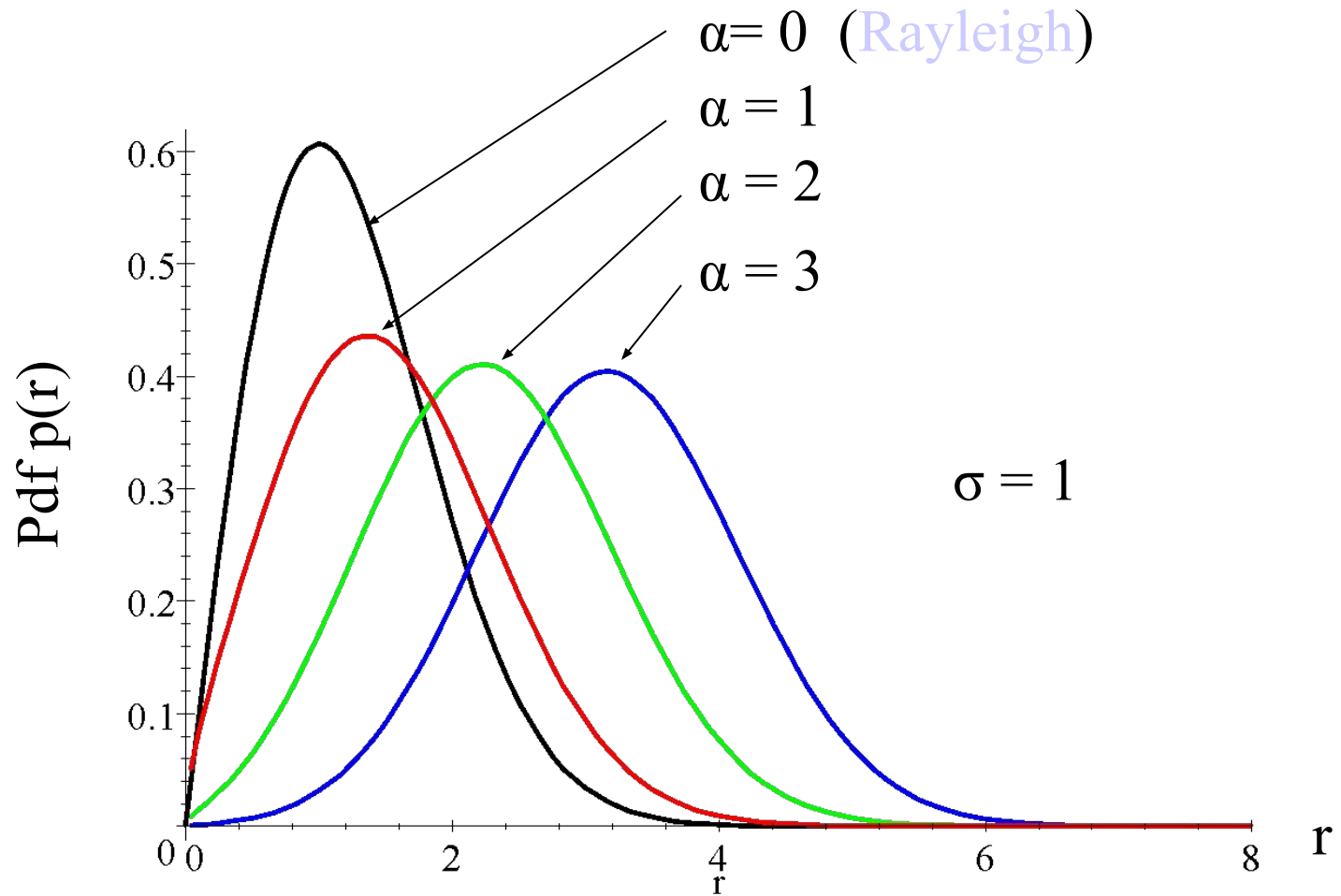
where

σ is the standard deviation,

$I_0(x)$ is the zero-order Bessel function of the first kind,

α is the amplitude of the direct signal

Rician Distribution



The pdf of the envelope variation

Characteristics of Instantaneous Amplitude

- Level Crossing Rate:
 - Average number of times per second that the signal envelope crosses the level in positive going direction.
- Fading Rate:
 - Number of times signal envelope crosses middle value in positive going direction per unit time.
- Depth of Fading:
 - Ratio of mean square value and minimum value of fading signal.
- Fading Duration:
 - Time for which signal is below given threshold.

Coherence bandwidth:

It is the range of frequencies over which two frequency components have a potential for amplitude correlation.

If two sinusoids with a frequency separation of greater than B_c are propagating in the same channel, they are affected quite differently by the channel.

Doppler Spread and Coherence Time:

Doppler spread and Coherence Time take into account the relative motion between mobile and base station, or by movements of objects in the channel.

They describe the time varying nature of the channel in a small scale region.

Doppler Spread B_d :

When a signal of frequency f_c is transmitted, the received signal spectrum, called the Doppler spectrum, will have components $f_c - f_d$ to $f_c + f_d$, where f_d is the Doppler shift.

Coherence time T_c :

It is used to characterize the time varying nature of the frequency dispersiveness of the channel in the time domain